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Global Materials

Aluminum: From Deficit to Oversupply

The combination of Middle East supply restarts and greenfield capacity ramp-ups points to a structurally looser aluminum market into 2027.

Aluminum: Faster Middle East restart, rising 2027 supply trimming exposure on recovery: We turn more cautious on aluminum equities into any near-term recovery, particularly among producers, where earnings and multiples remain most exposed to normalising LME aluminum prices and regional premiums. Key change in view is Middle East supply now returning faster than initially expected, while RoW supply additions are also progressing more quickly – shifting the aluminum market from a modest deficit in 2026 to a clearer surplus in 2027/28. The Middle East conflict removed around 3.5Mt of annualised aluminum capacity. This supply shock is now reversing faster than expected – most affected capacity projected to return by end-1H27. **Also, elevated profitability is incentivising new supply additions in RoW, which are also progressing more quickly – 2.1Mt of new capacity additions and 1.8Mt of global restarts in 2027, we estimate,** with Indonesia a notable contributor as some operators reallocate power from nickel to aluminum production. We now expect 866kt/1.2Mt/1.3Mt of new Indonesian production in 2026/2027/28, and 440kt/835kt/905kt new production from RoW.

As a result, market balance has deteriorated: 2026 deficit narrows; 2027 shifts into surplus. We now forecast a smaller 2026 deficit of 1.1 Mt (vs. 1.8 Mt prev.). More importantly, 2027 shifts into a surplus of more than 800kt, versus our previous forecast of approx. 800kt deficit. By 2028, the market moves from broadly balanced to a 1.8Mt surplus as projects currently in development ramp up. Our commodities team forecasts aluminum prices averaging US$3,150/t in 2H26 and falling to US$2800 by 4Q27, alongside easing regional premiums.

2027/28e Earnings cuts: Lower aluminum price assumptions, easing regional premiums and looser supply balance reduce earnings momentum for aluminum-exposed equities (Exhibit 3). Global aluminum equities have already corrected as concerns around Middle East conflict ease – share prices down 20-50% from recent peaks. Our "what's in the price" analysis suggests China aluminum names already discount aluminum prices 8–19% below spot, implying less downside risk than some global peers. However, with 2026 still in deficit but 2027/28 moving into oversupply, and with US rate-hike risk appearing to moderate after June jobs data, we would use any near-term bounce to trim exposure rather than add risk. Downgrade **AA** in US, **Nanshan** and **Tianshan** in China, from **OW** to **EW**.

Main upside risks to our more cautious view: China introducing tighter production controls; existing smelters forced to shut due to power-contract challenges, accidents or operational disruptions; higher copper prices driving substitution into aluminum, which could keep aluminum prices higher for longer.

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GREATER CHINA MATERIALS

Asia Pacific
Industry View **Attractive**

What's Changed?

	Currency	Rating New	Rating Old	PT New	PT Old	Upside/ Downside to PT
China Materials (Rmb/share)						
Chalco (H)	HKD	OW	OW	11.2	18.1	48%
Chalco (A)	CNY	OW	OW	10.8	17.5	31%
China Hongqiao	HKD	OW	OW	28.6	52.9	37%
Chuangxin Industries (H)	HKD	OW	OW	22.0	39.5	40%
Tianshan Alumnum (A)	CNY	EW	OW	13.3	23.0	17%
Nanshan Aluminium	CNY	EW	OW	4.6	7.3	12%
Shenhuo Coal and Power	CNY	OW	OW	31.0	42.3	42%
North Amrica (US$/share)						
Alcoa	USD	EW	OW	53.0	79.0	8%
India (Rs/share)						
Hindalco	INR	OW	OW	1140.0	1325.0	18%

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Investment Summary

We are turning more cautious on aluminum equities into any near-term recovery, particularly for producers, where earnings and multiples remain most exposed to normalising LME aluminum prices and regional premiums. The key change versus our prior view is not that the Middle East disruption was temporary — that was always our assumption — but that the restart now appears to be occurring faster than initially expected, while rest-of-world supply additions are also progressing more quickly. In our view, this shifts the aluminum market from a modest deficit in 2026 to a clearer surplus in 2027/28.

The Middle East conflict was the most important aluminum supply-side event in 2026. At its peak, the disruption removed around 3.5 Mt of annualised capacity, equivalent to 4.0–4.8% of global capacity. This supply shock is now reversing faster than expected, with most affected Middle Eastern capacity assumed to return by end-1H27. At the same time, elevated aluminum profitability is incentivising a new wave of greenfield and brownfield supply, particularly in Indonesia. With aluminum returns improving while nickel remains loss-making, some operators are reallocating power from nickel to aluminum. We now expect Indonesia to add 0.8Mt, 1.3 Mt and 1.3 Mt of production in 2026, 2027 and 2028, respectively. For the rest of the world, we forecast additions of 440 kt, 835 kt and 905 kt over the same period, with key 2027 projects including Saudi Arabia at 260 kt, India at 300 kt and Angola at 190 kt. In total, we expect around 2.1 Mt of new global supply in 2027, excluding Middle East restarts.

As a result, our commodities team expects the market balance to deteriorate materially beyond 2026. It now forecasts a smaller 2026 deficit of 1.1 Mt, but a surplus of more than 800kt in 2027, versus its previous forecast of an approximately 800 kt deficit. By 2028, the market shifts from broadly balanced to a 1.8 Mt surplus as projects currently in development ramp up. Our commodities team forecasts aluminum prices averaging US $3,150/t in 2H26 and US$2,850/t in 2027, alongside easing regional premiums.

The earnings impact is likely to be limited in 2026E, as the market remains in deficit and spot prices/premiums should remain supportive near term. The larger impact is in 2027/28E, where lower aluminum price assumptions, easing regional premiums and a more comfortable supply balance reduce earnings momentum for aluminum-exposed equities. We expect the greatest earnings pressure among high operating-leverage producers and names with more direct exposure to LME aluminum and regional premiums. We therefore reflect this through lower 2027/28E EPS assumptions for the most aluminum-sensitive names, while keeping 2026E revisions comparatively modest. See Exhibit 3 .

Some of this bearish view is already priced in. Global aluminum equities have corrected as the Middle East disruption has eased: Chinese aluminum stocks are down around 50% from their January peak, while Alcoa and Century have declinded around 40%, Norsk Hydro around 28%, and Hindalco around 20%. Our “what’s in the price” analysis suggests China A/H aluminum names are already discounting aluminum prices 8–19% below spot, implying less downside risk than some global peers. However, with the market remaining in deficit in 2026 but moving into oversupply in 2027/28, we would use any near-term rally to trim exposure rather than add risk.

Key upside risks to our more cautious view include China introducing tighter production controls, existing smelters shutting due to power-contract challenges or operational disruptions, and higher copper prices driving substitution into aluminum, which could keep aluminum prices higher for longer. We remain positive on copper, gold and a few other materials sub segments, so keep our wider Greater China Materials industry view as Attractive.

Global aluminum capacity and power contracts

Power cost and security are becoming an increasingly important considerations for global aluminum smelters. Across the industry, power-related risk is shifting from pure availability toward power price visibility, access to decarbonised power and contract duration. The key power contract renewal risk concentrated in the 2027–2029 window.

Exhibit 1: Alcoa and NHY power contract renewal details

Smelter	Capacity / exposure (Alcoa share)	Power Type	Contract expiry / renegotiation timing	Contract arrangement	Priority of renewal
			Alcoa		
San Ciprián, Spain	228kt capacity	Grid electricity (natural gas / market-based)	Electricity swaps expire **December 2027.** Long-term renewable PPAs are in place but depend on permitting and development of windfarms	Variable spot-market power; fixed-for-floating electricity swaps; two long-term renewable PPAs expected to supply ~40% of full-capacity power needs	**Highest risk.** The company does not provide information on protection once the swaps expire, with no disclosed replacement hedge plan, PPA pricing, firm project timing, or site-level power economics. Given San Ciprián's history of substantial losses primarily driven by high energy costs, failure to secure competitive post-2027 coverage could pressure margins, undermine restart economics, or increase curtailment risk.
Lista, Norway	95kt capacity	Hydroelectric / Renewable (purchased)	Coverage through 2027.	Several PPAs secure about 90% of required power through 2027; remaining power is bought at spot rates.	**Medium-high 2028 risk.** No post-2027 coverage is disclosed by the company. If Alcoa cannot extend or replace the PPAs, Lista could move from mostly contracted power to much higher spot exposure from 2028.
Mosjøen, Norway	200kt capacity	Wind (renewable) — converted to baseload via firming contract	Coverage through 2028.	Several long-term PPAs secure about 90% of required power through 2028, then 65% from 2029 through 2035; remaining power is spot.	**Medium risk.** Firming arrangement disclosed for July 2025 through Dec. 2028.
Bécancour, Quebec	350kt capacity	Hydroelectric (Manicouagan HPP / Hydro-Québec)	Contracts expire: **Dec. 31, 2029**	Hydro-Québec electricity contracts	**Medium risk.** Later-dated but important because Québec is a large part of Alcoa's smelting base.
Deschambault, Quebec	287kt capacity	Hydroelectric (Manicouagan HPP / Hydro-Québec)	Contacts expire: **Dec. 31, 2029**	Hydro-Québec electricity contracts	**Medium risk.** Later-dated but important because Québec is a large part of Alcoa's smelting base.
Fjarðaál, Iceland	351kt capacity	Hydroelectric / Geothermal (purchased)	Contract expires: 2048	Hydroelectric power supplied by Landsvirkjun, Iceland's national power company.	**Low risk.** Supply term is very long, but price renegotiation is effective from 2028 which is critical because economics can change at renegotiation.
Baie-Comeau, Quebec	324kt capacity	Hydroelectric (Manicouagan HPP / Hydro-Québec)	Hydro-Québec contracts expire **Dec. 31, 2029** ~25% of power from Manicouagan agreement expires **Feb. 2036.**	Hydro-Québec contracts; ~25% from Manicouagan Power Limited Partnership, which owns hydroelectric generation	**Low risk: Automatic renewal through Feb. 2036.** Large Canadian exposure, but later-dated. Main risk appears to be tariff / economics rather than physical availability.
			NHY		
Norweigan smelters	426.7 GWh/yr annual; 1.3 TWh total over 3 years	Renewable (PPA)	Delivery 2024-2027; contract coverage expires 2027.	Long-term power purchase with Statkraft for the NO3 area. Annual contracted volume about 426.7 GWh/yr.	Low risk. Expiring contract in NO3 area (2027) is covered by newer long-term contracts in the same area.
Norweigan smelters	150 GWh/yr annual; 900 GWh total over 6 years	Renewable (PPA)	Delivery 2021-2027; contract coverage expires 2027.	Long-term power purchase with Halsfund Eco for the NO5 area. Annual contracted volume about 150 GWh/yr.	Low risk. Expiring contract in NO5 area (2027) is covered by newer long-term contracts in the same area.
Norweigan smelters	196.7 GWh/yr annual; 1.2 TWh total over 6 years	Renewable (PPA)	Delivery 2021-2027; contract coverage expires 2027.	Long-term power purchase with Halsfund Eco for the NO1 area. Annual contracted volume about 196.7 GWh/yr.	Low risk. Expiring contract in NO1 area (2027) is covered by newer long-term contracts in the same area.
Norweigan smelters	113.3 GWh/yr annual; 680 GWh total over 6 years	Renewable (PPA)	Delivery 2022-2028; contract coverage expires 2028.	Long-term power purchase with Halsfund Eco for the NO3 area. Annual contracted volume about 113.3 GWh/yr.	Medium risk. Coverage rolls off in 2028; however, NO3 area has substantial replacement volume from other contracts.
Norweigan smelters	220 GWh/yr annual; 660 GWh total over 3 years	Renewable (PPA)	Delivery 2027-2029; contract coverage expires 2029.	Long-term power purchase with NTE for the NO3 area. Annual contracted volume about 220 GWh/yr.	Medium risk. Expiry falls in the 2029-2030 window; not immediate, but important for medium-term portfolio coverage.
Norweigan smelters	216.7 GWh/yr annual; 1.3 TWh total over 6 years	Renewable (PPA)	Delivery 2023-2029; contract coverage expires 2029.	Long-term power purchase with Halfslund Eco for the NO3 area. Annual contracted volume about 216.7 GWh/yr.	Medium risk. Expiry falls in the 2029-2030 window; not immediate, but important for medium-term portfolio coverage.
Norweigan smelters	750 GWh/yr annual; 6.8 TWh total over 9 years	Renewable (PPA)	Delivery 2021-2030; contract coverage expires 2030.	Long-term power purchase with Statkraft 1 for the NO3 area. Annual contracted volume about 750 GWh/yr. From other deal with	Medium risk. Expiry falls in the 2029-2030 window; not immediate, but important for medium-term portfolio coverage.
Norweigan smelters	500 GWh/yr annual; 5 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with BKK Produksjon AS for the NO5 area. Annual contracted volume about 500 GWh/yr. BKK in NO5 from reading	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	300 GWh/yr annual; 3 TWh total over 10 years	Hydro (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Eidesiva Vankraft for the NO1 area. Annual contracted volume about 300 GWh/yr. Eidevisa in NO1 from	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	250 GWh/yr annual; 2.5 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Axpo Trading AG for the NO2 area. Annual contracted volume about 250 GWh/yr. For Karmoy Pilot.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	500 GWh/yr annual; 5 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Adger Energi AS for the NO2 area. Annual contracted volume about 500 GWh/yr. Agder Energi located in	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	500 GWh/yr annual; 5 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Axpo Trading AG for the NO2 area. Annual contracted volume about 500 GWh/yr. From Elitsa.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	700 GWh/yr annual; 7 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Lyse for the NO2 area. Annual contracted volume about 700 GWh/yr. Stavanger Leverandar.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	1000 GWh/yr annual; 10 TWh total over 10 years	Renewable (PPA)	Delivery 2021-2031; contract coverage expires 2031.	Long-term power purchase with Adger Energi AS for the NO2 area. Annual contracted volume about 1000 GWh/yr. Agder Energi located	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	250 GWh/yr annual; 2 TWh total over 8 years	Renewable (PPA)	Delivery 2025-2032; contract coverage expires 2032.	Long-term power purchase with Skellefteå Kraft for the SE2 area. Annual contracted volume about 250 GWh/yr.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	147.5 GWh/yr annual; 1.2 TWh total over 8 years	Renewable (PPA)	Delivery 2024-2032; contract coverage expires 2032.	Long-term power purchase with Alpiq for the NO3 area. Annual contracted volume about 147.5 GWh/yr.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	67.5 GWh/yr annual; 540 GWh total over 8 years	Renewable (PPA)	Delivery 2025-2033; contract coverage expires 2033.	Long-term power purchase with Alpiq for the SE3 area. Annual contracted volume about 67.5 GWh/yr.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	263 GWh/yr annual; 2.6 TWh total over 10 years	Renewable (PPA)	Delivery 2026-2035; contract coverage expires 2035.	Long-term power purchase with Axpo Nordic for the SE2 area. Annual contracted volume about 263 GWh/yr.	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance.
Norweigan smelters	1040 GWh/yr annual; 14.6 TWh total over 14 years	Wind (PPA)	Delivery 2021-2035; contract coverage expires 2035.	Long-term power purchase with Nordic Wind Power DA 2 for the NO3 area. Annual contracted volume about 1040 GWh/yr. Annual baseload	Medium-low risk. Contract extends into the 2030s, but volume still matters for long-term portfolio balance. Large annual volume raises strategic importance.
Norweigan smelters	475 GWh/yr annual; 5.7 TWh total over 12 years	Renewable (PPA)	Delivery 2024-2036; contract coverage expires 2036.	Long-term power purchase with Fortum for the NO3 area. Annual contracted volume about 475 GWh/yr.	Low risk. Long-dated coverage.
Norweigan smelters	218.8 GWh/yr annual; 1.8 TWh total over 8 years	Renewable (PPA)	Delivery 2031-2038; contract coverage expires 2038.	Long-term power purchase with Alpiq for the NO3 area. Annual contracted volume about 218.8 GWh/yr.	Low risk. Long-dated coverage.
Norweigan smelters	440 GWh/yr annual; 6.6 TWh total over 15 years	Renewable (PPA)	Delivery 2024-2039; contract coverage expires 2039.	Long-term power purchase with Statkraft for the NO3 area. Annual contracted volume about 440 GWh/yr.	Low risk. Long-dated coverage.
Norweigan smelters	1000 GWh/yr annual; 8 TWh total over 8 years	Renewable (PPA)	Delivery 2031-2039; contract coverage expires 2039.	Long-term power purchase with Statkraft 2 for the NO3 area. Annual contracted volume about 1000 GWh/yr. From other deal with	Low risk. Long-dated coverage. Large annual volume raises strategic importance.
Norweigan smelters	740 GWh/yr annual; 2.2 TWh total over 3 years	Wind (PPA)	Delivery 2036-2039; contract coverage expires 2039.	Long-term power purchase with Nordic Wind Power DA 3 for the NO3 area. Annual contracted volume about 740 GWh/yr. Annual baseload	Low risk. Long-dated coverage. Large annual volume raises strategic importance.
Norweigan smelters	1650 GWh/yr annual; 29.7 TWh total over 18 years	Wind (PPA)	Delivery 2021-2039; contract coverage expires 2039.	Long-term power purchase with Markbygden Ett AB for the SE2 area. Annual contracted volume about 1650 GWh/yr. from map.	Low risk. Long-dated coverage. Large annual volume raises strategic importance.
Norweigan smelters	350 GWh/yr annual; 1.8 TWh total over 5 years	Renewable (PPA)	Delivery 2036-2040; contract coverage expires 2040.	Long-term power purchase with Hafslund Kraft AS for the NO3 area. Annual contracted volume about 350 GWh/yr.	Low risk. Long-dated coverage.
Norweigan smelters	350 GWh/yr annual; 3.5 TWh total over 10 years	Renewable (PPA)	Delivery 2031-2040; contract coverage expires 2040.	Long-term power purchase with Hafslund Kraft AS for the NO3 area. Annual contracted volume about 350 GWh/yr.	Low risk. Long-dated coverage.
Norweigan smelters	660 GWh/yr annual; 12.5 TWh total over 19 years	Wind (PPA)	Delivery 2022-2041; contract coverage expires 2041.	Long-term power purchase with Blakliden Fäbodberget Wind AB for the SE2 area. Annual contracted volume about 660 GWh/yr. from map.	Low risk. Long-dated coverage. Large annual volume raises strategic importance.
Norweigan smelters	700 GWh/yr annual; 17.5 TWh total over 25 years	Renewable (PPA)	Delivery 2020-2045; contract coverage expires 2045.	Long-term power purchase with Tonstad Vindpark AS for the NO2 area. Annual contracted volume about 700 GWh/yr. from map.	Low risk. Long-dated coverage. Large annual volume raises strategic importance.

Source: Company data, Morgan Stanley Research

What's in the prices on global aluminum equities

Exhibit 2: What's in the price

Aluminum	Chalco (Rmb)	Hongqiao (Rmb)	Chuangxin (Rmb)	Tianshan (Rmb)	Shenhhuo (Rmb)	Norsk Hydro (NOK)	Alcoa (USD)
Stock Price (HK$)	7.58	20.92	15.72	11.36	21.78	86	48.68
shs (mn)	17097	9494	2075	4590	2213	1965	266
Equity Value(HK$bn)	130	199	33	52	48	168990	12945
Equity Value (Rmb bn)	113	173	28	52	48	168990	12945
Net Debt '26e	67	-11	7	3	-2	31020	285
Enterprise Value	180	162	35	55	47	200010	13231
Alumina %	10%	7%	10%				22%
Aluminum %	90%	86%	90%	96%	93%		87%
Aluminum EV	162	139	32	52	43	69037	11492
Assumed Norm. EV/EBITDA multi.	6	5	8	8	7	6	6
Implied aluminum EBITDA (Rmb bn)	27	28	4	7	6	11506	1915
'26 aluminum eqv. vol (mt)	8	7	1	1	2	2	3
Implied Norm. EBITDA/t (Rmb/t)	3343	4259	5262	5007	3808	5848	715
ACP (Rmb/t)	13919	12832	11742	11755	12211	20439	3180
Implied aluminium price (Rmb/t / US$/t)	17263	17091	17003	16763	16019	2672	3895
SHFE Aluminium Price (Rmb/t excl. VAT / US$/t)	20173	20173	20173	20173	20173	3090	3090
	-14%	-15%	-16%	-17%	-21%	-14%	26%
Aluminium Price (LME + Regional Premium US$/t)							4,294
							-9%

Source: Morgan Stanley Research estimates. Price as of July 7, 2026

Exhibit 3: Old. vs. New Changes

	2026e			2027e			2028e		
	New	Old	% Chg	New	Old	% Chg	New	Old	% Chg
China Materials (Rmb/share)									
Chalco (H)	**1.27 e**	1.45 e	-13%	**0.73 e**	1.25 e	-41%	**0.91 e**	1.38 e	-34%
Chalco (A)	**1.27 e**	1.45 e	-13%	**0.73 e**	1.25 e	-41%	**0.91 e**	1.38 e	-34%
China Hongqiao	**3.33 e**	4.09 e	-19%	**2.44 e**	4.35 e	-44%	**2.45 e**	4.58 e	-46%
Chuangxin Industries (H)	**2.28 e**	2.78 e	-18%	**2.20 e**	3.91 e	-44%	**2.62 e**	4.53 e	-42%
Tianshan Alumnum (A)	**1.78 e**	2.14 e	-17%	**1.21 e**	2.27 e	-47%	**1.25 e**	2.33 e	-46%
Nanshan Aluminium	**0.44 e**	0.48 e	-8%	**0.35 e**	0.53 e	-34%	**0.36 e**	0.57 e	-36%
Shenhuo Coal and Power	**3.94 e**	4.26 e	-8%	**2.81 e**	4.14 e	-32%	**2.98 e**	4.46 e	-33%
Europe									
Metlen (EURc)	**391 e**	386 e	1%	**529 e**	525 e	1%	**693 e**	657 e	5%
Norsk Hydro (NOKc)	**841 e**	953 e	-12%	**872 e**	982 e	-11%	**735 e**	925 e	-21%
Rio Tinto (US$c)	**829 e**	876 e	-5%	**757 e**	852 e	-11%	**708 e**	778 e	-9%
North Amrica (US$/share)									
Alcoa	**7.27 e**	8.65 e	-16%	**4.69 e**	8.38 e	-44%	**2.69 e**	5.31 e	-49%
India (Rs/share)									
	F2027e			F2028e			F2029e		
Hindalco	**116.16 e**	115.54 e	1%	**115.35 e**	131.70 e	-12%	**126.91 e**	136.95 e	-7%

Note: Metlen is followed by MS analyst Ioannis Masvoulas, Norsk Hydro by Alain Gabriel, Rio Tinto – AX, by Rahul Anand, Rio Tinto – LDN by Alain Gabriel, Alcoa by Carlos De Alba and Hindalco by Rahul Gupta. Source: Morgan Stanley Research estimates

Aluminum: Rebalancing Comes Quicker

Middle East events have driven a large 2026 deficit and higher smelter margins

Exhibit 4: We estimate around 3.5 mln tonnes of Middle East outages at peak

Smelter	Company	Country	Capacity (ktpa)	Cuts (kt) (annualised)
Al Taweelah	EGA	UAE	1575	1575
Qatalum	QatarEnergy/Norsk Hydro	Qatar	638	383
Alba	Alba	Bahrain	1620	1134
Total			3833	3092
Unconfirmed	-	Iran	650	450
Total			4483	3542

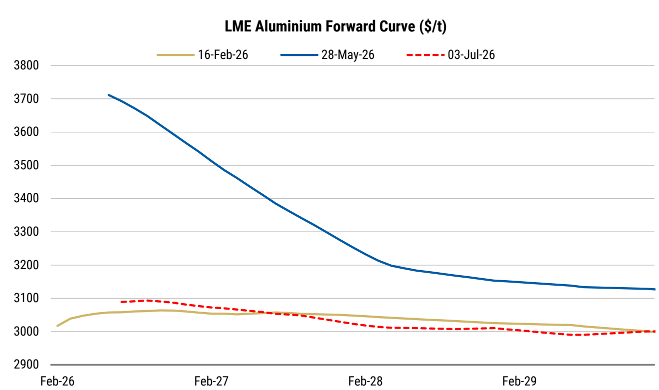
Source: Morgan Stanley Research

Middle East outages reached ~3.5 mtpa at peak: We estimate that around 3.5Mtpa of Middle East smelting capacity was impacted by the conflict, (>4% of global supply), driven by strikes and raw material shortages (summarised in Exhibit 4). This includes unconfirmed reports of disruptions in Iran, with Wood Mackenzie noting a direct attack on one smelter as well as supply chain issues. This materially tightened an already constrained market, with South32's 500 ktpa Mozal smelter also shutting in March, taking prices and premiums higher.

Stronger China output has provided some offset: China's aluminum output has annualised 45.4 Mt YTD in 2026, according to NBS data (or 45.6 based on MySteel, 45.1 on IAI and 45.25 on SMM). All of these estimates imply production at or slightly above the 45.4 mtpa capacity cap. We note that smelters can temporarily overproduce versus their rated capacity by increasing amperage, although this is unlikely to be sustainable over an extended period. We model China's output at 45.9 Mt for 2026, remaining broadly flat thereafter.

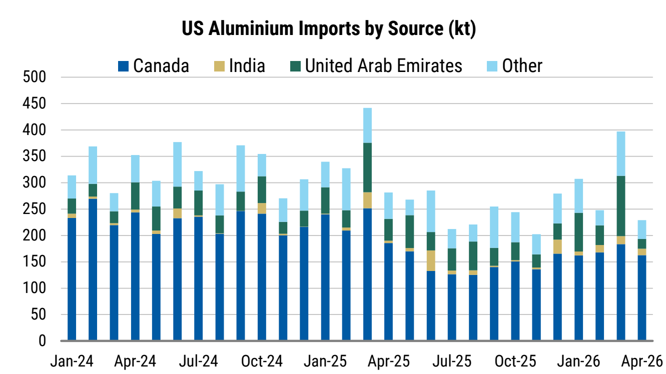
And so has destocking: Much of the supply tightness has been reflected in aluminum timespreads than in prices directly, with the curve moving into a steep backwardation through April/May. This indicates near-term tightness but also impacts inventory decisions. While this can contribute to the "roll yield" for financial participants, it makes holding excess inventory punitive for end users, as the value of inventory declines over time. Platts notes that this is weighing on spot market activity, particularly in the US and Japan, as end users de-stock. We note US imports dropped significantly in April from the recent run rate, suggesting continued de-stocking. While this is unlikely to persist indefinitely, in our view, it is helping keep the market looser than it otherwise would be.

Exhibit 5: The aluminum forward curve in backwardation discouraged inventory holding



Source: Bloomberg

Exhibit 6: US imports fell sharply in April

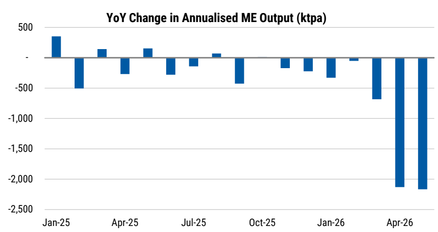


Source: TradeMap

But Middle East tightness is starting to reverse

Stockpiled material is beginning to be released: Following the initial agreement between the US and Iran to end the conflict on June 24 aluminum has given back some of its gains, with record net-long investor positioning into the event likely a key driver here. Some stockpiled material is starting to be shipped out too (although we understand some has been exported by truck since the conflict began, with AlCircle noting that 40-60% of Alba's exports are being re-routed via Jeddah port, and Emirates Global Aluminium (EGA) is also shifting logistics through the Sohar port). Based on one month of production at Al Taweelah (prior to the drone strike), four months of Jebel Ali (still running), partial production at Alba, and four months at 60% utilisation for Qatalum, we estimate that a maximum of ~870 kt of aluminum could have accumulated behind the Strait. However, after accounting for volumes already moved by truck, the figure is more likely 400-500kt. The release of this material could provide temporary relief to the aluminum market and further ease near-term tightness.

Exhibit 7: The YoY change in annualised Middle East production has been smaller than expected versus peak smelter outages



Source: IAI

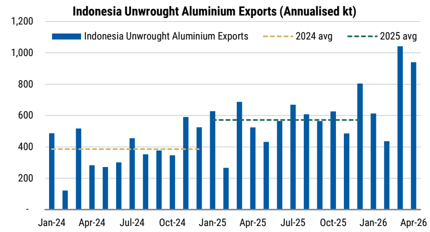
And restarts appear to be happening more quickly: According to IAI data, Middle East output annusalised around 4Mt in April/May, down from 6.1 Mt in Q2 2025, implying a smaller production impact of ~2.1 Mt, suggesting that some of the 3.5 Mt of production that was reportedly impacted at peak disruption may have recovered more quickly. Woodmac notes that most of the decline can be attributed to Al Taweelah (1.6 Mt) with "all other smelters recovering faster than expected… [while] market intelligence points to Al Taweelah restarting ahead of EGA's 12-month guidance and Qatalum increasing utilisation beyond 60%." Alba is reportedly already running at close to 80%, from 30% 4-6 weeks ago. EGA estimated a full restoration of output at the Al Taweelah smelter could take 12 months, but Woodmac noted that some pots are being "jump started" - this can shorten the lifespan of cells, but EGA is also reportedly looking to purchases Chinese steel pot shells to replace damaged units. Taken together, these developments point to a quicker restart. We continue to model Middle East aluminum output declining by 1.8Mt YoY, before recovering by 1.6 Mt in 2026 and rising by a further 750 kt in 2026 as new capacity coming online in Saudi Arabia.

We continue to model a 1.1 Mt deficit for 2026, but this is lower than the 1.8 Mt we previously modelled: In our view, the Middle East supply losses (particularly from the 1.6 Mt Al Taweelah smelter) are likely to outpace any offset from Chinese smelters maximising utilisation and Indonesia ramping faster in 2026. This is likely to keep the market in a deficit of 1.1 Mt. This has been reflected in aluminum timespreads and regional premiums, but LME price action has been held back by destocking at end users, in our view. This means prices could still hold above $3000/t through 2H 2026, particularly if there are any more smelter outages or we see a restock cycle, but are likely to weaken into 2027.

High margins are starting to drive a supply response elsewhere, pointing to a quicker rebalancing in 2027

Exhibit 8: Indonesia aluminum exports have been rising

Indonesia: Significant growth ahead: Indonesia has been building significant aluminum capacity, largely driven by Chinese investment, but production was constrained by power availability. However, changes to nickel policy have reduced the attractiveness of some nickel operations, prompting certain producers, including Tsingshan to reallocate power



Source: TradeMap

capacity toward aluminum, allowing aluminum production growth to be brought forward. We model output rising from 1.55 Mt in 2026 to 2.84 Mt in 2027 and 4.2 Mt in 2028.

Other investments outside of China are accelerating too: We now identify three potential smelters being built by Chinese companies in Angola with a combined capacity of >500 ktpa, where low cost hydropower is available. Start up dates for these three new projects could come as early as 2027, with the country's first 120 ktpa smelter ramping up at the moment.

Chinese companies are also ramping up investment in Kazakhstan, with East Hope Group looking to build a vertically integrated smelter complex (1 Mtpa), while construction continues at the Chuangxin International smelter in Saudi Arabia (500 ktpa, potential start up in 2027).

Elsewhere, EGA and Century Aluminum have outlined plans for a new 750 ktpa smelter in the US, although we think first production is unlikely before 2030 and would depend heavily on securing a long-term, economical power contract. Zoning approvals are currently paused with local pushback. 75 ktpa of the 165 ktpa Slovalko smelter is also due to restart (after shutting in 2022), supported by compensation for indirect carbon costs under the EU Emissions Trading System (ETS) and a long-term power purchase agreement (PPA). Mag7 will also restart 75 kt of its Missouri smelter, which was shut in 2024, supported by US aluminum tariffs. We are also watching a planned Indian smelter by IHC and Adani, which could ultimately add 2 mtpa capacity.

Exhibit 9: We model 3.9 Mt of supply growth in 2027, with faster Middle East resumptions as well as accelerating capacity additions in Indonesia, India and other markets

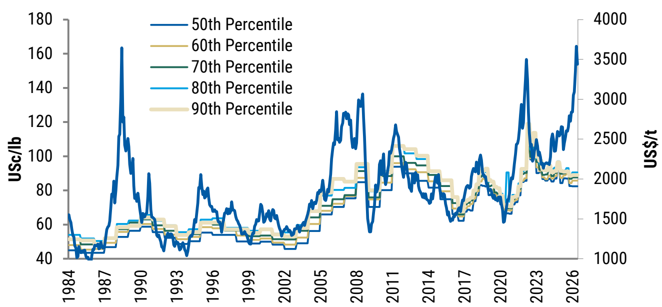
Source: Morgan Stanley Research estimates

Power costs rising, while alumina remains soft

New smelters are raising the cost curve through higher power costs: Aluminum is currently trading well above the 90th percentile of the cash cost curve (Exhibit 10), a marked divergence from historical norms. In particular, stronger demand for electricity-conducting metals like copper and aluminum, coupled with a tighter supply backdrop in recent years as China approached its capacity cap, has supported this divergence.

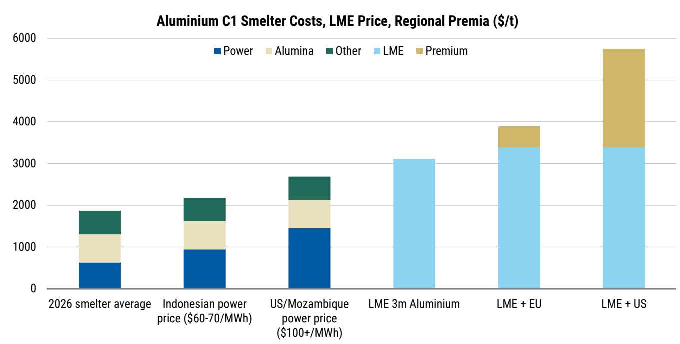
With the market is set to tip into surplus again from 2027, this could put pressure on aluminum to trade closer to its cost curve. However, we do not view the next wave of supply growth as necessarily deflationary. The average global smelter pays $43/MWh for electricity, and each tonne of aluminum requires 14.5 MWh of power. As such, electricity contributes ~$600/t or around ⅓ of the average smelter cost of ~$1900/t (90th percentile at $2050/t). New smelters in Indonesia, the largest source of supply growth, are likely to pay $60-70/MWh, adding >$320/t to costs at the midpoint, while US / Mozambique power costs are $100/MWh+ (implying a $820/t increase to power costs) in new contracts. Recent increases in regional premiums should help offset these higher costs, but they also imply a higher floor for aluminum prices than historically, and a reversal in prices/premiums could bring additional challenges.

Exhibit 10: Aluminum has historically hugged the cost curve, but has broken away more recently



Source: Bloomberg, Wood Mackenzie, Morgan Stanley Research

Exhibit 11: New smelters are coming in with slightly higher power prices than existing ones, on average

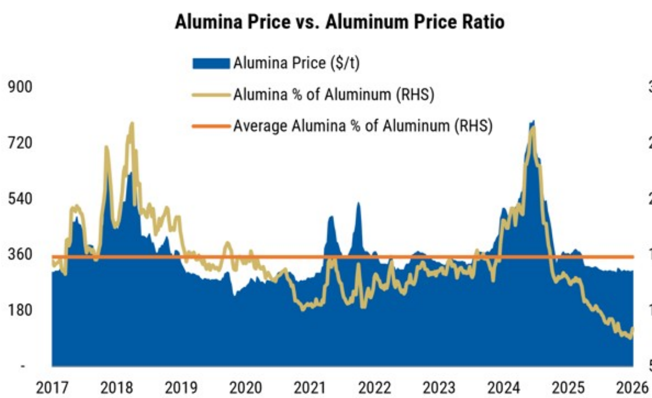


Source: Bloomberg, Wood Mackenzie, Morgan Stanley Research

However, Alumina has become unusually cheap relative to aluminum. The alumina-to-aluminum price ratio has fallen to 8-9%, well below its historical average of 17% since 2017 (Exhibit 12). This does not mean alumina prices are at historical lows in absolute terms. Rather, it is because alumina has normalized sharply from the 2024 spike, while aluminum prices have remained supported as Middle East aluminum production cuts reduced alumina demand while tightening aluminum markets. The result is a historically wide aluminum-alumina spread, improving the raw-material backdrop for smelters. Since Jan-25, alumina prices have fallen by over 50%, reaching close to $300/t, while aluminum prices have increased 23% to $3,100/t, vs. $2,500/t in early 2025 (Exhibit 13).

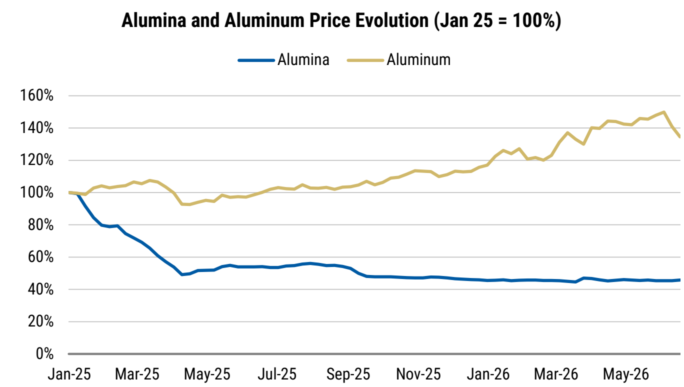
For primary aluminum producers, lower alumina costs support margins. Alumina is one of the largest input costs in primary aluminum production, accounting for approximately one-third of smelting costs. Thus a lower alumina-to-aluminum ratio directly improves smelter economics. This matters especially at a time when power, freight and carbon costs remain elevated.

Exhibit 12: Alumina has fallen to an unusually low share of aluminum prices...



Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 13: ...as alumina has reset lower while aluminum has remained supported



Source: Bloomberg, Platts, Morgan Stanley Research

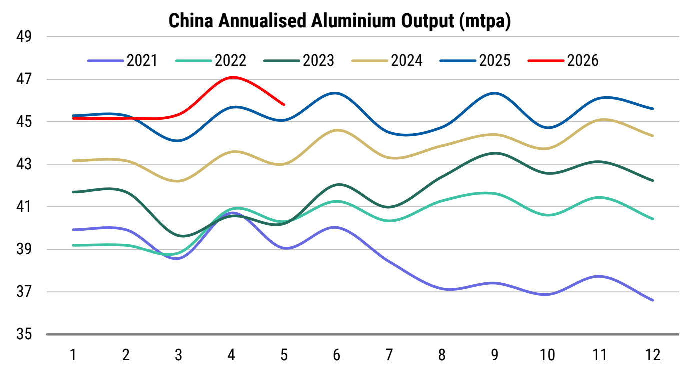
The outlook for demand remains constructive

Demand for aluminum has appeared softer due to de-stocking: Much of the supply tightness has been reflected in aluminum timespreads than in prices directly, with the curve moving into a steep backwardation since the start of the conflict. This indicates near-term tightness but also impacts inventory decisions. For example, in recent months, Dec'26 aluminum was trading as much as ~$160/t below spot. While this can contribute to the 'roll yield' for financial participants, it makes holding excess inventory punitive for end users as the value of inventory declines over time. Platts notes that this has been weighing on spot market activity, particularly in the US and Japan as end users destock. While this is unlikely to persist indefinitely, in our view, it is helping keep the market looser than it otherwise would be. It also creates potential upside demand risk later in 2026 or in 2027 should restocking become necessary.

China demand slows but still positive: We estimate China's aluminum demand will rise 1.1% in 2026, a slowdown from ~5.0% growth in 2025 but remaining in positive territory. Key drivers are likely to be rising grid spend (e.g. with China's "six networks" infrastructure push), exports of EVs, solar panels, and appliances, as well as strong domestic and export markets for stationary storage batteries. However, grid spend may be skewed more toward copper-heavy distribution this year, after strong transmission investments in 2025. Solar installations are slowing significantly, after front loading in 2025, but adding ESS is helping improve project economics.

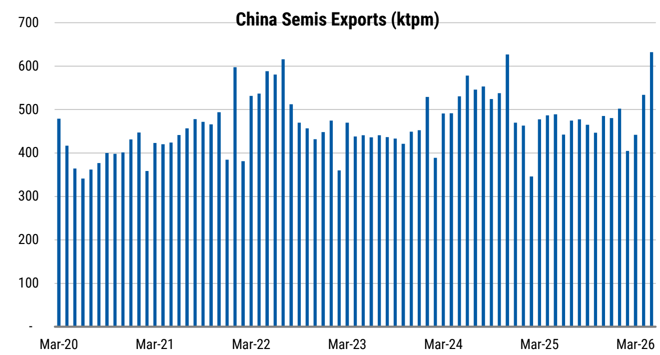
But China's semis exports are likely to increase further in the near term: With strong domestic production and China remaining a net exporter of aluminum, the SHFE price has lagged the LME price, pushing the LME-SHFE spread to record levels. This should encourage exports. However, China imposes a 30% export tariff on aluminum ingot exports, making direct exports of unwrought aluminum uneconomic and helping explain the build-up in SHFE aluminum inventories. By contrast, aluminum wire still qualifies for the 13% VAT rebate, meaning much of the increase in semis exports has been in the form of pure aluminum wire. As a result, Chinese exports are helping keep Asian aluminum markets relatively well supplied.

Exhibit 14: China aluminum production up 3% YoY YTD



Source: Bloomberg

Exhibit 15: This is also supporting exports, particularly of semis

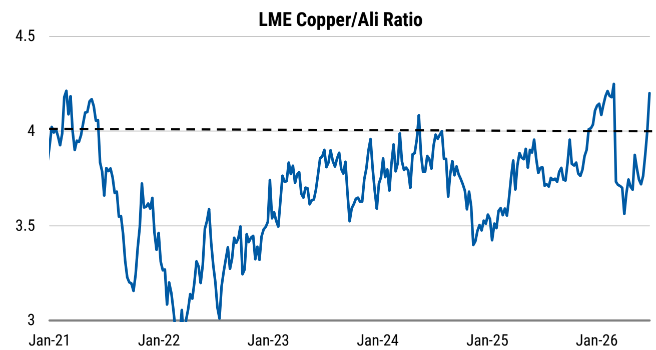


Source: Antaike, SMM

Aluminum is the key substitute for copper: With copper prices near record highs, incentives to seek alternatives are increasing. While a range of materials compete with copper in specific applications (e.g., fibre-optics in telecommunications and plastics in plumbing), aluminum remains the most widely recognised substitute in heating/cooling and electrical applications. To deliver equivalent electrical performance, an aluminum cable requires a cross-sectional area around 1.6x that of a copper cable, which increases conductor size and complicates insulation design. However, weight is frequently cited as the main factor supporting adoption, alongside lower material costs. Copper has traded at least 3x the price of aluminum for the past 15 years, suggesting that much of the easier substitution has likely already occurred, but we do see scope for further substitution in selected applications, particularly air conditioning, with the LME copper/aluminum ratio now at 4.2x. For example, Bloomberg reports that Wanbao and JD.Com (covered by Eddy Wang) launched China's first aluminum AC unit, at a 20% discount to comparable copper-based units. Reuters has also highlighted some examples in the autos sector and cabling here.

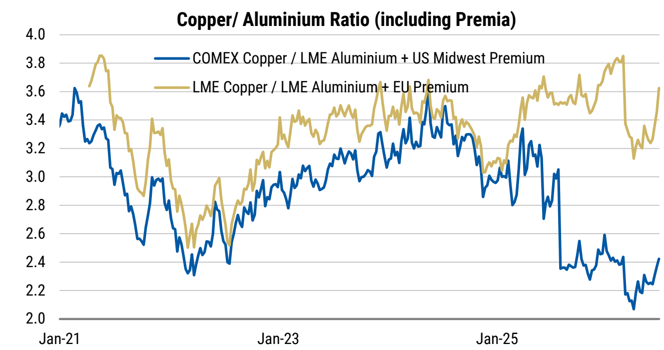
But the LME Copper/Aluminum spread does not tell the full story - high premiums have hurt aluminum's relative competitiveness in some regions: The LME Copper/Aluminum ratio is near the upper end of its recent range, but incorporating regional premiums shows a different picture. In the US, the 50% aluminum tariff has made aluminum less competitive relative to copper (even with a 15% copper tariff under consideration from 2027). In Europe, the all-in aluminum price has actually strengthened versus copper since the Middle East disruption, although this trend is now beginning to reverse.

Exhibit 16: The LME Copper/Aluminum Ratio is near the upper end of the range



Source: Bloomberg, Morgan Stanley Research

Exhibit 17: But looks less elevated once aluminum premiums are considered too



Source: Bloomberg, Morgan Stanley Research

Aluminum also faces substitution risk from steel: Aluminum has generally been seeing more use in vehicles globally as lightweighting becomes more important. However, rising aluminum prices may prompt some manufacturers to reconsider steel as an alternative, while steel can also be used in solar module mounting systems. Following the fire at Novelis Oswego's plant (here), Cleveland-Cliffs's CEO said "I have never seen so much momentum in substituting aluminum with steel," and that the company has been shifting more of its equipment previously used for aluminum production to steel to meet growing domestic demand in automotive, building products, appliances and truck and trailer sectors. We note, however, that the Novelis Oswego plant has since restarted (WSJ).

Aluminum in data centres: Aluminum is also expected to benefit from data centre build out. Wood Mackenzie forecasts aluminum demand from data centres increasing from 550 kt in 2026 to 740 kt by 2030. Our estimates are slightly higher, reflecting Morgan Stanley's higher data centre addition assumptions (809 kt in 2026, rising to 1.2 mln tonnes by 2030). Aluminum demand should also increase with associated infrastructure e.g. busways, structural housings, generator enclosures, distribution systems etc, which could more than double the aluminum demand.

Impact on balances: Smaller 2026 deficit, 2027 moves into surplus, and 2028 surplus increases

With a faster recovery in Middle Eastern production, a quicker ramp up in Indonesian supply and and additional supply growth from other regions (some of which has moved from project status to committed in our model), we now see 2027 in a surplus of 830 kt, compared with our previous forecast of an 807kt deficit. We also expect the market to move from broadly balanced in 2028 to a 1.8Mt surplus as projects currently under development ramp up.

Exhibit 18: We now see the aluminum market tipping into surplus in 2027

	unit	2021	2022	2023	2024	2025	2026e	2027e	2028e	2029e	2030e
World Bauxite Production	Mt	**378**	**380**	**396**	**417**	**408**	**437**	**446**	**446**	**446**	**445**
World Smelter Grade Alumina Production	Mt	**131**	**133**	**137**	**138**	**147**	**150**	**161**	**164**	**168**	**170**
Chemical-grade production	Mt	8.1	8.0	8.3	8.4	8.9	9.1	9.7	9.9	10.1	10.3
total production	Mt	139	141	145	147	156	159	171	174	178	181
YoY change	%	*3.7%*	*1.6%*	*2.7%*	*1.1%*	*6.6%*	*1.9%*	*7.2%*	*2.0%*	*2.3%*	*1.6%*
World Metallurgical Alumina Consumption	Mt	**131**	**134**	**138**	**142**	**145**	**146**	**154**	**159**	**162**	**166**
YoY change	%	*2.4%*	*1.9%*	*3.2%*	*3.1%*	*2.0%*	*0.7%*	*5.2%*	*3.5%*	*1.7%*	*2.6%*
Apparent Alumina Surplus/(Deficit)	Mt	**-0.2**	**-0.5**	**-1.2**	**-3.9**	**2.3**	**4.2**	**7.4**	**5.1**	**6.2**	**4.6**
Average Spot Alumina Prices	US$/t	**$332**	**$376**	**$341**	**$493**	**$362**	**$314**	**$325**	**$350**	**$350**	**$380**
Australia'a average contract price	US$/t	$375	$411	$338	$354	$396	$505	$456	$448	$480	$480
World Primary Aluminium Production	Mt	**67.2**	**68.5**	**70.7**	**72.8**	**74.3**	**74.8**	**78.7**	**81.5**	**82.9**	**85.0**
YoY change		*2.4%*	*1.9%*	*3.2%*	*3.1%*	*2.0%*	*0.7%*	*5.2%*	*3.5%*	*1.7%*	*2.6%*
China primary production	Mt	38.5	40.5	42.1	43.9	45.0	45.9	45.9	45.9	45.9	45.9
YoY change		*2.4%*	*5.1%*	*4.0%*	*4.4%*	*2.4%*	*2.0%*	*0.0%*	*0.0%*	*0.0%*	*0.0%*
non-China primary production	Mt	28.7	28.0	28.6	28.9	29.3	28.9	32.8	35.6	37.0	39.1
YoY change		*2.3%*	*-2.3%*	*1.9%*	*1.1%*	*1.4%*	*-1.3%*	*13.5%*	*8.5%*	*3.8%*	*5.9%*
World Primary Aluminium Demand	Mt	**68.8**	**69.45**	**70.85**	**72.52**	**75.03**	**75.89**	**77.88**	**79.68**	**81.42**	**83.47**
YoY change		*6.6%*	*1.0%*	*2.0%*	*2.4%*	*3.5%*	*1.1%*	*2.6%*	*2.3%*	*2.2%*	*2.5%*
Regional Demand Breakdown											
China	Mt	39.5	39.9	41.8	43.0	45.2	45.7	46.8	47.8	48.7	49.7
India	Mt	2.2	2.3	2.7	3.0	3.1	3.3	3.5	3.7	4.0	4.3
USA	Mt	5.6	5.7	5.5	5.6	5.5	5.4	5.6	5.7	5.7	5.9
Europe	Mt	9.9	9.8	9.1	8.9	8.9	9.0	9.2	9.3	9.5	9.7
Japan	Mt	1.9	1.9	1.8	1.9	1.9	1.9	1.9	1.9	2.0	2.0
ROW	Mt	9.6	9.8	9.9	10.2	10.4	10.6	10.9	11.2	11.6	12.0
Primary Aluminium Market Balance (before inventory)	Mt	**-1.57**	**-0.97**	**-0.20**	**0.30**	**-0.73**	**-1.08**	**0.84**	**1.82**	**1.44**	**1.58**
Reported inventories	Mt	2.79	2.03	2.38	2.13	1.86					
Change in reported inventories	Mt	-2.82	-0.76	0.36	-0.25	-0.27					
Apparent change in off-warrant inventories	Mt	1.25	-0.21	-0.56	0.56	-0.46					
Inventory-to-Consumption Ratio	wks	**2.11**	**1.52**	**1.75**	**1.53**	**1.29**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Primary Aluminium Market Balance	Mt	**-1.57**	**-0.97**	**-0.20**	**0.30**	**-0.73**	**-1.08**	**0.84**	**1.82**	**1.44**	**1.58**
Price	US$/t	**$2,497**	**$2,740**	**$2,254**	**$2,362**	**$2,638**	**$3,265**	**$2,850**	**$2,800**	**$3,000**	**$3,000**
	US$/lb	**$1.13**	**$1.24**	**$1.02**	**$1.07**	**$1.20**	**$1.48**	**$1.29**	**$1.27**	**$1.36**	**$1.36**
US Mid-West premium	US$/t	**$575**	**$654**	**$501**	**$411**	**$1,290**	**$2,380**	**$2,113**	**$838**	**$661**	**$331**
'all-in' US price	US$/t	**$3,072**	**$3,394**	**$2,755**	**$2,773**	**$3,928**	**$5,644**	**$4,963**	**$3,638**	**$3,661**	**$3,331**

*Total exchange stocks, unwrought producer, consumer, port and merchant stocks at period end as reported by IAI & WBMS

Source: Woodmac, IAI, Morgan Stanley Research Estimates

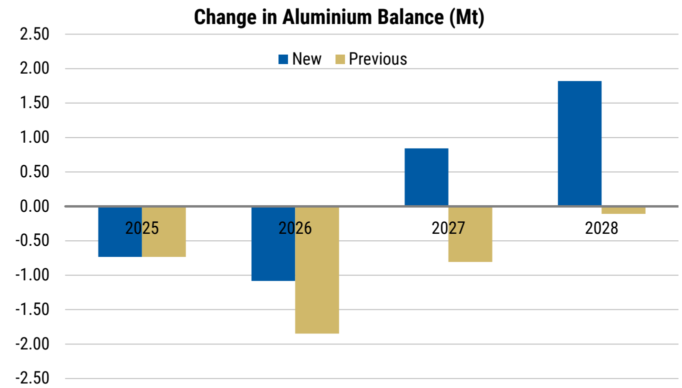
Lowering 2027-28 price forecasts

With the market balance shifting into surplus, we lower our aluminum price forecasts. We still see scope for some tightness in 2H2026, especially if there is a restocking cycle. However, we see limited further upside for LME aluminum prices given already elevated net-long positioning, with part of any remaining tightness potentially expressed through regional premiums.

However, we note that much of the new supply is being added at higher power costs than those currently incurred by the existing smelter base, suggesting aluminum prices are unlikely to fall back to the cost curve. In addition, aluminum's position as the primary substitute for copper should help limit downside risk, particularly if copper prices remain elevated.

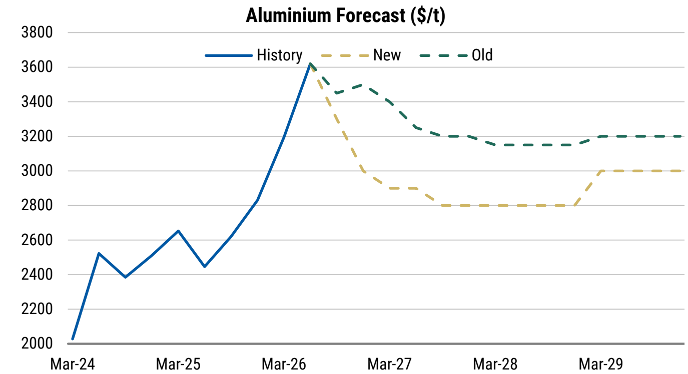
We see aluminum averaging $3,150/t in 2H 2026 and $2,850/t in 2027.

Exhibit 19: 2027 and 2028 flip to surplus on our updated modelling



Source: IAI, WoodMac, Morgan Stanley Research estimates

Exhibit 20: We cut our 2027 and 2028 aluminum price forecasts by 12.6% and 11% respectively



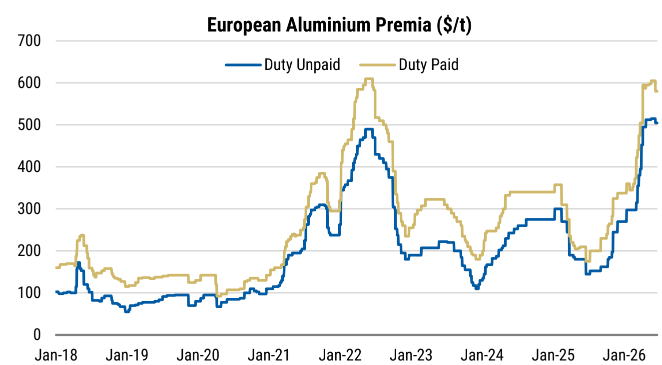
Source: Bloomberg, Morgan Stanley Research estimates

Regional premiums could continue to reflect regional tightness

Regional premiums have come under pressure since the US-Iran MoU (e.g. Europe duty paid falling from $605/t to $572/t, and the Platts Midwest falling from $1.18/lb to $109/lb). However, premiums may continue to stay above historical levels as regional imbalances become more significant.

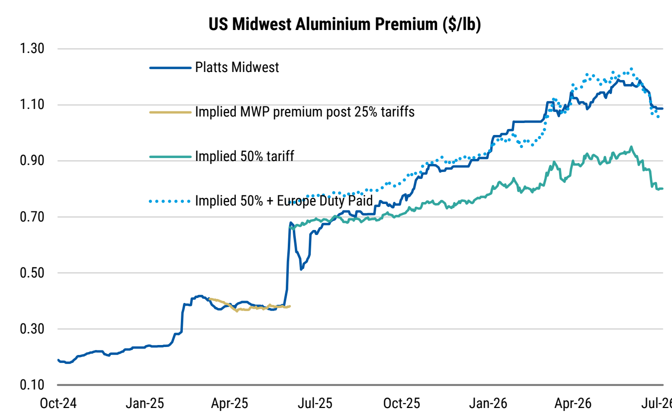
The Midwest premium (dark blue line in Exhibit 22) remains well above the tariff cost (green line), with the difference roughly equivalent to the European Duty Paid premium. Canadian metal can be shipped duty-free to Europe, meaning US consumers are effectively competing with European buyers for supply, requiring an additional premium on top of the tariff cost.

Exhibit 21: The European premium has pulled back slightly after the US-Iran MOU



Source: Bloomberg

Exhibit 22: The MWP has also moderated reflecting pass through of lower aluminum prices and European premiums



Source: Platts, Bloomberg

Bull/Bear Outlook

Upside risks

With the LME copper/LME aluminum ratio already near the upper end of its historical range, further upside in copper prices could also provide support for aluminum prices.

Middle East restarts may stall, while faster ramp-ups could increase the risk of smelter damage.

Switching power from nickel to aluminum in Indonesia may prove more challenging than expected, with the risk that producers may shift capacity back towards nickel if Indonesia backtracks on some of its nickel policies.

Elevated utilisation rates at Chinese smelters may become unsustainable, or smelters may prove less willing to push operating rates beyond capacity if aluminum prices pull back. China may constrain output growth.

If the forward curve flattens, we may see a re-stocking cycle among manufacturers, supporting demand and prices.

The macro backdrop could improve. If the Fed avoids further rate hikes and the USD weakens, metals prices would likely find broader support.

Downside risks

If copper falls (e.g., if the US rules out copper tariffs), aluminum is likely to come under pressure.

The copper/aluminum ratio may structurally re-rate higher, allowing copper to outperform on tighter supply fundamentals and data centre-related demand, while a faster aluminum supply response limits aluminum price appreciation.

If the macro backdrop worsens, the US Fed hikes interest rates and overall demand sentiment softens, aluminum could weaken further.

Middle East smelters may return faster than expected, while Indonesia output could exceed expectations. If demand remains soft, we could see large inventory builds.

Chinese producers' overseas projects are accelerating

Capacity expansion in Indonesia has been gaining more focus

Chinese aluminum smelters has been seeking capacity expansion opportunities in past few years, given limited growth potential in the domestic market following the implementation of China's smelting capacity cap in 2017. The scale and speed of capacity expansion in Indonesia has been the market focus as most of the projects are long planned capacities. Meanwhile, Indonesia has abundant bauxite resources, established alumina capacity, and ample thermal coal supply, making aluminum production more competitive than in many other regions.

Previously, based on our channel checking trip, we had expected the capacity in Indonesia to gradually come online eventually, with the pace constrained by power availability. However, the conflict in Middle East has changes the outlook. The disruption has resulted in ~3.5Mt capacity suspension in the region, leading to tight market supply globally and supporting higher aluminium prices. This has improved smelter economics globally, particularly for Indonesian producers, which benefit from cheaper cost of alumina.

What's more, The Middle East disruption has also tightened the global sulfur market. Sulfur is a key upstream material for nickel production that utilizes hydrometallurgical process. The surge in sulfur prices and new price mechanism in Indonesia has lifted the production cost for nickel in Indonesia, and led to capacity suspension for some players, resulting in potential extra power supply to aluminum smelters in the country. Nickel production typically consumes around 30-35,000 Kwh of power per tonne of metal, compared with approximately 13,000 kWh per tonne for aluminum.

Several projects are now progressing faster than initially expected, supported by strong industry profitability. Based on industry feedback and our discussions with companies, Tsingshan and Xinfa's projects in Halmahera and Sulawesi islands have both commenced production ahead of initial schedule. Xianfeng's Halmahera project, originally scheduled to start operations in August, began production in May, while the Taijing project on Sulawesi also started production in May this year. At the same time, Tsingshan announced an additional 800kt aluminum capacity at Halmahera. The latest update indicates that the phase I of 400kt capacity is expected to complete construction and potentially commence production by year-end 2026, supported in part by reallocating power from its RKEF nickel operations to aluminum production.

Reflecting these timeline changes we now expect Indonesia's aluminum constructed capacity to increase by 925kt, driven mainly by Tsingshan & Xinfa, Adaro, and Nanshan. Together with the 800kt of capacity already in operation at Huaqing and Inalum, **we expect Indonesia's aluminum capacity to reach 2.3mnt in 2026**. Factoring in phased start-ups and ramp-ups, operating capacity could rise further to 3.5mnt by the end of 2027.

We expect Indonesia's aluminum production to increase by ~866kt in 2026, with growth still being constrained by power supply. Capacity additions scheduled by Adaro and Nanshan this year still await power availability, which depends on the completion of captive thermal power plants. **We expect aluminum production in Indonesia to increase**

by ~1.26mnt in 2027 despite limited capacity additions, as recently commissioned facilities ramp up operations.

Exhibit 23: Capacity expansion and volume increase estimates in Indonesia

MS estimate of Indonesia aluminum capacity and prodution

Capacity (kt)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Xinfa Halmahera (small K)				250	250	250	400	600	600	800	800	1000	1000	1200	1200	1400	250	600	1000	1400
Xinfa Sulawesi (large K)							150	300	600	600	600	600	800	1000	1200	1200		300	600	1200
Adaro				100	250	420	420	500	500	500	500	500	500	500	500	500	100	500	500	500
Nanshan BAI								125	125	125	250	375	375	500	500	500		125	375	500
Bosai Aluminum											100	250	250	250	250	400			250	400
Tsing Shan & Huaqing	250	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500
Inalum	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300	300
East Hope																				
PT Dahma Inti Bersama - Harita													250	500	750	1000				1000
Sub total	**550**	**800**	**800**	**1400**	**1550**	**1720**	**1770**	**2325**	**2625**	**2825**	**3050**	**3525**	**3975**	**4750**	**5200**	**5800**	**1400**	**2325**	**3525**	**5800**
																		925	1200	2275

Production (kt)	1Q25	2Q25	3Q25	4Q25	1Q26e	2Q26e	3Q26e	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Xinfa Halmahera (small K)					63	63	81	125	125	175	200	225	250	275	300	325		331	725	1150
Xinfa Sulawesi (large K)							19	56	113	150	150	150	175	225	275	300		75	563	975
Adaro					63	105	105	125	125	125	125	125	125	125	125	125		398	500	500
Nanshan BAI									31	31	63	94	94	125	125	125			219	469
Bosai Aluminum											13	44	63	63	75	81			56	281
Tsing Shan & Huaqing	63	125	125	125	125	125	125	125	125	125	125	125	125	125	125	125	438	500	500	500
Inalum	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	300	300	300	300
East Hope																				
PT Dahma Inti Bersama - Harita													31	94	156	219				500
Sub total	**138**	**200**	**200**	**200**	**325**	**368**	**405**	**506**	**594**	**681**	**750**	**838**	**938**	**1106**	**1256**	**1375**	**738**	**1604**	**2863**	**4175**
Incremental YoY (kt)					188	168	205	306	269	314	345	331	344	425	506	538		866	1259	1313

Source: Company data, ALD, Morgan Stanley Research estimates

However, we see additional ~400kt upside to 2027 Indonesia production if further nickel curtailments free up power for aluminum. Sulfur prices continues to increase, increasing production costs for Indonesian nickel producers and raising the likelihood of additional capacity curtailments due to weaker margins. In such a scenario, we see the possibility that Tsingshan could reallocate more power to its phase II 600kt project located on Sulawesi island, which is currently targeted for construction completion by year-end 2026. If this happens, Indonesian aluminum production growth could be 1.66mnt YoY in 2027 vs. our base-case forecast of 1.26Mt YoY.

New capacity in other regions is also progressing

In addition to Indonesia, Chinese producers are pursuing aluminum expansion projects across several regions, including Central Asia, the Middle East, and Africa. Supported by strong aluminum prices and healthy smelter margins outside China, we also expect some of the those projects to accelerate.

Middle East: The region had 6.7mnt of aluminum capacity as of 2025. Capacity in Abu Dhabi, Bahrain, Dubai, and Saudi Arabia accounted for ~76% of the region's total capacity. In 2025, Chuangxin Industries announced an aluminum capacity plan in Saudi Arabia, targeting 1mnt of aluminum smelting capacity over the long term. According to the company, it has secured stable power supply from the Saudi national grid at a US$0.032/kWh. Construction of phase I, with 500kt of capacity, has started and is likely to complete and begin operations in 1Q27. Currently, the construction progress is going on schedule with no impact from the ongoing tension in the region.

Central & East Asia: Benefiting from its proximity to both China and Europe and its abundant bauxite resources, Kazakhstan has emerged as a major aluminum producer in Asia. According to data from Henan Geological Bureau, Kazakhstan holds 292mnt of proven bauxite reserves with a grade of ~40%, mostly located in the Kostanay region. In 2007, China Nonferrous Metal Industry's Foreign Engineering and Construction Co. Ltd completed construction and commissioned an aluminum plant in Kazakhstan, which currently operates with capacity of approximately 260ktpa.

Attracted by stronger profitability and constrained by domestic capacity quotas, several Chinese players have announced plans to build new overseas capacity:

- **East Hope**: The company signed an agreement with the Kazakhstan government in June 2025 to build an integrated green aluminum industrial park. The long-term target is to build 3mnt of aluminum smelting capacity and 6mnt of bauxite mining capacity. The project will be divided into three phases. Phase I includes 1mnt of aluminum, 2mnt of alumina capacity, and 3GW of wind and solar power capacity. The company is pushing the project forward and targets commencement of Phase I in 2028. However, given the current status, we think the project is likely to be postponed to 2029 or afterward.
- **Xinfa Group**: Similar to East Hope, Xinfa plans to build an integrated aluminum supply chain spanning upstream bauxite mining to downstream aluminum product processing. The company has announced a total investment of US$15bn with a long-term smelting capacity target of 3mnt.

Africa: In June 2024, Huatong Wires & Cable Group started construction of its aluminum capacity in Angola, with the first 120kt of capacity successfully commencing production in January 2026. Power supply poses less risk for this project, as the company has secured a 10-year power supply contract with the local government. The contracted power tariff stands at US$0.01-0.15/kWh for the first year, US$0.015-.025/kWh for the second year, and US$0.035/kWh thereafter. These rates are significantly lower than China's Rmb0.4-0.5/kWh tariff, implying higher aluminum margins in Angola under the current elevated price environment. The company targets a total capacity of 500kt over the long term. However, Phase II (240kt) and Phase III (140kt) face constraints from local power availability and will depend on the commissions of a local hydro power station.

Huatong's initial move has attracted several other Chinese producers, including Baitong Energy and Tianyuan Chemical, to announce capacity expansion plans in Angola. Both projects has made progress, with projects from Baitong and Tianyuan Chemical scheduled to start operation in 1Q28e and 1Q27e, respectively. If these projects proceed as scheduled, aluminum production growth in Africa could reach approximately 290kt in 2027, above our previous estimate. However, both projects remain dependent on power availability in Angola, and therefore execution risk remains elevated.

Beyond Angola, we also see rising interest and a growing pipeline of projects across other African countries, including Ghana, Morocco, Madagascar, and Guinea, largely driven by Chinese producers. However, given relatively weak infrastructure in these markets, we think it will be challenging for these planned capacities to come onstream in the near term, despite the availability of upstream bauxite resources.

Overall, based on the current projects tracking, **we estimate aluminum supply outside China (and excluding Middle East changes) will increase by 1.3mnt in 2026.** Together with the ~3.5mnt capacity suspended due to the Middle East conflict and curtailments at some smelters that have been unable to secure competitively priced power contracts, this should result in a market deficit in 2026. **As the commenced capacity gradually ramps up, we expect ex-China aluminum supply growth to be ~2mnt in 2027**.

Exhibit 24: Volume growth in overseas markets besides Indonesia

MS estimate of capacity increase outside China and Indonesia

Production (kt)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	1Q27e	2Q27e	3Q27e	4Q27e	1Q28e	2Q28e	3Q28e	4Q28e	2025	2026e	2027e	2028e
Malaysia																				
Press Metals						19	19	19	19	19	19	19	19	19	19	19		56	75	75
Vietnam																				
Dak nong							19	38	38	38	19	75	75	75	75	94	0	56	169	319
Vinacomin																				
India																				
Angul - Nalco																				
Jharsuguda II - Vedanta																4				4
Balco - Vedanta				8	25	45	63	90	109	109	109	109	116	121	126	131	8	223	435	495
Aditya - Hindalco													15	25	35	48				123
Mahan - Hindalco																				
Odisha - Hindalco																				
Rio Tinto & AMG																				
Central Asia																				
Kazakhstan - Xinfa Group																				
Kazakhstan - East Hope																				
Tajikistan - Anshun Zhongsai Xinbo																				
Mongolia																				
Middle East																				
Bahrain - Alba																				
Saudi Arabia - Chuangxin Industries										38	100	125	125	125	125	125			263	500
Oman - CMOC																				
Iran - Salco-Asalouyeh																				
Europe																				
Russia - Boguchansky (UC Rusal)																				
Itlay - Portovesme																				
Finland - (Rio Tinto)																				
North America																				
Canada - Arvida (Rio Tinto)							2	9	14	14	14	14	14	14	14	14		11	55	55
US - Century Ali + EGA																				
Africa																				
Ethiopia - (UC Rusal)																				
Egypt - Egyptalum																				
Angola - Huatong					15	30	30	30	30	30	45	60	60	60	75	75		105	165	270
Angola - Baitong Energy													13	38	50	50				150
Angola - Tianyuan Chemical										25	50	50	50	50	50	50			125	200
Sub-total				**8**	**40**	**94**	**133**	**185**	**209**	**271**	**355**	**451**	**486**	**526**	**569**	**610**	**8**	**451**	**1286**	**2191**
YoY (kt)																		444	835	905

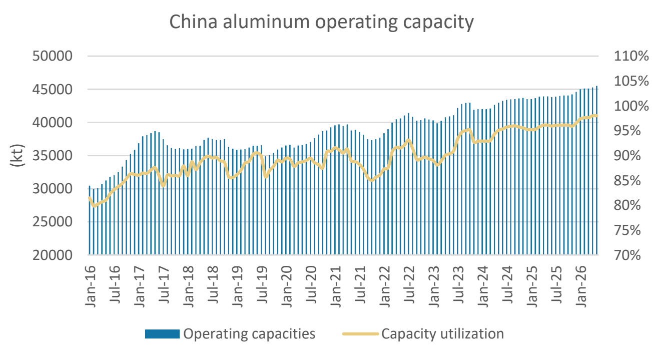
Source: Company data, ALD, Morgan Stanley Research estimates

China domestic production increase is very limited in 2027

China domestic operating capacity has reached the ceiling imposed by China's government: The government set a ~45mnt capacity ceiling for the aluminum industry during the first round of supply-side reforms in 2017. After years of expansion, total operating capacity reached 45.5mnt as of May 2025, at the cap set by the Central Government with limited room for increases.

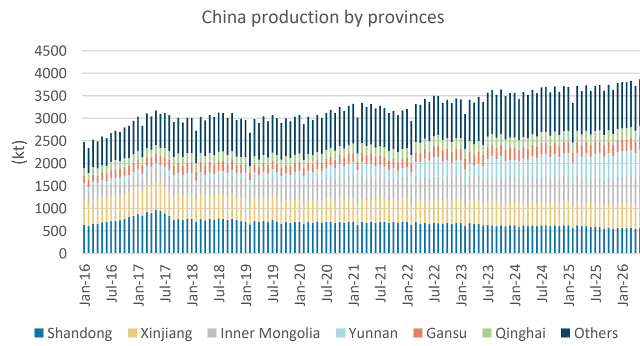
Attracted by the high margins currently recorded, **while not materially, the domestic market is seeing overproduction**. China's aluminum production remained elevated at 3.9mnt in April. Total aluminum production YTD is up 2.4% YoY. Given the better profitability, some producers overproduced by increasing capacity utilization rates. However, overall overproduction volume is limited for smelters. Meanwhile, long periods of overproduction would be harmful to pot lines and result in instability, leading to accidents. Thus, we think the overall capacity utilization in China could reach 102-103%. Meanwhile, the emission and energy consumption data is connected to provincial government authorities in real time, which will make overproduction difficult.

Exhibit 25: Aluminum operating capacity reached 45.5mnt in May



Source: SMM, Morgan Stanley Research

Exhibit 26: Domestic aluminum production remain elevated



Source: SMM, Morgan Stanley Research

For 2026, we expect ~545kt of new capacity additions in the domestic market (within the ~45mnt cap). Based on our channel checks and Baiinfo data, a total of 3.22mnt of capacity is scheduled to commence operations in 2026. Of this, 2.68mnt relates to capacity relocation, including Hongqiao's move from Shandong to Yunnan, Qiya Metal's shift from Yunnan province to Xinjiang, and additional relocations from Henan to Xinjiang. This implies that total 545kt of new capacity will come onstream in 2026, mainly located in Inner Mongolia and Xinjiang.

Meanwhile, we saw 0.3mnt of previously suspended capacity in Liaoning province to resume production in 2Q26, supported by the currently attractive profitability levels.

Considering the timing for start-ups of both incremental and resumed capacity as well as the higher capacity utilization, we expect China's aluminum production to reach 45mnt in 2026, up 2% YoY from 43.9mnt in 2025.

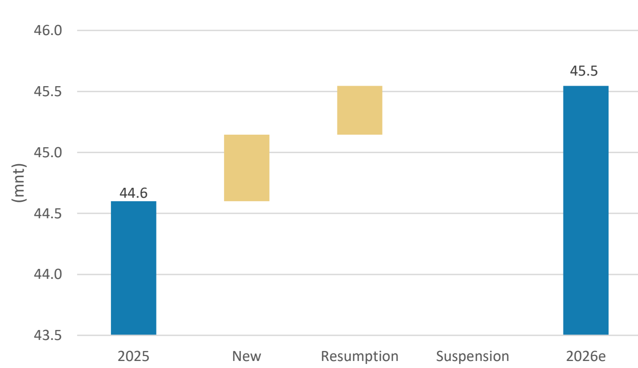
Once all new capacity comes onstream during 2026, total industry operating capacity should reach 45.2mnt in 2026, leaving no room for capacity expansion in the coming years. However, production could still see a marginal increase in 2027 as output ramps up from capacity commissioned in 2026. We expect total aluminum production in 2027 to rise slightly, by 0.5% YoY, to 45.1-45.2mnt and remain broadly stable thereafter.

Exhibit 27: Capacity in pipeline to start operation in 2026

Province	Company	Capacities Planned (mnt)	Able to commence within the year	Already commenced (mnt)	Remaining to start within the year (mnt)
Xinjiang	Xinjiang Nonliushi Aluminum	0.28	0.28	-	0.28
Xinjiang	Xinjiang Huazhang Aluminum	0.20	0.44	-	0.44
Xinjiang	Xinjiang East Hope	0.16	0.16	-	0.16
Xinjiang	Xinjiang Qiya Aluminum	0.21	0.21	-	0.21
Xinjiang	Xinjiang Tianshan Aluminum Jinlian	0.12	0.12	-	0.12
Inner Mongolia	Inner Mongolia Dongshan Aluminum	0.44	0.44	-	0.44
Inner Mongolia	Inner Mongolia Huomeihongjun Aluminum	0.35	0.35	0.02	0.33
Inner mongolia	Inner Mongolia Jinlian Aluminum	0.02	0.02	-	0.02
Yunnan	Yunnan Hongtai Aluminum	0.23	0.23	-	0.23
Yunnan	Yunnan Honghe Aluminum	0.50	0.50	-	0.50
Shanxi	Shanxi Zhaofeng Aluminum	0.29	0.29	-	0.29
Qinghai	Qinghai Qiaotou Aluminum	0.10	0.10	-	0.10
Guangxi	Guangxi Longlin Baikuang Aluminum	0.08	0.08	-	0.08
Sub total		**2.98**	**3.22**	**0.02**	**3.20**

Source: Company data, Baiinfo, Morgan Stanley Research

Exhibit 28: Total operating capacity in 2026 year end is expected to be 45.5mnt



Source: Baiinfo, Company data, Morgan Stanley Research

Demand still muted YTD in China

Demand for aluminum in China is likely to come from a few major industries, including property, transportation and power as well as home appliances and the like. Demand in the domestic Chinese market has been relatively muted year to date.

No obvious improvement in demand from the property sector: While the recent strong secondary sales rebound may suggest the worst has passed in select high-tier cities, limited improvement in macro and housing indicators, divergent trends between new and secondary homes, and still-fragile home-buyer sentiment lead us to doubt the sustainability of the rebound, which will hinge on replacement demand from home sellers. Our view on property new starts and housing completions for 2026 is unchanged – down by 15.5% YoY and 15.7% YoY, respectively. Aluminum demand from the property market remains weak, but do not see it being a further drag on overall demand.

Rigid demand from the power industry, while solar installation slips: To better handle the increasing demand from energy transition globally, grid upgrades as well as power consumption from AI, power investment has been increasing steadily in the past few years. In early this year, State Grid announced that the total investment in 15FYP will reach Rmb4trn. This implies a 40% increase compared to the Rmb2.85trn investment in 14FYP. Meanwhile, there has also been a material increase in investment in Southern Grid. These investments, especially for high-voltage power projects, will drive up aluminum demand.

Solar demand slipped on weaker installations and lower unit consumption. In 2025, solar installation was supported by favorable government policies, including a change in electricity pricing mechanisms, which triggered a rush in solar installation in 1H25, and the removal of the module export tax rebate, which resulted a boost in exports in 1Q26. Reflecting the higher base and weak demand, we think solar installations this year will see a major decline. Based on NEA data, total solar installations in 4M26 were down 51% YoY to just 50.9GW. This translated into weak demand for aluminum for the solar sector as well. Meanwhile, we the per GW consumption for aluminum decreased gradually in past year. Combined, we estimate these factors will lead to a ~35% YoY decline in aluminum consumption this year.

Healthy energy storage demand partially offsets weak demand from solar: Energy storage installation has seen robust growth thanks to declining costs and increasing demand in the past few years. We expect total energy storage installation to see another >300GWh increase in 2026. Note that for every 100GWh of storage, aluminum consumption is ~160kt. We expect ~500kt incremental consumption volume from ESS this year.

Solid exports mitigate weakness in auto demand in China in 2026: Geopolitical factors have changed rapidly this year, and tensions in the Middle East have boosted EV demand overseas. Based on this, our auto analyst, Tim Hsiao, estimates NEV sales will increase by 13% YoY with 60% new car penetration driven primarily by strong exports, which he expects to be up by 88% YoY. However, dragged by prolonged weakness in the domestic market, total domestic PV sales volume is expected to drop by 2% YoY in 2026. Aluminum consumption in China's auto industry is mainly to meet the lightweight requirements for EV. Better EV sales volume and the increasing per unit usage should therefor continue to drive mild a increase in aluminum demand.

Demand from home appliance slips on the high base in 2025: We think overall demand from home appliances will drop by 5% YoY in 2026.

Exhibit 29: We think the aluminum consumption in China will increase by ~1% in 2026

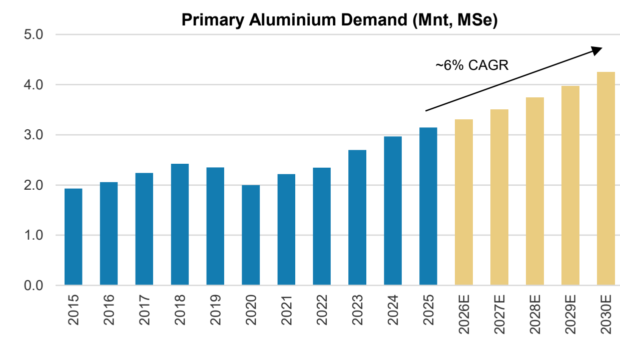
(kt)	2020	2021	2022	2023	2024	2025e	2026e	2027e
Property	9760	9990	8530	9120	8620	8189	7861	7547
% YoY		2%	-15%	7%	-5%	-5%	-4%	-4%
Electricity	5950	6250	6910	8200	8830	9244	8760	9804
% YoY		5%	11%	19%	8%	5%	-5%	12%
- Solar industry		1553	2612	4200	3877	3548	2323	2788
- Other electricity		4697	4298	4000	4953	5696	6437	7016
% YoY			-8%	-7%	24%	15%	13%	9%
Transportation	7770	7760	7960	8650	9220	10193	10412	11088
% YoY		0%	3%	9%	7%	11%	2%	6%
-Passenger car	2706	3067	3694	4361	5283	5862	5865	6314
-Other transportation	5064	4693	4266	4289	3937	4330	4547	4774
% YoY		-7%	-9%	1%	-8%	10%	5%	5%
Packaging	3730	4050	4250	4380	4430	4563	4700	4841
% YoY		9%	5%	3%	1%	3%	3%	3%
Machinery	2590	2720	2800	2880	2930	3223	3384	3486
% YoY		5%	3%	3%	2%	10%	5%	3%
White goods	3160	3300	3430	3650	3730	3805	3614	3614
% YoY		4%	4%	6%	2%	2%	-5%	0%
Others	980	1080	1130	1200	1210	1246	1284	1322
% YoY		10%	5%	6%	1%	3%	3%	3%
Aluminum product export	4860	5620	6604	5679	6706	6134	7054	6349
% YoY		16%	18%	-14%	18%	-9%	15%	-10%
Total	**38800**	**40770**	**41614**	**43759**	**45676**	**46597**	**47069**	**48051**
		5%	2%	5%	4%	2%	1%	2%

Source: Morgan Stanley Research estimates

India: Strong Demand Is Driving Capacity Expansion, Albeit on a Low Base

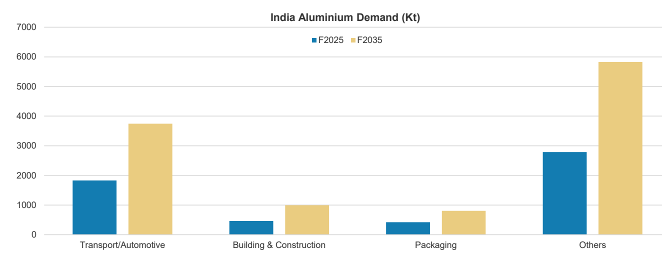
Strong aluminum demand… India's aluminum demand outlook is structurally strong, underpinned by a multiyear capex and consumption cycle. Demand is being driven by a broad-based expansion across infrastructure, manufacturing, transportation and energy transition applications – particularly renewables, grid build-out, EVs and urbanisation-led construction. India remains under-penetrated on a per capita basis, and we expect domestic primary aluminum demand growth to remain strong over the next few years, significantly outpacing global trends. This creates a sustained demand runway rather than a cyclical spike, driving continued investments in capacity expansion.

Exhibit 30: India aluminum demand to remain strong over the next few years...



Source: IPAI, WBMS, Wood Mackenzie, Morgan Stanley Research estimates

Exhibit 31: ... And double over the next 10 years, led by all major segments



Source: Company data, Morgan Stanley Research estimates

… is driving strong capacity additions… This strong demand trajectory is already translating into a tangible capacity response from large domestic players. With the industry having operated at near full utilisation in recent years, and demand seeing a ~9% CAGR over the past four years, incremental supply is required to maintain balance. As a result, leading players are accelerating brownfield expansions – Hindalco (HALC.NS) is targeting ~375kt of additional capacity by end-2028 (with more pipeline potential), Vedanta (VDAN.NS, NC) continues to both add new capacity and debottleneck existing capacity to expand (moved from 2.4mnt in 2024 to ~2.8mnt now and should get to ~3.1mnt by end-2027), and Nalco (NALU.NS, NC) has articulated plans to double capacity to ~1mnt over time. In addition, Adani (ADEL.NS) recently announced entry into a fully integrated aluminum value chain (4mnt alumina, 2mnt aluminum smelting and 1mnt aluminum downstream capacity), further strengthening India's long-term aluminum supply ambitions, although meaningful production impact remains several years away given the greenfield nature of the investment.

The key point is that capacity addition is no longer speculative, but is a direct response to a visible domestic demand upcycle. Given upcoming capacity additions are brownfield in nature, we see limited room for delays and expect new capacities to ramp-up fast (both Vedanta and Hindalco new capacities can fully ramp-up within a year of capacity addition).

Exhibit 32: India: Aluminum smelting capacity snapshot

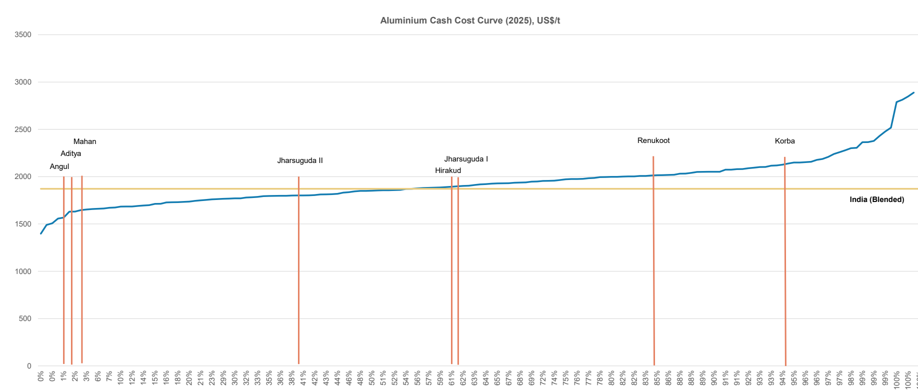
in kt	2024	2025	2026	2027	2028
Vedanta	2400	2835	2835	3100	3100
Hindalco	1357	1357	1357	1538	1731
Nalco	460	460	460	460	460
Total Capacity	**4217**	**4652**	**4652**	**5098**	**5291**
Change		*435*	*0*	*446*	*193*

Source: Company data, Morgan Stanley Research

... also supported by cost advantage from backward integration: What differentiates India's aluminum expansion versus that in other regions is the high degree of backward integration, which reinforces both the pace and sustainability of capacity build-outs. Indian producers benefit from abundant domestic bauxite reserves (~650mnt) and integrated operations spanning mining, refining and smelting, allowing them to operate lower on the global cost curve. In Hindalco's case, for reference, full backward integration in bauxite and alumina keeps them in the second quartile of overall global cash cost curve. This, coupled with strong focus on increasing captive energy sourcing, structurally reduces cost volatility and ensures raw material security. This backward integration coupled with balance sheet strength is enabling Indian players to invest in strong capacity expansion.

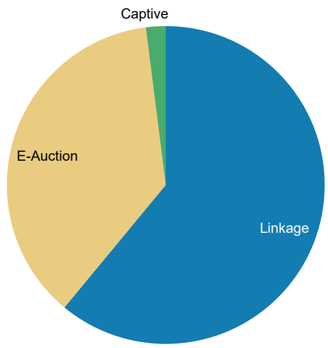
Near term, Indian aluminum smelters remain relatively insulated from energy cost volatility, with Hindalco and Vedanta benefiting from captive power generation and ~75-85% of coal requirements secured through long-term Coal India linkages, where coal prices have remained broadly stable.

Exhibit 33: India aluminum's cash cost curve lies in the second quartile on a blended basis



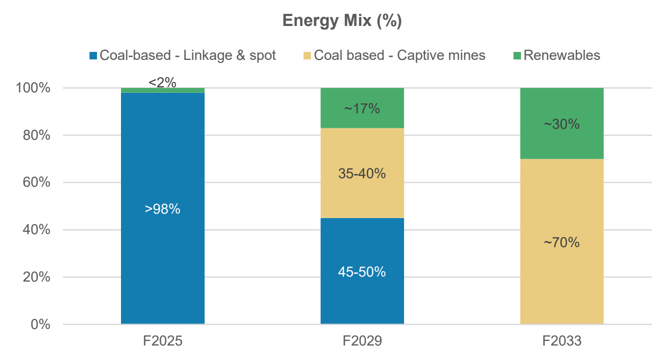
Source: Wood Mackenzie, Morgan Stanley Research

Exhibit 34: India aluminum smelters have high reliance on relatively cheap linkage coal (Hindalco's Power Mix)...



Source: Company data, Morgan Stanley Research

Exhibit 35: ... But with acquisition of captive coal mines and investments in building renewables capacities, large Indian smelters are shifting toward self sufficiency (Hindalco's Energy Mix)



Source: Company data, Morgan Stanley Research

What's in the price?

Based on prevailing market prices and implied valuations and as well as our earnings estimates for Novelis and the copper businesses, the current stock price implies US $2,807/t LME aluminum prices on average for 12 months ending June 2028 (~10% discount to current prices and in-line with our revised estimates).

Exhibit 36: Hindalco: What's in the price analysis – Current prices and valuations imply LME aluminum prices of US$2,807/t on average for 12 months ending June 2028

What's in the Price Analysis	Hindalco
CMP (Rs)	967
No. of Shares (Bn)	2.2
Current Equity Value (Rs Bn)	2147
Less: Investments in group companies (Rs Bn)	143
Add: Net Debt, adjusted for CWIP (June-2027e) (Rs Bn)	709
Implied EV (Rs Bn)	2713
2Y Forward EV/EBITDA (Current)	6.2x
--% Premium/Discount to 5Y Averages	*7%*
12M-ending June-2028E EBITDA (Rs Bn)	436
Less: Novelis EBITDA (Rs Bn)	226
Less: Copper EBITDA (Rs Bn)	33
12M-ending June-2028E India Aluminium EBITDA (Rs Bn)	177
12M-ending June-2028E India Aluminium Opex (Rs Bn)	316
12M-ending June-2028E India Aluminium Revenues (Rs Bn)	493
12M-ending June-2028E India Aluminium Revenues Per ton Basis (US$)	3699
Adjusted for VAP	3217
Adjusted for Tariffs & Regional Premia	**2807**

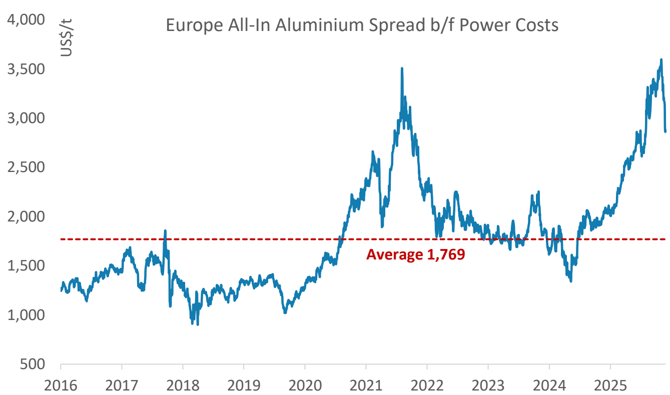
Source: Morgan Stanley Research estimates

Europe & EMEA: Restart Economics Improve as GCC Supply Shock Starts to Normalise

Europe – From supply tightness to restart incentives

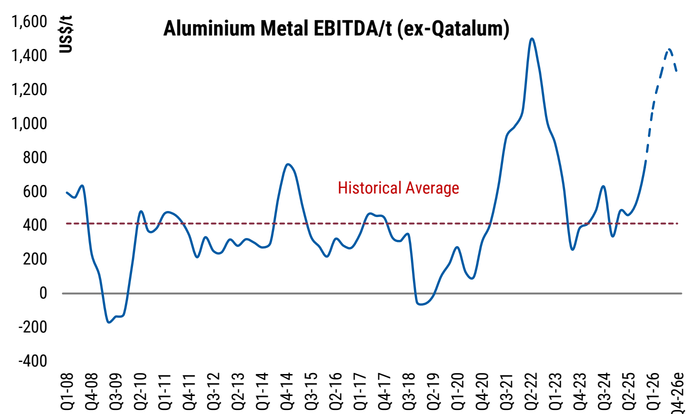
Europe's all-in aluminum economics have improved materially as premiums moved to cyclical highs, LME prices remained well underpinned and alumina reset lower. The spread between all-in aluminum prices and alumina has widened to ~US$2,884/t (Exhibit 37), lifting smelter margins despite elevated power, freight and carbon costs. On our estimates, these conditions imply that some European smelters are earning 4x their historical EBITDA/t (Exhibit 38), making idled capacity economical again. This is in contrast to the past several years where Europe has been losing primary capacity to power-price shocks, becoming more reliant on imports. As such, the current spread environment has become a catalyst for smelter restarts where power availability is economical.

Exhibit 37: The spread between all-in aluminum prices and alumina has widened to ~US$2,884/t



Source: Morgan Stanley Research, S&P Platts, Bloomberg

Exhibit 38: Primary aluminum EBITDA/t has sharply increased on spot to approximately 4x its historical average



Source: Company Data, Morgan Stanley Research estimates

Exhibit 39:

An overview of capacity restarts in Europe

Owner	Country	Smelter	Capacity (kt)			
			2025	**2026**	**2027**	**2028**
Century	Iceland	Grundartangi	275	215	280	320
Alcoa	Spain	San Ciprian	30	120	220	220
Norsk Hydro	Slovakia	Slovalco	0	0	75	75
Total			**305**	**335**	**575**	**615**
Change				*30*	*240*	*40*

Source: Company Data, Morgan Stanley Research estimates

Exhibit 40: An overview of EU aluminum smelter capacity

Owner	Smelter	Country	Capacity (kt)
Aluminij Industries	Mostar	Bosnia	135
Alba	Dunkerque	France	280
Trimet	St.Jean de Maurienne	France	145
Trimet	Essen	Germany	160
Trimet	Hamburg	Germany	135
Trimet	Voerde	Germany	115
Metlen	Distomon	Greece	185
Alcoa	Fjardaal	Iceland	300
Century	Grundartangi	Iceland	320
Rio Tinto	Straumsvik (ISAL)	Iceland	200
Sider Alloys	Portovesme	Italy	147
Uniprom	Podgorica	Montenegro	120
Norsk Hydro	Ardal	Norway	211
Norsk Hydro	Hoyanger	Norway	65
Norsk Hydro	Karmoy	Norway	250
Alcoa	Lista	Norway	64
Alcoa	Mosjoen	Norway	200
Norsk Hydro	Sunndalsora	Norway	405
Norsk Hydro	Husnes	Norway	185
Alro	Slatina	Romania	65
Norsk Hydro / Penta	Slovalko	Slovakia	165
Alcoa	San Ciprian	Spain	220
Rusal	Kubikenborg (Sundsvall)	Sweden	125
Eti Aluminyum	Seydisehir	Turkey	80
Alvance	Lochaber	UK	40
		Total	**4,182**

Source: Company Data, Morgan Stanley Research estimates

The ramp-up of San Ciprián

San Ciprián is an example of European restarts translating into additional near-term supply. Alcoa (covered by Carlos de Alba) originally curtailed its ~220ktpa smelter for two years beginning in January 2022, citing Spain's high energy costs, while committing to a restart from January 2024, and to long-term power procurement. In 2023, the company and its workforce supported an updated plan that consisted of a phased restart from January 1, 2024, with full production targeted by October 2025, with a minimum production of 75% of capacity from October 2025 through end-2026.

The restart was later supported by the creation of the Alcoa-IGNIS JV, effective March 31, 2025, with Alcoa retaining 75% and IGNIS taking a 25% stake in the smelter. The process was temporarily interrupted by the April 28, 2025, Spain-wide power outage, but subsequently resumed and a full restart is planned by mid-2026. We note that the San Ciprián restart and ramp-up through 2026 is a driver of higher EU smelter output.

Slovalco restart back in focus

The Slovalco smelter (owned 55.3% by Norway's Hydro and 44.7% by Central Europe-focused Penta Investments Group) is a more relevant example of how widening spreads can change the optionality and economics of idled European capacity. Norsk Hydro has confirmed that it will restart part of Slovalco's primary aluminum production, following an agreement with the Slovak government that establishes the long-term framework conditions needed for aluminum production, including compensation for indirect carbon costs under the EU ETS. Slovalco has also entered into a long-term commercial power purchase agreement with Vodohospodárska výstavba, addressing the key power-cost issue that led to the smelter's curtailment in 2022.

The restart remains conditional on the European Commission approval of Slovakia's updated scheme for indirect carbon cost compensation. Subject to that approval, Slovalco will invest €100mn to restart production, supporting more than 200 jobs, with aluminum output expected to begin in 4Q26. The first phase will cover 75ktpa of Slovalco's total 175ktpa capacity, while a restart of the remaining 100ktpa will depend on post-2030 framework conditions and additional power contracts.

GCC: from supply shock to gradual normalisation

Peak Middle East outages reached ~3.5mtpa at the height of the conflict, including EGA's Al Taweelah, Qatalum, Alba and Iranian smelters – more than 4% of global supply. With ~23% of world aluminum supply ex-China, the Middle East is a key piece of global supply.

Among those smelters that had been disrupted or partly damaged, **Qatalum** (50%-owned by Norsk Hydro with total capacity of 648ktpa) appears to have the fastest route to normalisation. Norsk Hydro announced on March 3, 2026, that the smelter had started a controlled shutdown due to gas shortages, noting that a full restart from full closure could take six to 12 months. By March 12, the company said further curtailment had been halted and production was being maintained at around 60% after QatarEnergy continued reduced gas supply. The controlled curtailment improved conditions for a future restart.

EGA's recovery path still appears slower, despite restoration progressing ahead of earlier expectations. Reuters reported on July 2 that EGA was restoring production sooner than expected at its Al Taweelah complex (~1.5mtpa) after an emergency shutdown of the site in late March, while its Jebel Ali smelter (~1mtpa capacity) continued operating at full capacity. The casthouse produced its first cast metal on May 4, the recycling plant resumed cast metal production in early May, and first alumina refinery production is expected early in 3Q, subject to bauxite supply chains. However, primary aluminum recovery remains more complex, as EGA must progressively restart 1,262 reduction cells; the first cell was restarted on May 26, and 89 cells had been brought back online by early July, but hot metal output may still take up to a year to return to previous levels.

On March 15, 2026, **Alba** (Aluminium Bahrain, not covered) initiated a controlled shutdown of potlines 1-3, representing 19% of its 1.623mtpa capacity, to manage raw material and transit disruptions linked to the Strait of Hormuz, while focusing feedstock on lines 4-6. The company later confirmed its facility had been subject to an attack and was assessing damage. Although we do not have enough visibility on the extent of damage to some of the potlines, it appears that the remaining asset remains in production and could resume output normalisation over the course of the year.

In summary, we expect GCC output to gradually normalise over the course of the next 12 months. The Qatalum smelter is likely to recover first, followed by Alba, and EGA's Al Taweelah. This setup should allow European premiums to ease gradually. That said, a "residual" risk premium may remain in place until the market sees a clearer path to the Al Taweelah ramp-up.

Exhibit 41: An overview of capacity restarts in Europe

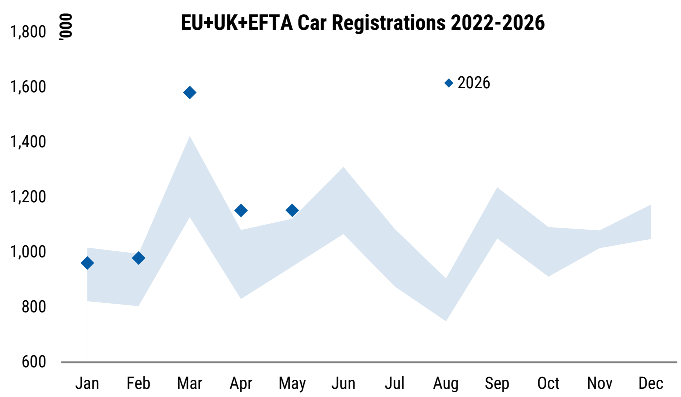
Owner	Region	Smelter	Capacity (kt)			
			2025	2026	2027	2028
EGA	Abu Dhabi	Al Taweelah	1,575	394	1,200	1,575
Alba	Bahrain	Alba	1,620	1,283	1,500	1,620
Norsk Hydro/Qatar Energy	Qatar	Qatalum	638	513	630	638
Chuangxin Industrial Group	Saudi Arabia	Chuangxin	0	0	250	500
Total			**3,833**	**2,189**	**3,580**	**4,333**
Change				*-1,644*	*1,391*	*753*

Source: Company Data, Morgan Stanley Research estimates

European demand… a slow cyclical recovery

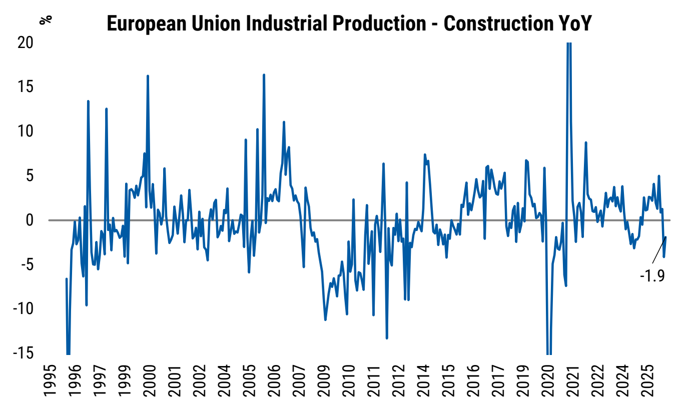
We maintain that the European demand recovery remains fragile. Industrial activity remains subdued and a construction recovery remains soft. This is partly offset by some growth in automotive demand and an accelerated shift toward more aluminum-intensive BEV/hybrid vehicles. For context, CRU sees 2026 demand growth of ~1% for Europe (extrusions).

Exhibit 42: EU car registrations tracking ahead of four-year average in 2026



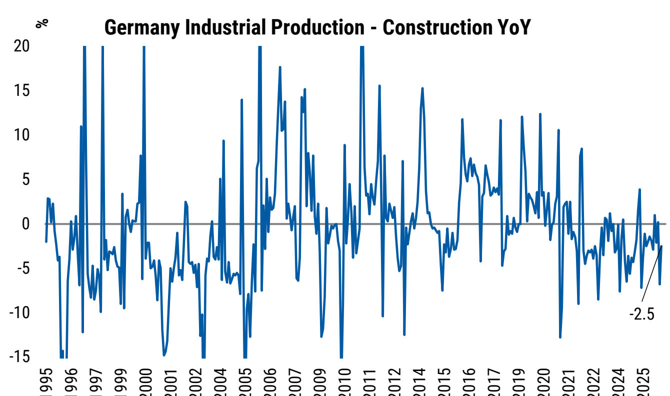
Source: Morgan Stanley Research, VDA

Exhibit 43: EU construction output still contracting at -1.9% YoY



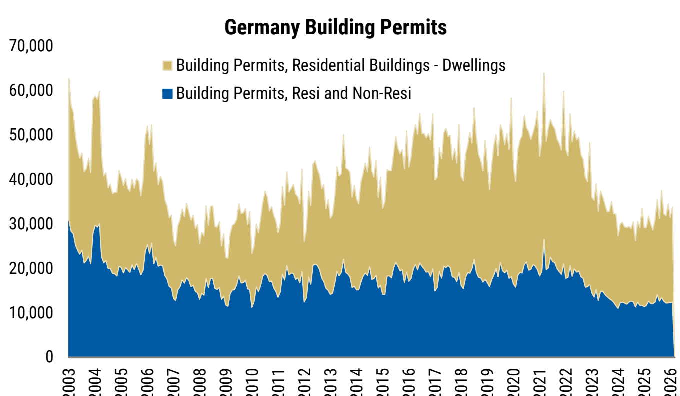
Source: Morgan Stanley Research, Eurostat

Exhibit 44: German construction output remains weak at -2.5% YoY



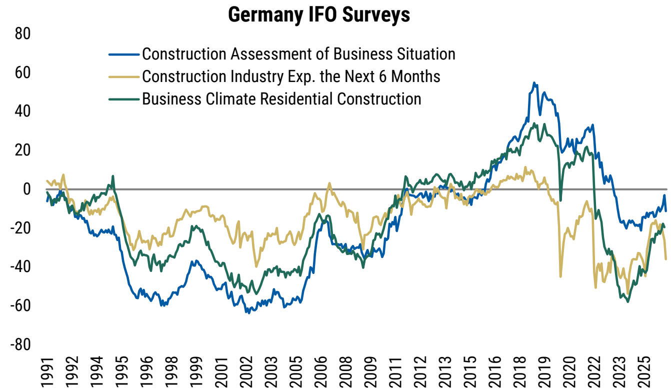
Source: Morgan Stanley Research, Federal Statistic Office of Germany

Exhibit 45: German building permits remain near multi-decade lows



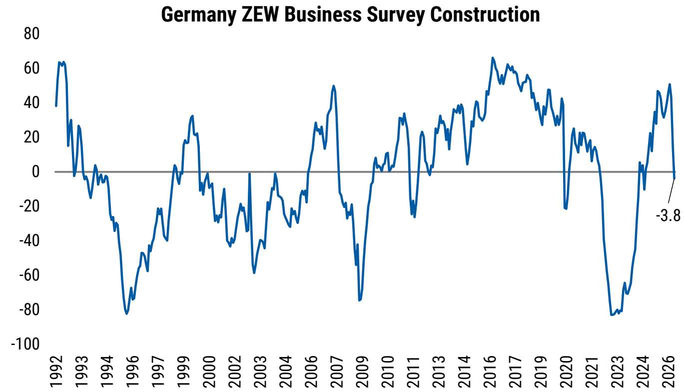
Source: Federal Statistical Office Germany, Workspace, Morgan Stanley Research

Exhibit 46: German IFO construction surveys improving, but still negative



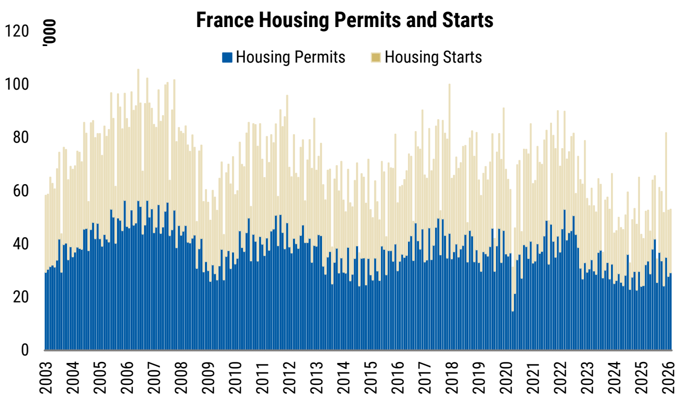
Source: Morgan Stanley Research, IFO

Exhibit 47: German ZEW construction sentiment back near neutral at -3.8



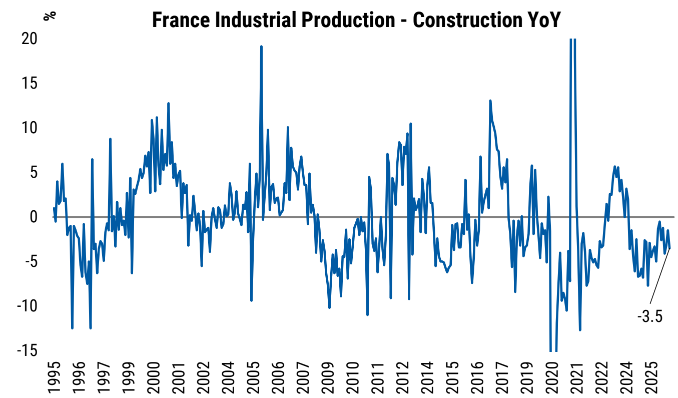
Source: Morgan Stanley Research, ZEW

Exhibit 48: French housing permits and starts remain below pre-Covid levels



Source: Morgan Stanley Research, Ministere de l'Equipement/DAEI-SES, Ministere de l'Ecologie

Exhibit 49: French construction output continues to contract at -3.5% YoY



Source: Morgan Stanley Research, INSEE

Scrap export levy: longer-dated secondary supply relief

The European Union is reportedly considering a 15% levy on aluminum scrap exports, after EU scrap exports reached a record ~1.27mt in 2025, up ~50% vs. 2019, with most volumes directed to Asia. The Commission has delayed formal measures until September 2026, and we expect any new policy to be introduced no earlier than January 1, 2028.

Exhibit 50: Europe's aluminum scrap exports have been on the rise since 2018

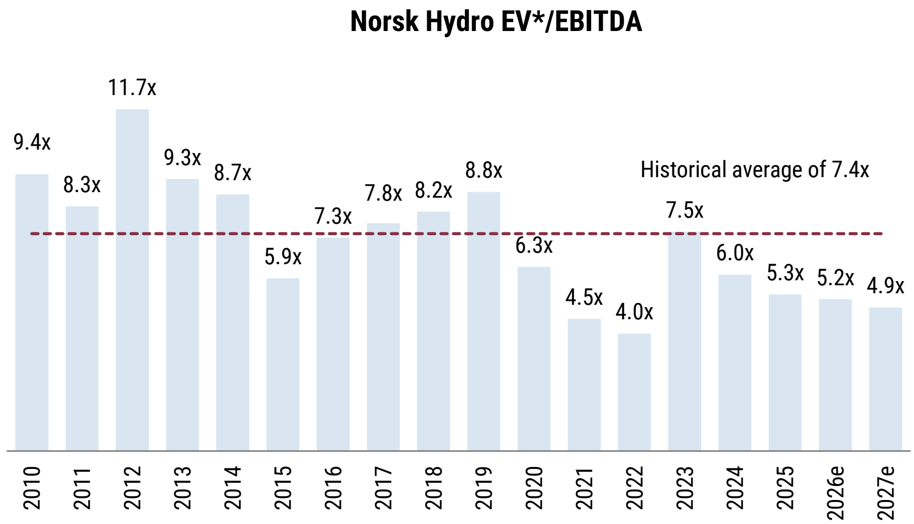
Source: Eurostat, Morgan Stanley Research. 2026 YtD (Jan-April) annualised

Nonetheless, we do not expect the levy to materially change the 2026-27 European premium profile, but it would create a medium-term domestic supply lever. By narrowing the export arbitrage, more European scrap should remain locally, improving feedstock availability for underutilised recycling and remelt assets and lifting secondary supply over time. This should add another medium-term source of pressure on domestic premia, alongside primary restarts and GCC normalisation.

Norsk Hydro – Quality intact, but valuation framework increasingly vulnerable

Norsk Hydro remains one of the highest-quality aluminum producers; the group combines a first-quartile smelting position, advantaged Norwegian power exposure, meaningful backward integration into renewable power and strong operating leverage to high-for-longer European regional premiums. Following the steep re-rating following the recent supply shock in the Middle East and the accompanying energy price inflation, the end of the conflict has underpinned a pull-back in the shares (-28% vs peak on June 2nd), which we now argue has a more balanced risk-reward in the context of the current price environment. Shares now trade at ~4.9x 2027e EV/EBITDA, or ~5.1x on spot, below the long-term average of ~7.4x.

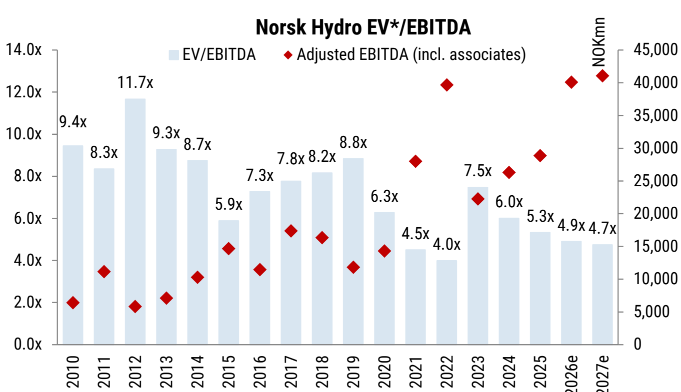
Exhibit 51: Shares are trading at 4.9x 2027e EV/EBITDA vs. a long-term average of ~7.4x



* broad definition of EV. Source: Company Data, Morgan Stanley Research estimates

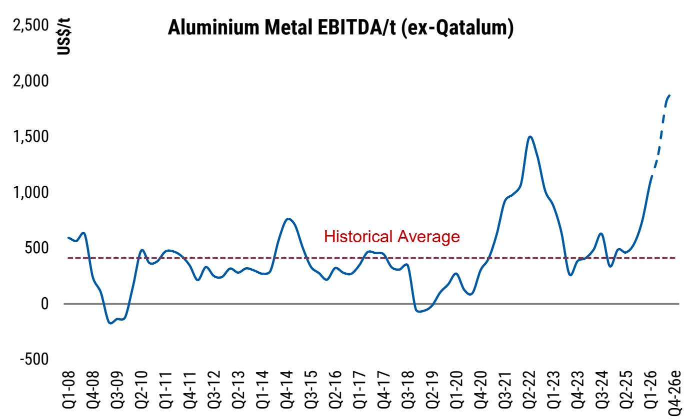
Nonetheless, we argue that the market is increasingly capitalising today's spot margin environment as if it were more durable. The company's primary aluminum spot EBITDA/t run-rate for 2027 stands at ~US$1,600/t, around 4x the historical average of ~US$400/t, reflecting the combination of high LME prices, elevated European premiums and lower alumina costs.

Exhibit 52: The market is capitalising a higher-for-longer profits, consistent with previous cyclical highs



* broad definition of EV. Source: Source: Company Data, Morgan Stanley Research estimates

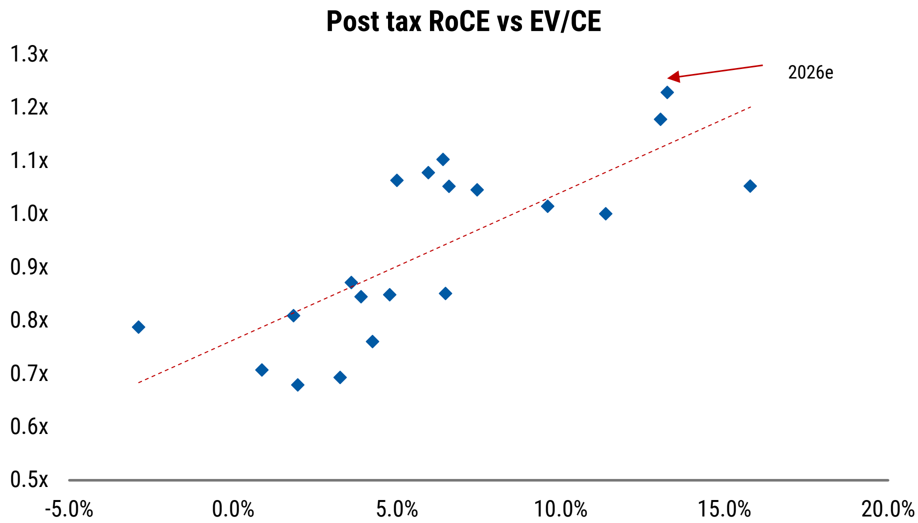
Exhibit 53: Primary Aluminum EBITDA/t has sharply increased on spot to approximately 4x its historical average



Source: Company Data, Morgan Stanley Research estimates

We acknowledge the merits of Norsk Hydro's strategic power cost position and low carbon exposure, which would help rebase group margins/earnings higher; nonetheless, we argue that the spot margin uplift is not fully structural. As GCC supply normalises, European premiums ease and aluminum-alumina spreads compress, the downside skew to earnings momentum becomes more prevalent in the context of the investment case. In our view, this makes the equity risk-reward more balanced. Further upside now requires either another leg of aluminum tightening or clearer evidence that current spot margins can persist. **Valuation framework:** Shares are already trading at ~1.4x EV/capital employed versus 2026-27e post-tax RoCE of 12-13% and a ~9% cost of capital.

Exhibit 54: The shares are already trading on an EV/capital employed of ~1.2x vs. a 2026-27 post-tax RoCE of 13% vs a cost of capital of ~9% on MSe (base case)



Source: Company Data, Morgan Stanley Research estimates

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United States

Global balances may loosen into 2027-28, but the US remains a structurally short, import-reliant market, with limited domestic supply relief before the potential 2030+ Oklahoma step-change.

The US primary aluminum system has only four operating smelters...: Despite being the world's second-largest consumer of primary aluminum on a market level, the US currently has only four operating smelters, split between two companies: Alcoa (AA.N) and Century (CENX.O, NC). Alcoa runs Massena West in New York and Warrick in Indiana, with 130ktpy and 215ktpy of capacity, respectively. Century runs Sebree in Kentucky and Mt. Holly in South Carolina, with 220ktpy and 230ktpy of capacity, respectively. Together, these four assets total 795ktpy of US primary aluminum nameplate capacity (Exhibit 31)

Exhibit 55: The US primary aluminum footprint is concentrated in only four operating smelters, split between Alcoa and Century...

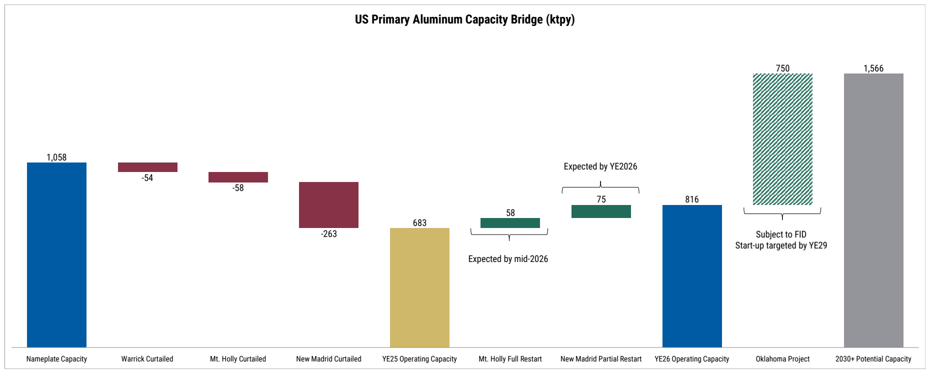
Company	Plant	State	Nameplate Capacity	Curtailed Capacity (ktpy)	Operating Capacity
Alcoa	Warrick	IN	215	54	161
	Massena West	NY	130	0	130
Century	Sebree	KY	220	0	220
	Mt. Holly	SC	230	58	172
Magnitude 7 Metals	New Madrid	MO	263	263	0
Total			**1,058**	**375**	**683**

Source: Company Data, Morgan Stanley Research

... And currently faces some idled capacity. Alcoa has 54ktpy of curtailed capacity at Warrick, with no plans to restart it. Century idled its Hawesville, Kentucky smelter in 2022 due to higher energy costs following Russia's invasion of Ukraine, but rather than attempt a restart the company opted to sell the site to Terrawulf for data center development in Feb-26 (here). Meanwhile, Century's Mt. Holly also has idled capacity, but is ramping up. The first phase of its restart lifted active capacity to 172.5ktpy, or 75% of the smelter's 230ktpy nameplate capacity. In August 2025, Century announced a ~US$50m project to restart more than 50ktpy of idled capacity, bringing Mt. Holly back to full production for the first time since 2015. In April 2026, the company said production of first hot metal had begun from the expansion and remained on track to bring the full expansion online by end-June 2026. Beyond Mt. Holly, Magnitude 7 Metals has announced plans to restart one potline at its New Madrid, Missouri smelter, potentially adding 75ktpy of capacity before year-end (here). However, the project remains at an early stage, with limited disclosure around funding, long-term power supply and environmental permits. Should the restarts at Mt. Holly and New Madrid be completed successfully and on time, they would increase US operating capacity from ~683ktpy at year-end 2025 to ~816ktpy by year-end 2026.

The Oklahoma JV offers a real step-change in domestic primary supply, but in the long term: Century and EGA plan to build a 750ktpy primary aluminum smelter in Inola, Oklahoma, with EGA owning 60% and Century 40% of the JV. It would be the first new US primary aluminum smelter in 50 years, and would effectively double the size of the domestic industry (Exhibit 56). The project also carries strategic policy support, including a US Department of Energy (DOE) grant of up to US$500m. However, despite support at the federal level, local government has been less receptive to the project with the Oklahoma Attorney General filing a petition on June 2, 2026 (here), to block the proposed smelter, citing foreign majority ownership, high electricity demand and environmental concerns amongst other objections. The timeline will depend on final energy contracting, detailed engineering and FID, which Century expects to make in 4Q26; first hot metal production targets end-2029.

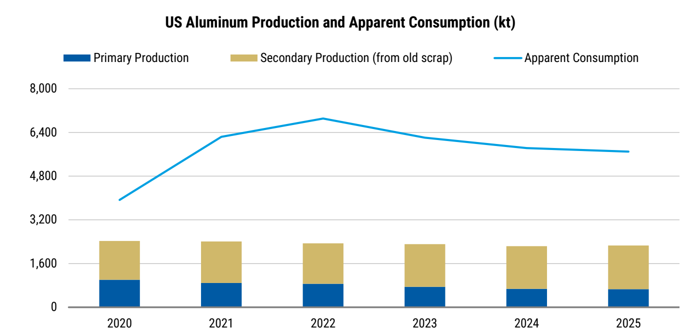
Exhibit 56: ... But could see capacity nearly double if the Century/EGA Oklahoma JV goes online



Source: Company Data, Morgan Stanley Research

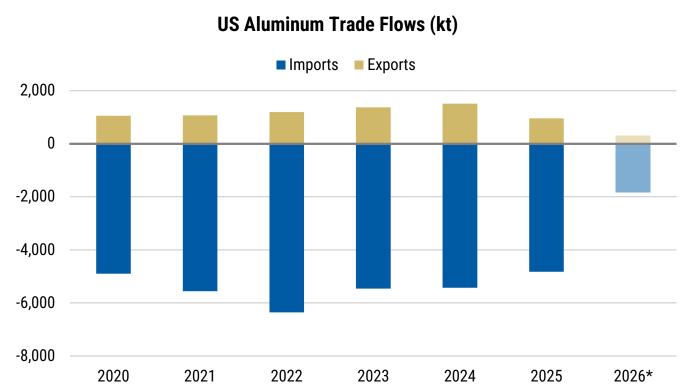
The US is a net aluminum importer, and lack of new capacity shows this will not change in the short to medium term: US apparent aluminum consumption remains well above domestic production, even after including secondary output. Apparent consumption peaked at 6.9mt in 2022 and normalized to ~5.7mt in 2025, but total domestic production stayed closer to 2.2-2.4mt over the same period (Exhibit 57). The country's aluminum imports peaked at ~6.3mt in 2022 and declined to 4.8mt in 2025, broadly tracking the normalization in apparent consumption (Exhibit 58). Imports have remained far larger than exports across the period, leaving the US consistently dependent on external supply. Considering the lack of primary capacity coming online before Oklahoma, the country will remain a large net importer of primary metal.

Exhibit 57: Even considering secondary output, US aluminum production falls well short of consumption...



Source: USGS, Morgan Stanley Research

Exhibit 58: ...Leaving imports to clear the structural supply gap

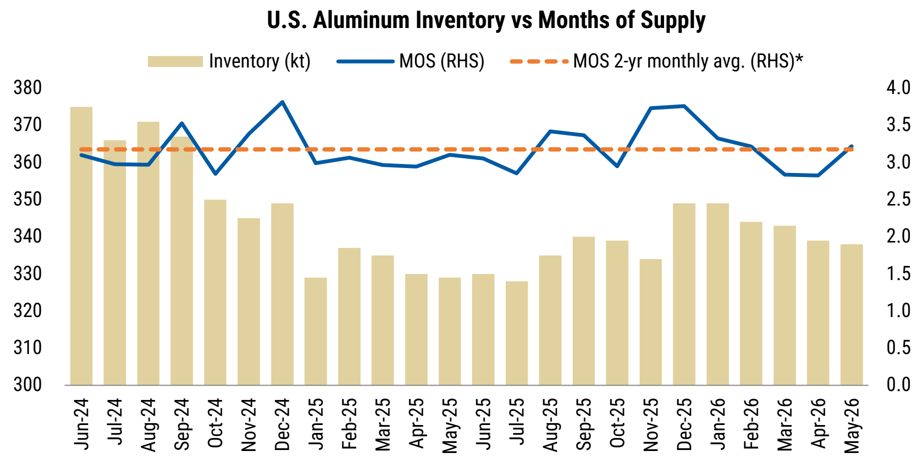


Source: International Trade Administration, Morgan Stanley Research.
(*) 2026 data up to April

Inventories offer limited cushion for the US market: MSCI distribution channel data show US aluminum inventories near 340kt in May 2026, equivalent to approximately three months of supply and broadly in line with the recent average. Despite four months of consecutive drops since January 2026, inventories at distributors remain above every single month in 2025 (Exhibit 59). This suggests the US has normal working inventories in the physical distribution channel, but not a large excess buffer.

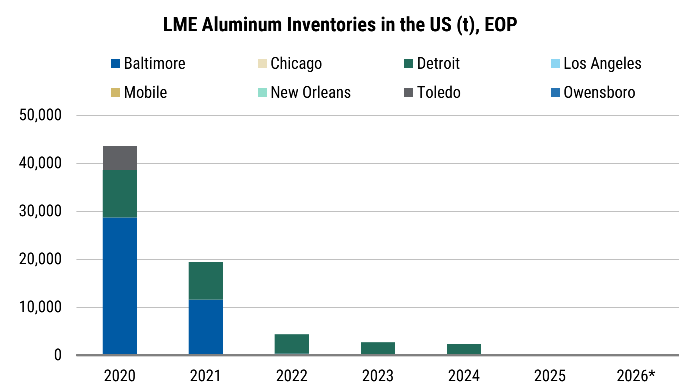
At the same time, exchange-visible inventories have been largely depleted: LME and COMEX aluminum stocks in US locations have fallen from ~75-80kt in 2020 to virtually zero by 2025-26 (Exhibit 60 and Exhibit 61) . Together, the data reinforce that the US market cannot rely on domestic primary capacity or visible inventories to balance supply, again highlighting the reliance on imported metal.

Exhibit 59: Distribution channel inventories remain near normal working levels...



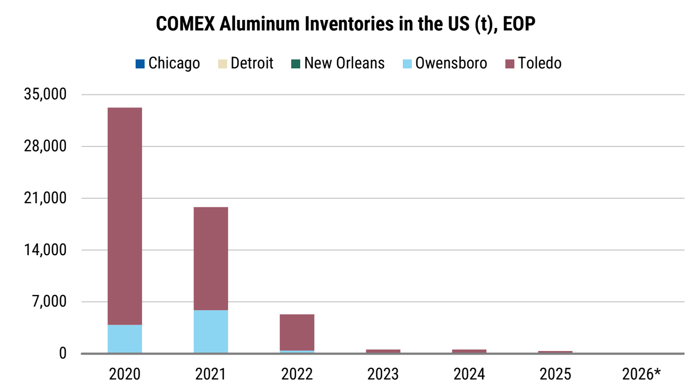
Source: MSCI, Morgan Stanley Research. (*) Average of last 24 consecutive months.

Exhibit 60: ...While LME US stocks have been largely depleted...



Source: Bloomberg, Morgan Stanley Research. (*) 2026 data as of June

Exhibit 61: ...And COMEX inventories provide little additional cushion



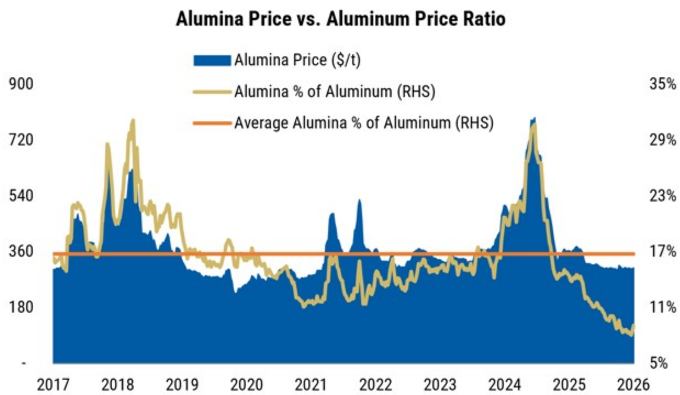
Source: Bloomberg, Morgan Stanley Research. (*) 2026 data as of June

Alumina discount supports smelter margins

Alumina has become unusually cheap relative to aluminum: The alumina-to-aluminum price ratio has fallen to 8-9%, well below its historical average of 17% since 2017 (Exhibit 62). This does not mean alumina prices are at historical lows in absolute terms; rather, it is because alumina has normalized sharply from the 2024 spike while aluminum has remained supported. The result is a historically wide aluminum-alumina spread, improving the raw-material backdrop for smelters. Since Jan-25, alumina prices have fallen by over 50%, reaching close to US$300/t, while aluminum prices have increased 35% to US $3,400/t vs. US$2,500/t in early 2025 (Exhibit 63).

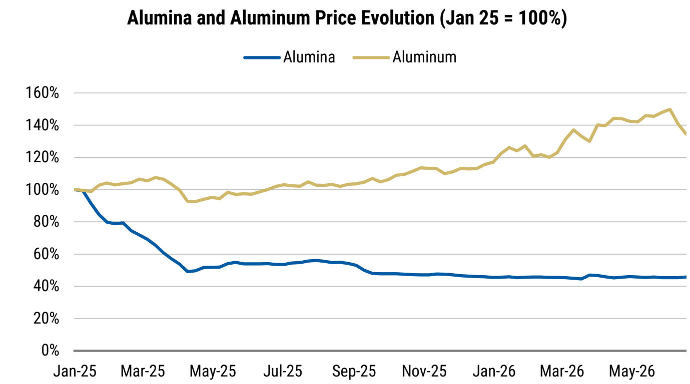
For primary aluminum producers, lower alumina cost burden supports margins: Alumina is one of the largest input costs in primary aluminum production, accounting for approximately one-third of smelting costs. Thus, a lower alumina-to-aluminum ratio directly improves smelter economics. This matters especially while power, freight and carbon costs remain elevated.

Exhibit 62: Alumina has fallen to an unusually low share of aluminum prices...



Source: Bloomberg, Platts, Morgan Stanley Research

Exhibit 63: ...As alumina has reset lower while aluminum has remained supported



Source: Bloomberg, Platts, Morgan Stanley Research

Australia

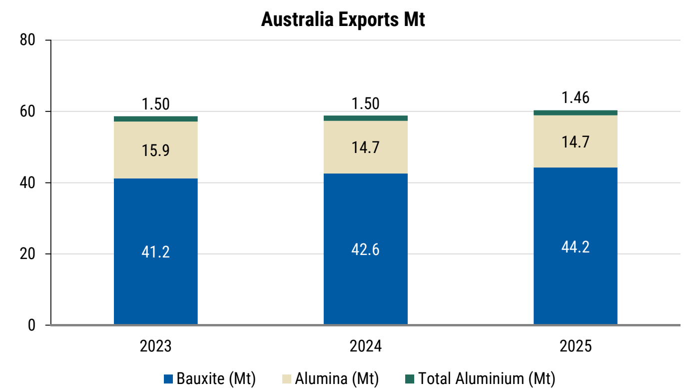
Australia has only four smelters currently operating: Bell Bay (Rio Tinto; located in Tasmania), Boyne Island (Rio Tinto; Queensland), Portland (Victoria; Alcoa), and Tomago (CSR, Hydro Aluminium; NSW). Rio Tinto also operates the ~335ktpa Tiwai point smelter in New Zealand. Australia also has a number of alumina and bauxite mines – all of which have varying degrees of integration with aluminum smelters. We expect combined Australia and New Zealand output to be broadly flat at ~2.0Mtpa from 2025 through 2030e, although Bell Bay and Tomago smelters are dependent on securing durable power arrangements to operate beyond 2026 and 2028, respectively.

Exhibit 64: Australia has four operating smelters, with Rio Tinto also operating the Tiwai point smelter in New Zealand

Smelter	2025	2026	2027	2028
Bell Bay	190	190	190	190
Boyne Island	>500	>500	>500	>500
Portland	358	358	358	358
Tomago	590	590	590	590
Tiwai Point	>335	>335	>335	>335
Total Capacity (ktpa)	**~1,973**	**~1,973**	**~1,973**	**~1,973**

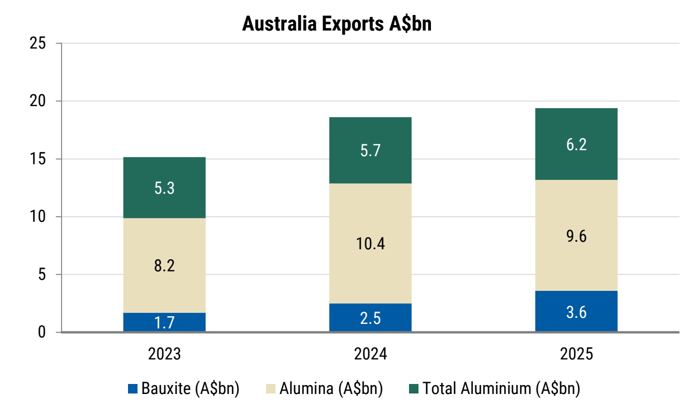
Source: Company reports, Morgan Stanley Research

Exhibit 65: Australia exports the majority of its aluminum produced; by volume, bauxite accounts for the most exports



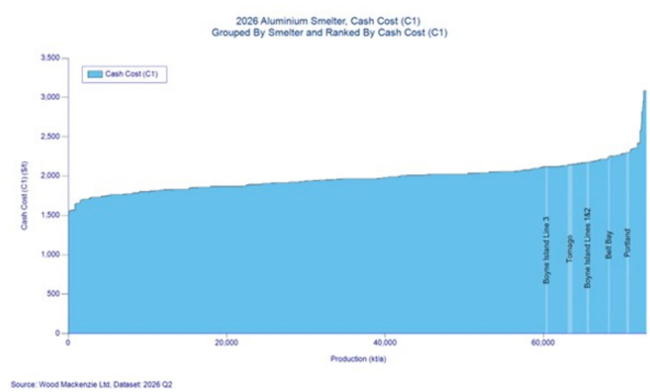
Source: Australia Aluminium Council. Morgan Stanley Research

Exhibit 66: Australia exports the majority of its aluminum produced; by value, alumina accounts for the most exports



Source: Australia Aluminium Council. Morgan Stanley Research

Exhibit 67: Most of Australia's Aluminum operations are in the high cost fourth quartile



Source: Wood Mackenzie. Morgan Stanley Research

Power is the main constraint for Australian smelters: Australia's smelter outlook is dominated by power contract negotiations. Smelting economics are highly sensitive to electricity at around 30-40% of the cost base, and Australia's smelters are moving from older base-load contracts into a renewable-heavy power system. Continued economic viability relies on whether utilities, operators and governments can structure long-duration contracts that keep assets internationally competitive, while supporting grid transition. Tomago, operated by Rio Tinto, accounts for ~590ktpa of aluminum production, making it the single-largest Australian smelter. Its current energy contract expires in 2028, and the Federal and NSW governments are working with Tomago on a long-term renewable energy solution to support viability beyond that date. Given Tomago's scale, the outcome remains a risk to our flat Australian supply assumption from 2029 onward. Bell Bay has agreed in principle to a 12-month extension beyond the December 31, 2025, end date of its existing power contract, enabling continued production while Rio Tinto, Hydro Tasmania and the Tasmanian Government negotiate a new 10-year agreement. The smelter still needs a durable pricing and supply framework to remain in production through the decade.

Boyne Island has secured a long-term partnership with the Queensland and Commonwealth governments intended to support the smelter beyond its current power contract and preserve international competitiveness (see here). Portland has also secured longer-dated power supply with AGL Energy (AGL.AX), including a 300MW agreement and an additional 287MW agreement, both effective from July 2026 and running to June 30, 2035. For Rio Tinto's Tiwai Point, 20-year electricity agreements were signed in May 2024, covering an aggregate 572MW of supply and securing the smelter through at least 2044 and also preserving flexible industrial load for the New Zealand grid.

Implications for supply: Our base case assumes that power negotiations for Tomago and Bell Bay are resolved sufficiently to hold Australian production flat through the decade. Failed or uneconomic outcomes, although unlikely, in our view, would remove supply up to ~780kt and partly offset the loosening expected from Middle East restarts and new Indonesian capacity.

South32's Mozal smelter shows the risks from failed power negotiations…: Mozal was placed on care and maintenance in March 2026 after South32 was unable to secure sufficient affordable power beyond contract expiry, which were exacerbated by drought conditions impacting hydroelectric generation at Hidroeléctrica de Cahora Bassa (HCB), removing ~550ktpa of capacity from the market. South32 said the formal Eskom offer was around ~US$100/MWh, well above the US$51/MWh that South32 required, while noting that less than 1% of ex-China smelters pay above US$50/MWh. Restart at Mozal therefore looks unlikely near term and would require a durable power solution, with 2030+ a more realistic window and restart costs likely >US$200m, in our view.

…However, Hillside risk looks more contained: Hillside, the ~720ktpa South African smelter is South32's major remaining Southern African aluminum asset, the largest aluminum smelter in the Southern Hemisphere and Eskom's largest industrial customer, consuming ~10.3TWh pa, or ~5% of Eskom power sales. Its current contract runs to 2031, and South32/Eskom are already advancing talks on a new long-term solution from 2031, including additional renewable energy and firming options. We see much lower risk of a similar outcome to Mozal given Hillside's direct importance to Eskom base-load demand, and larger South African industrial/employment implications.

Chalco

Exhibit 68: Financial Summary

Key Drivers	2024	2025	2026e	2027e	2028e
Aluminum Prod ('000 tonne)	7,610	8,080	8,069	8,069	8,069
Alumnium ASP (Rmb/tonne)	19,900	20,700	23,724	20,308	20,627
Aluminum Cost (Rmb/tonne)	16,071	14,613	13,919	13,819	13,826
Alumina Prod'n ('000 tonne)	16,870	17,350	18,926	19,333	19,333
Alumina ASP (Rmb/tonne)	3,800	3,228	2,659	2,644	2,856
Alumina Cost (Rmb/tonne)	2,411	2,295	2,228	2,225	2,224
Income Statement (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Under IFRS					
Net Sales	237,109	241,125	254,100	225,273	227,010
Gross Profit	34,260	39,852	61,552	37,681	44,380
EBITDA	37,446	40,542	62,256	40,156	47,336
Consensus EBITDA	39,847	40,301	43,823 e	44,848 e	#Refresh#
Diff. to Consensus EBITDA	*-6%*	*1%*	*42%*	*NA*	*NA*
EBIT	24,120	27,322	50,644	29,424	36,054
Pre-tax Profit	22,312	25,840	49,260	28,395	35,286
Net Profit	**12,395**	**12,674**	**21,753**	**12,539**	**15,583**
EPS (Rmb)	**0.72**	**0.74**	**1.27**	**0.73**	**0.91**
DPS (Rmb)	0.22	0.27	0.47	0.27	0.33
BVPS (Rmb)	4.03	4.37	5.17	5.63	6.21
Under PRC GAAP					
Net profit	**12,395**	**12,674**	**21,753**	**12,539**	**15,583**
EBITDA	37,446	40,542	62,256	40,156	47,336
EPS (Rmb)	**0.72**	**0.74**	**1.27**	**0.73**	**0.91**
DPS (Rmb)	0.22	0.27	0.47	0.27	0.33
BVPS (Rmb)	4.03	4.37	5.17	5.63	6.21
Cash Flow (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
EBITDA	37,446	40,542	62,256	40,156	47,336
-Taxes Paid	(2,940)	(4,315)	(12,315)	(7,099)	(8,822)
-Working Capital	325	286	(7,952)	(4,584)	(5,696)
-Others	(2,321)	(3,746)	(2,065)	2,617	(473)
Operating cash flow	**32,809**	**34,092**	**47,425**	**37,221**	**38,388**
-Capex.	(10,369)	(9,244)	(15,754)	(13,967)	(14,075)
FCF	22,440	24,848	31,671	23,254	24,314
Balance Sheet (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Cash & Equivalents	22,210	29,238	50,893	67,855	85,025
Receivables	5,231	5,172	5,451	4,832	4,869
Inventories	24,421	24,022	22,937	22,388	21,782
Total Assets	**215,942**	**227,022**	**255,849**	**270,954**	**291,334**
Payables	22,234	21,720	20,740	20,243	19,695
Borrowings	46,302	46,153	46,153	46,153	46,153
Total Liabilities	**103,887**	**104,450**	**104,284**	**102,676**	**102,288**
Shareholders Equity	69,204	74,925	88,727	96,682	106,569
Minority Interest	42,850	47,647	62,838	71,595	82,478
Total Liabilities and Equity	**215,942**	**227,022**	**255,849**	**270,954**	**291,334**
Per Share Data (Rmb)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
EPS for Consensus	0.72	0.74	1.27	0.73	0.91
Consensus EPS			0.92 e	0.92 e	0.92 e
Diff. to Consensus EPS (%)			37.4	(20.8)	(1.6)
DPS	0.22	0.27	0.47	0.27	0.33
BVPS	4.03	4.37	5.17	5.63	6.21

Profitability Ratios %	2024	2025	2026e	2027e	2028e
ROE	20%	20%	23%	22%	21%
EBITDA Margin	16%	17%	25%	18%	21%
EBIT Margin	10%	11%	20%	13%	16%
Pre-tax Profit Margin	9%	11%	19%	13%	16%
Net Profit Margin	5%	5%	9%	6%	7%
Net Debt / Equity %	30%	21%	3%	(8%)	(16%)
Z score (<1.8, higher risk)	2.2	2.2	2.4	2.0	2.0
Interest Cover (EBITDA) (x)	13.56	17.31	26.56	17.13	20.20
Working Capital % of Sales	(1.0)	(1.6)	(0.8)	1.2	(0.2)
Days Sales AR outstanding	8	9	10	10	10
Days in Inventory	43	44	45	44	44
Days Payable outstanding	(24)	(23)	(23)	(23)	(23)
Cash Conversion	27	30	32	31	31

Valuation	2024	2025	2026e	2027e	2028e
P/E	5.8	12.2 e	9.2 e	8.4 e	8.0 e
P/BV	1.0	2.1 e	1.8 e	1.6 e	1.5 e
EV/Sales	0.8	1.0 e	0.9 e	0.9 e	0.9 e
EV/EBITDA	4.9	5.9 e	4.7 e	4.2 e	3.8 e
FCF Yield %	31.0%	11.3% e	15.0% e	17.3% e	19.1% e
Dividend Yield %	3.9%	2.2% e	2.7% e	3.0% e	3.8% e

EV/EBITDA Analysis	2024	2025e	2026e	2027e	2028e
Average Stock price (HK$)	4.86	6.62	11.20	11.20	11.20
Eq Shares outstanding (in Mn)	17,046	17,097	17,097	17,097	17,097
Equity Value	78,846	104,867	171,606	172,712	176,665
Net Debt	33,173	25,910	4,254	(12,707)	(29,877)
Minority Interest	42,850	47,647	62,838	71,595	82,478
Enterprise Value	154,869	178,423	238,699	231,600	229,265
EBITDA (in Millions)	37,446	40,542	62,256	40,156	47,336
Associate Income	871	680	680	680	680
EV/EBITDA incl. asocte inc	4.0	4.3	3.8	5.7	4.8

Sensitivity analysis: Impact of a 1% change in the following					
2026e Rmb					% impact
(Rmb mn, EPS in Rmb)	EBITDA	EBIT	Net Income	EPS	on EPS
Aluminum price (up 1%)	3%	3%	3%	1.31	3%
Alumina price (up 1%)	1%	1%	1%	1.29	1%
Aluminum volume (up 1%)	1%	1%	1%	1.29	1%

Quarterly Financials	2Q24	3Q24	4Q24	2024	2025
Revenue	61,763	63,060	63,330	237,109	55,789
Gross Profit	10,835	6,633	10,002	34,260	8,047
EBIT	8,622	4,962	5,153	24,121	6,689
Net Income	4,786	1,998	3,383	12,395	3,535
EPS (Rmb)	0.28	0.12	0.20	0.73	0.21

Source: Company data, Morgan Stanley Research estimates

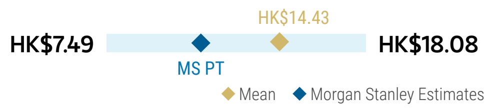
Risk Reward – Aluminum Corp. of China Ltd. (2600.HK)

Resilient demand and disrupted supply on capped capacity

PRICE TARGET HK$11.20

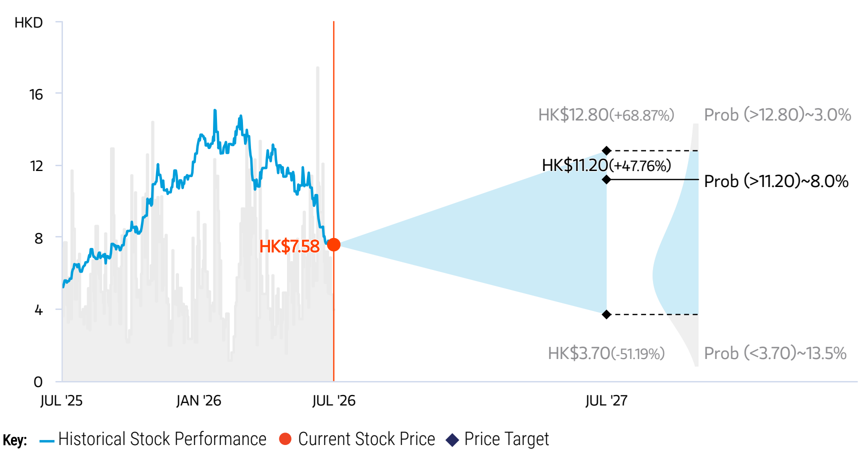
Base case, residual income model. Key assumptions: A cost of equity of 12.1% (as calculated from a 1.3 beta, 2.3% risk-free rate and 7.5% equity risk premium), a steady-state growth rate of 2% and long-term ROE of 11.0%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)

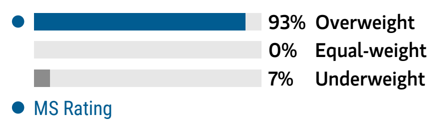


Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 7 Jul 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology here

OVERWEIGHT THESIS

- Chalco is one of our favored plays on the aluminum price trends that we expect.
- As a result of tightened carbon emission controls, an increased cost curve and rising demand driven by the solar and auto segments as well as large-scale equipment and consumer goods trade-ins, we expect aluminum prices to stay high.
- Improving bauxite self-sufficiency rates should help secure stable high-quality raw material supply. Together with alumina capacity commenced in Guangxi Fangchenggang, this should create synergies in the company's alumina business.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE HK$12.80

2.5x 2026e BVPS

Higher margins and strong demand: Downstream demand grows more strongly in China and overseas, driven by the solar and auto segments, large-scale equipment and consumer goods trade-ins. We apply our bull-case aluminum price forecast of Rmb27,308/t in 2026.

BASE CASE HK$11.20

2.2x 2026e BVPS

Base case pricing: We apply our base-case aluminum price forecast of Rmb23,724/t in 2026. We expect aluminum production to remain at a high level but demand should increase, driven by the solar and auto segments as well as large-scale equipment and consumer goods trade-ins.

BEAR CASE HK$3.70

0.7x 2026e BVPS

Aluminum prices fall with excess supply: Demand decrease in the property sector outweighs increases in the solar and auto segments, leading to a bear-case aluminum price forecast of Rmb18,518/t in 2026 amid volume decreases.

Risk Reward – Aluminum Corp. of China Ltd. (2600.HK)

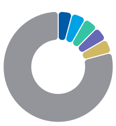
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Reported aluminium ASP (Rmb/t, in VAT)	20,700.0	23,724.0	20,307.6	20,627.5
Aluminium Production Volumes (kt)	8,080.0	8,068.5	8,068.5	8,068.5
Reported Alumina ASP (Rmb/t, in VAT)	3,228.0	2,659.3	2,644.1	2,856.1
Aluminum COGS/t (ex-VAT) (Rmb/ton)	14,613.2	13,919.2	13,819.0	13,826.2
Thermal coal price (US$/t)	106.7	135.8	131.2	123.8

INVESTMENT DRIVERS

- Demand recovery, with the government stimulus increasing demand for aluminum.
- Slower-than-expected capacity increase in China.

GLOBAL REVENUE EXPOSURE



- 0-10% APAC, ex Japan, Mainland China and India
- 0-10% Europe ex UK
- 0-10% India
- 0-10% Japan
- 0-10% North America
- 90-100% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Demand is better than we expect.
- Supply cut is greater than we expect.

RISKS TO DOWNSIDE

- Faster-than-expected production resumption.
- Faster commencement of new swapped capacity.
- Weaker-than-expected demand.

OWNERSHIP POSITIONING

Inst. Owners, % Active 52.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (Rmb, mn)	236,751.0	254,099.7	257,938.4	278,368.0
EBITDA (Rmb, mn)	47,918.0	62,396	59,517.1	70,733.0
Net income (Rmb, mn)	16,461.2	21,753	22,379.6	29,339.0
EPS (Rmb)	1.0	1.3	1.3	1.7

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

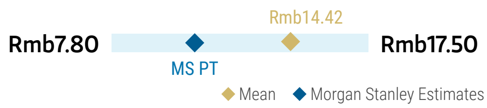
Risk Reward – Aluminum Corp. of China Ltd. (601600.SS)

Resilient demand and disrupted supply on capped capacity

PRICE TARGET Rmb10.80

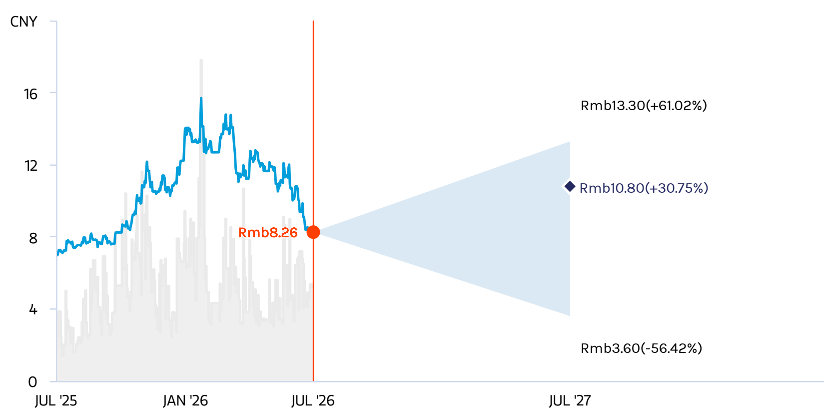
We derive our A-share price target by applying a 10% premium to our H-share price target, and adjusting for our estimated CNYHKD exchange rate of 1.14.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



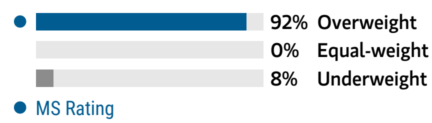
Key: Historical Stock Performance, Current Stock Price, Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Chalco is one of our favored plays on the aluminum price trends that we expect.
- As a result of tightened carbon emission controls, an increased cost curve and rising demand driven by the solar and auto segments as well as large-scale equipment and consumer goods trade-ins, we expect aluminum prices to remain high.
- Improving bauxite self-sufficiency rates should help secure stable high-quality raw material supply. Together with alumina capacity commenced in Guangxi Fangchenggang, this should create synergies in the company's alumina business.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb13.30

2.6x 2026e BVPS

Higher margins and strong demand: Downstream demand grows more strongly in China and overseas, driven by the solar and auto segments, large-scale equipment and consumer goods trade-ins. We apply our bull-case aluminum price forecast of Rmb27,308/t in 2026.

BASE CASE Rmb10.80

2.1x 2026e BVPS

Base case pricing: We apply our base-case aluminum price forecast of Rmb23,724/t in 2026. We expect aluminum production to remain at a high level but demand should increase, driven by the solar and auto segments as well as large-scale equipment and consumer goods trade-ins.

BEAR CASE Rmb3.60

0.7x 2026e BVPS

Aluminum prices fall with excess supply: Demand decrease in the property sector outweighs increases in the solar and auto segments, leading to a bear-case aluminum price forecast of Rmb18,518/t in 2026 amid volume decreases.

Risk Reward – Aluminum Corp. of China Ltd. (601600.SS)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Reported aluminium ASP (Rmb/t, in VAT)	20,700.0	23,724.0	20,307.6	20,627.5
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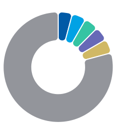
Thermal coal price (US$/t)	106.7	135.8	131.2	123.8

INVESTMENT DRIVERS

- Slowdown in auto and related downstream segments, leading to slower aluminum demand.
- Resumption of production from overseas capacity.

GLOBAL REVENUE EXPOSURE



- 0-10% APAC, ex Japan, Mainland China and India
- 0-10% Europe ex UK
- 0-10% India
- 0-10% Japan
- 0-10% North America
- 90-100% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

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1/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

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- Supply cut is greater than we expect.

RISKS TO DOWNSIDE

- Faster-than-expected production resumption.
- Faster commencement of new swapped capacities.
- Weaker-than-expected demand.

OWNERSHIP POSITIONING

Inst. Owners, % Active 88.9%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (Rmb, mn)	236,751.0	254,099.7	258,600.8	278,368.0
EBITDA (Rmb, mn)	47,918.0	62,396	59,099.7	70,733.0
Net income (Rmb, mn)	15,655.0	21,753	21,844.0	29,339.0
EPS (Rmb)	1.0	1.3	1.3	1.7

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation and earnings changes

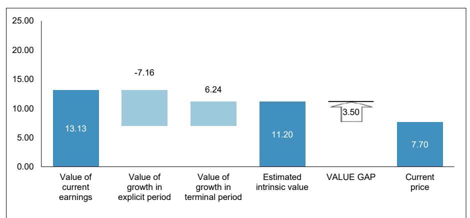
We update our model to reflect the latest aluminum and alumina outlook. Reflected the lower prices estimates, our EPS decreases by 41% and 34% compare to our previous estimates. Dragged by the lower earnings, our PT lowers to HK$11.2 for Chalco H and Rmb10.8 for Chalco A.

Chalco is the largest aluminum producers in China. Despite lower earnings estimates, we think the increasing bauxite self-sufficiency and strategic layout of alumina plants would offer some cost reduction for aluminum. Meanwhile, the company is progressing on acquiring Brazil Aluminum. Once completed, Chalco would have some oversea volume increase with green power energy. Furthermore, management team targets to gradually increase its payout in next few years. Based on all these reasons, we stay OW on the name.

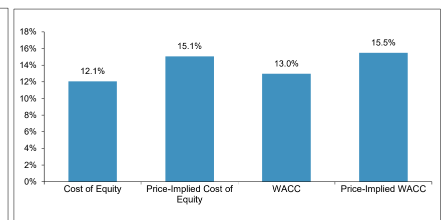
Exhibit 69: Intrinsic Valuation

INTRINSIC VALUE

2600.HK



Source: Morgan Stanley Research, ModelWare



Source: Morgan Stanley Research, ModelWare

Intrinsic Value Model

Inputs:		**Outputs:**		
		Price		**7.70**
Cost of equity	**12.1%**	**IV Per Share (12 Month), Ex. Div**	**Jul-27**	**11.20**
Long-term ROE on new investments	**11.0%**	Expected share price return		45.45%
Years to reach steady-state growth	**10**	Expected dividend yield		6.04%
Steady-state revenue growth rate (%)	**2%**	**Expected total return**		**51.49%**
Shares Outstanding	**17,158**			
Steady-state borrowing cost (net of tax)	**5.0%**			
Steady-state leverage (Net debt/Equity)	**50%**	WACC		13.0%
Price target horizon (months)	**12**	Long-term RNOA on new investments		9.0%
Conv. factor - Model to traded Ccy	**1.15**	Explicit forecast period (years)		7
Decimals	**2**	Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):		**DCF (operating):**	
Beginning NOA	**139,529**		
Sum of PVRI	71,732	Sum of PVFCFO	211,261
Beginning NNOAL	(64,604)	Beginning NNOAL	(64,604)
ROI equity value	**146,657**	**DCF Equity Value**	**146,657**
Residual Income (equity):		**DDM (Equity):**	
Beginning equity	74,925	Sum of div & net rep	146,656
Sum of PVRI	71,732	**DDM equity value**	**146,656**
RI equity value	**146,656**		

Source: Morgan Stanley Research estimates

China Hongqiao

Exhibit 70: Financial Summary

Key Drivers	2024	2025	2026e	2027e	2028e
Production (kt)					
Aluminum alloy products	6,500	6,545	6,524	6,524	6,524
Alumina	20,470	20,138	20,138	20,138	20,138
Aluminum fabrication	1,000	1,006	1,006	1,006	1,006
ASP (Rmb/t)					
Aluminum alloy products	19,830	20,585	23,573	20,373	20,587
Alumina	3,865	3,276	2,684	2,644	2,686
Aluminum fabrication	20,328	23,306	25,636	24,354	22,893
Income Statement (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Net Sales	156,169	162,354	177,781	159,438	159,533
Gross Profit	42,163	41,505	55,072	40,157	39,838
EBITDA	42,435	43,222	58,627	46,283	47,097
Consensus EBITDA			58,958 e	58,371 e	59,248 e
Diff. to Consensus EBITDA			-1%	-21%	-21%
EBIT	36,161	36,530	51,198	38,282	38,526
Pre-tax Profit	32,797	33,046	48,088	35,172	35,415
Net Profit	22,372	22,636	31,633	23,137	23,297
Cash Flow (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
EBITDA	42,435	43,222	58,627	46,283	47,097
-Taxes Paid	-8,252	-8,893	-12,940	-9,465	-9,530
-Working Capital	-4,459	5,360	-2,008	2,928	-451
-Others	4,258	-694	-4,281	-2,063	-3,118
Operating cash flow	33,983	38,995	39,398	37,684	33,998
-Capex.	-12,609	-10,657	-15,645	-12,755	-12,763
FCF	21,374	28,339	23,753	24,928	21,235
Balance Sheet (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Cash & Equivalents	44,770	51,187	54,666	64,761	71,060
Receivables	16,376	9,240	10,118	9,074	9,080
Inventories	37,344	36,636	37,200	36,161	36,286
Total Assets	229,165	245,379	260,282	271,354	282,012
Payables	14,931	11,189	11,189	11,361	11,044
Borrowings	54,084	49,907	49,907	49,907	49,907
Total Liabilities	110,552	103,669	103,669	103,841	103,523
Shareholders Equity	107,800	132,593	143,981	152,310	160,697
Minority Interest	10,814	9,117	12,632	15,203	17,792
Total Liabilities and Equity	229,165	245,379	260,282	271,354	282,012

Per Share Data (Rmb)	2024	2025	2026e	2027e	2028e
EPS for concensus	2.36	2.38	3.33	2.44	2.45
Consensus EPS			3.4 e	3.4 e	3.4 e
Diff. to Consensus EPS (%)			-0.57	-27.45	-28.21
DPS	1.61	1.65	2.29	1.68	1.69
BVPS	11.38	13.97	15.17	16.04	16.93
Profitability Ratios %	**2024**	**2025**	**2026e**	**2027e**	**2028e**
ROE	21%	17%	22%	15%	14%
EBITDA Margin	27%	27%	33%	29%	30%
EBIT Margin	23%	23%	29%	24%	24%
Pre-tax Profit Margin	21%	20%	27%	22%	22%
Net Profit Margin	14%	14%	18%	15%	15%
Net Debt / Equity %	3%	-6%	-8%	-14%	-17%
Interest Cover (EBITDA) (x)	13	12	19	15	15
Working Capital % of Sales	25%	21%	20%	21%	22%
Days Sales AR outstanding	38	21	21	21	21
Days in Inventory	120	111	111	111	111
Days Payable outstanding	48	34	34	34	34
Cash Conversion	110	98	98	98	98
Valuation	**2024**	**2025**	**2026e**	**2027e**	**2028e**
P/E	4.7	10.1 e	7.9 e	6.9 e	8.0 e
P/BV	1.0	2.2 e	2.0 e	1.8 e	1.9 e
EV/Sales	0.9	1.7 e	1.6 e	1.5 e	1.6 e
EV/EBITDA	3.3	6.3 e	5.2 e	4.5 e	4.9 e
FCF Yield %	18.7%	4.6% e	10.2% e	11.6% e	10.8% e
Dividend Yield %	8.9%	6.3% e	8.1% e	9.2% e	7.9% e
EV/EBITDA Analysis	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Average Stock price (HK$)	10	20	29	29	29
Eq Shares outstanding (CN)	9476	9494	9494	9494	9494
Equity Value	96425	192434	271529	271529	271529
Net Debt	3532	-7776	-11255	-21350	-27649
Minority Interest	10814	9117	12632	15203	17792
Enterprise Value	110771	193776	272907	265383	261672
EBITDA (in Millions)	42435	43222	58627	46283	47097
Sensitivity analysis: Impact of a 1% change in the following					
2026e Rmb					
(Rmb mn, EPS in Rmb)	EBITDA	EBIT	Net Income	EPS	
Volume (up 1%)	1.0%	1.1%	1.2%	1.2%	
Price (up 1%)	2.3%	2.6%	2.8%	2.8%	
Cost (up 1%)	-1.1%	-1.2%	-1.3%	-1.3%	

Source: Company data, Morgan Stanley Research estimates

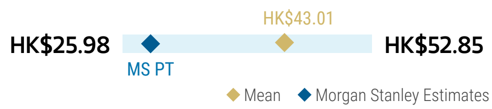
Risk Reward – China Hongqiao Group (1378.HK)

Key beneficiary of structural shift in industry demand and constrained supply

PRICE TARGET HK$28.60

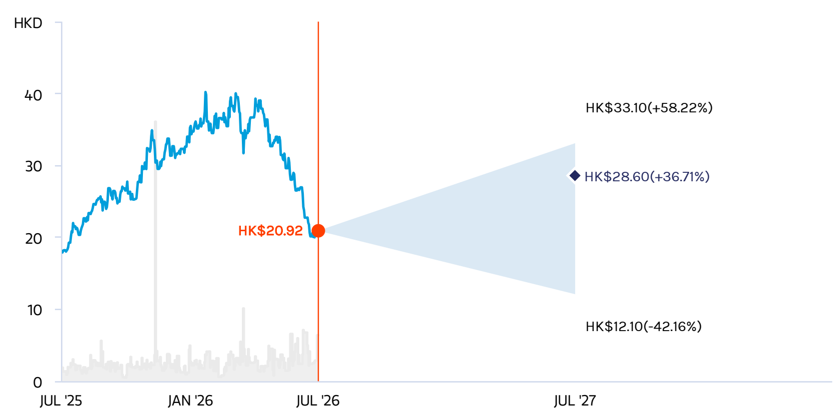
We derive our price target using a residual income model given the relatively high visibility of its long-term earnings. We apply a cost of equity of 10.5%, calculated from a beta of 1.18, a 1.6% risk-free rate and a 7.5% equity risk premium. We assume a steady state revenue growth rate of 2% for Hongqiao.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



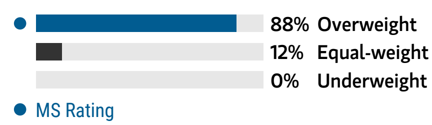
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Supply is limited by a capacity ceiling of 45mnt imposed by the central government. Meanwhile, supply disruption may continue in southwest China.
- Hongqiao's margins are likely to improve thanks to lower costs on lower power tariff from capacity relocation and coal price declines.
- Integrated business into upstream raw materials and downstream processing should help bring solid earnings momentum.
- Valuation remains attractive despite outperformance; yield is also appealing.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Renewable Energy: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE HK$33.10

9.9x 2026e P/E

Strong demand and severe supply disruption: Aluminum price is higher than our expectation due to the slow volume release from capacity overseas and potential supply disruptions. This would secure strong earnings for Hongqiao.

BASE CASE HK$28.60

8.6x 2026e P/E

Demand shift and continued supply disruption on power constraints: We assume the aluminum price increases by 15% YoY in 2026, to Rmb23,573/t, while the alumina price falls 18% to Rmb2684/t.

BEAR CASE HK$12.10

3.6x 2026e P/E

Sluggish demand and sufficient supply: The supply increase overseas is quicker than our expectations. This, together with weaker-than-expected demand, pressures aluminum prices and is a drag on Hongqiao's earnings.

Risk Reward – China Hongqiao Group (1378.HK)

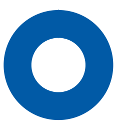
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Aluminum Alloy Production	6,545.0	6,523.6	6,523.6	6,523.6
Alumina Production	20,137.8	20,137.8	20,137.8	20,137.8
LME price (US$/lb) (Rmb)	1.20	1.48	1.29	1.27
Global Alumina price (Rmb)	364	314	325	350

INVESTMENT DRIVERS

- More stable volume compared with peers because of better performance on environmental protection
- Better-than-expected demand from downstream industries
- Cost savings through lower power tariff

GLOBAL REVENUE EXPOSURE



100% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST | 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected demand
- Decreasing raw material and energy prices
- More plants in maintenance than expected

RISKS TO DOWNSIDE

- Slowdown in demand globally
- Rising raw material and energy prices
- Industry overcapacity

OWNERSHIP POSITIONING

Inst. Owners, % Active 51.9%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue (Rmb, mn)	164,867.0	177,781.3	175,191.0	195,602.0
EBITDA (Rmb, mn)	50,102.0	58,323	59,788.8	70,459.0
Net income (Rmb, mn)	26,602.0	31,633	33,463.7	42,618.0
EPS (Rmb)	2.7	3.3	3.4	4.3

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

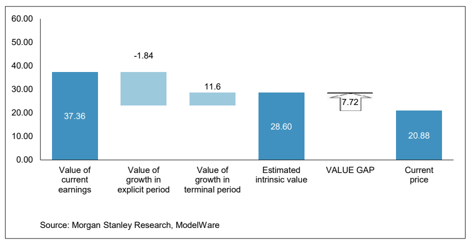
Valuation and earnings changes

We update our model using our latest aluminum and alumina price estimates from global commodity team. Reflecting the lower aluminum price outlook, our EPS in 2027-2028 decreases by 44% and 46% compare to our previous estimates. Our PT lowers to HK $28.6/share. Under current price, the valuation and dividend yield for China Hongqiao looks attractive. The company pays out a 63% dividend, the highest among all the aluminum players in China. Meanwhile, the constant share buyback from the company would also buffer the volatility in stock price. We stay OW on the name.

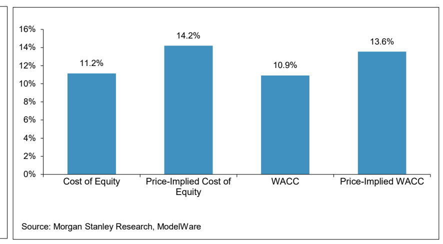
Exhibit 71: Intrinsic valuation

INTRINSIC VALUE

1378.HK



Source: Morgan Stanley Research, ModelWare



Source: Morgan Stanley Research, ModelWare

Intrinsic Value Model

Inputs:	
Cost of equity	11.2%
Long-term ROE on new investments	12.0%
Years to reach steady-state growth	10
Steady-state revenue growth rate (%)	2%
Shares Outstanding	9,494
Steady-state borrowing cost (net of tax)	7.0%
Steady-state leverage (Net debt/Equity)	50%
Price target horizon (months)	12
Conv. factor - Model to traded Ccy	1.15
Decimals	2

Outputs:		
Price		**20.88**
IV Per Share (12 Month), Ex. Div	**Jul-27**	**28.60**
Expected share price return		36.97%
Expected dividend yield		10.21%
Expected total return		**47.19%**
WACC		10.9%
Long-term RNOA on new investments		10.3%
Explicit forecast period (years)		7
Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):	
Beginning NOA	**158,838**
Sum of PVRI	87,024
Beginning NNOAL	(26,245)
ROI equity value	**219,617**

Residual Income (equity):	
Beginning equity	132,593
Sum of PVRI	87,024
RI equity value	**219,617**

DCF (operating):	
Sum of PVFCFO	245,863
Beginning NNOAL	(26,245)
DCF Equity Value	**219,617**

DDM (Equity):	
Sum of div & net rep	219,617
DDM equity value	**219,617**

Source: Morgan Stanley Research estimates

Chuangxin Industries

Exhibit 72: Financial Summary

Key Drivers	2024	2025	2026e	2027e	2028e
Production (kt)					
Aluminum	755	752	772	1,072	1,257
Alumina	1,540	2,550	2,784	2,880	2,880
ASP (Rmb/t)					
Aluminum	17,120	18,107	20,861	18,029	18,218
Alumina	3,707	2,869	2,375	2,340	2,377
Income Statement (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Revenue	15,163	18,681	21,077	23,244	26,870
Gross Profit	4,276	4,625	7,786	7,938	9,349
EBITDA	4,660	5,035	8,352	8,610	10,090
Consensus EBITDA			8,026	8,674	9,209
Diff. to Consensus EBITDA			4%	-1%	10%
EBIT	4,043	4,161	7,313	7,416	8,746
Pre-tax Profit	3,281	3,481	6,747	6,912	8,242
Net Profit	2,056	2,731	4,734	4,562	5,440
Cash Flow (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
EBITDA	4,660	5,035	8,352	8,610	10,090
-Taxes Paid	-651	-562	-1,687	-1,728	-2,061
-Working Capital	-880	313	-369	224	156
-Others	333	139	121	121	121
Operating cash flow	3,462	4,925	6,416	7,227	8,306
-Capex.	-3,103	-2,613	-2,951	-2,789	-2,687
FCF	359	2,312	3,466	4,438	5,619
Balance Sheet (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Cash & Equivalents	176	5,091	5,563	7,157	9,483
Receivables	525	172	245	265	299
Inventories	1,578	2,436	2,303	2,653	3,036
Total Assets	18,320	28,349	30,704	34,279	38,473
Payables	2,945	5,091	4,813	5,544	6,346
Borrowings	10,948	12,404	12,404	12,404	12,404
Total Liabilities	15,994	18,176	17,898	18,629	19,431
Shareholders Equity	1,255	9,623	11,929	14,152	16,802
Minority Interest	1,072	550	877	1,499	2,240
Total Liabilities and Equity	18,320	28,349	30,704	34,279	38,473

Per Share Data (Rmb)	2024	2025	2026e	2027e	2028e
EPS for concensus	1.37	1.75	2.28	2.20	2.62
Consensus EPS			2.46 e	2.97 e	2.79 e
Diff. to Consensus EPS (%)			-7.3%	-26.0%	-6.0%
DPS	0.00	0.77	1.17	1.13	1.34
BVPS	1.55	6.54	6.17	7.54	9.18
Profitability Ratios %	**2024**	**2025**	**2026e**	**2027e**	**2028e**
ROE	164%	28%	40%	32%	32%
EBITDA Margin	31%	27%	40%	37%	38%
EBIT Margin	27%	22%	35%	32%	33%
Pre-tax Profit Margin	22%	19%	32%	30%	31%
Net Profit Margin	14%	15%	22%	20%	20%
Net Debt / Equity %	859%	76%	57%	37%	17%
Interest Cover (EBITDA) (x)	6	7	15	17	20
Working Capital % of Sales	-6%	-13%	-11%	-11%	-11%
Days Sales AR outstanding	13	3	3	3	3
Days in Inventory	53	63	63	63	63
Days Payable outstanding	99	132	132	132	132
Cash Conversion	-33	-66	-66	-66	-66
Valuation	**2024**	**2025**	**2026e**	**2027e**	**2028e**
P/E		8.9 e	8.5 e	8.8 e	7.4 e
P/BV		2.4 e	3.1 e	2.6 e	2.1 e
EV/Sales		1.9 e	2.5 e	2.3 e	1.9 e
EV/EBITDA		7.1 e	6.4 e	6.1 e	5.0 e
FCF Yield %		8.3% e	7.6% e	9.7% e	12.3% e
Dividend Yield %		4.3% e	5.3% e	5.1% e	6.1% e
EV/EBITDA Analysis	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Average Stock price (HK$)		18	22	22	22
Eq Shares outstanding (CN)	1,500	1,556	2,075	2,075	2,075
Equity Value	-	27,837	45,650	45,650	45,650
Net Debt	10,772	7,313	6,841	5,247	2,921
Minority Interest	1,072	550	877	1,499	2,240
Enterprise Value	11,843	35,700	53,368	52,395	50,811
EBITDA (in Millions)	4,660	5,035	8,352	8,610	10,090
Sensitivity analysis: Impact of a 1% change in the following					
2026e Rmb					
(Rmb mn, EPS in Rmb)	Revenue	EBIT	Net Income	EPS	
Volume (up 1%)	0.7%	1.3%	1.4%	1.4%	
Price (up 1%)	0.8%	2.6%	2.8%	2.8%	
Cost (up 1%)	0.0%	-0.8%	-0.9%	-0.9%	

Source: Company data, Morgan Stanley Research estimates

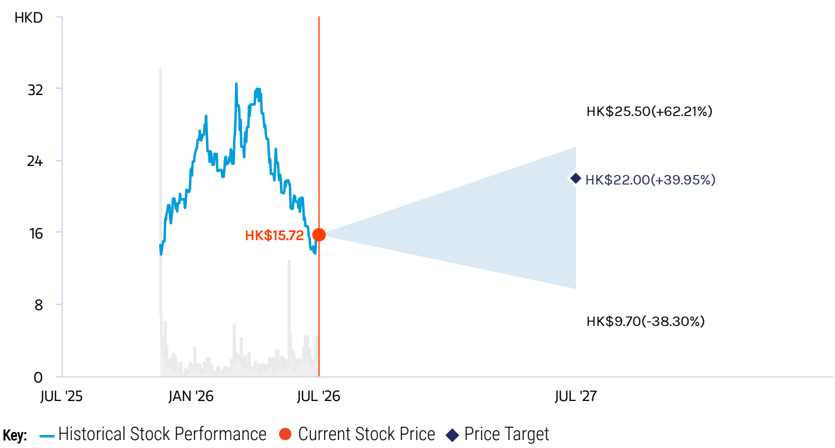
Risk Reward – Chuangxin Industries Holdings Ltd. (2788.HK)

A leading domestic aluminum smelter with overseas expansion potential

PRICE TARGET HK$22.00

Base case, derived from residual income valuation model. We discount our earnings forecasts through 2037 and then normalize earnings. Key assumptions include a cost of equity of 9.2% (risk-free rate of 2.3%, risk premium of 7%, beta of 0.98), a long-term ROE of 10% and a steady-state growth rate of 4%.

RISK REWARD CHART

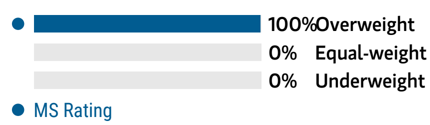


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect overseas capacity expansion to drive volume growth in the next few years. Constrained by China's capacity cap and seeking growth, Chuangxin plans to build a total of 1mnt of aluminum capacity in Saudi Arabia, with construction of the first 500kt phase already underway. The new capacity is targeted to commence operations in 1Q27.
- Costs are declining. Along with a rising share of renewables, we expect the total power tariff to fall to Rmb0.27/kWh (excluding VAT) in 2026 and settle at Rmb0.25-0.26/kWh (including VAT) over the long term.
- We are OW relative to our coverage given the solid earnings expected and volume growth overseas.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE HK$25.50

11.2x 2026e P/E

Bull-case aluminum price assumption and greater-than-expected improvement in cost competitiveness: Demand recovery is stronger than expected. Together with slower-than-expected aluminum production release due to Middle East tensions, we assume aluminum prices reach US$1.70/lb in 2026. Lower power tariffs on rising green power contribution and integrated raw material supply chain could support Chuangxin's overall profitability.

BASE CASE HK$22.00

9.6x 2026e P/E

Base-case aluminum price assumption and steady improvement in cost competitiveness: We expect aluminum prices in China to remain high against the backdrop of supply disruption from ongoing tensions in the Middle East. As an aluminum producer, Chuangxin should benefit from higher processing fees given potential disruption in aluminum production overseas. Rising green power contribution and an integrated raw material supply chain should further lower the company's production costs.

BEAR CASE HK$9.70

4.3x 2026e P/E

Bear-case aluminum price assumption and slower-than-expected capacity release: Chuangxin plans to build 1mnt of aluminum capacity in Saudi Arabia with construction of Phase I (500kt) already underway. Ongoing tensions in the Middle East could delay capacity release.

Risk Reward – Chuangxin Industries Holdings Ltd. (2788.HK)

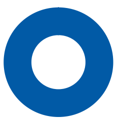
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GPM (%)	24.8	36.9	34.1	34.8

INVESTMENT DRIVERS

- Volume from capacity expansion in Saudi Arabia
- Costs along with increasing wind power contribution in Inner Mongolia.
- Aluminum demand.

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

- Quicker-than-expected capacity release
- Stronger-than-expected demand from major downstream such as power, auto and ESS industries.

RISKS TO DOWNSIDE

- Slower-than-expected capacity release due to ongoing tensions in the Middle East for Chuangxin
- Slowdown in demand and infrastructure spending

OWNERSHIP POSITIONING

Inst. Owners, % Active 100%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

EBIT (Rmb, mn) ◆ 7,313
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (Rmb, mn) ◆ 8,473
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (Rmb) ◆ 2.3
Note: There are not sufficient brokers supplying consensus data for this metric

DPS (Rmb) ◆ 1.2
Note: There are not sufficient brokers supplying consensus data for this metric

ROE (%) 49.2; range 38.0 – 72.0; 50.7

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation and earnings changes

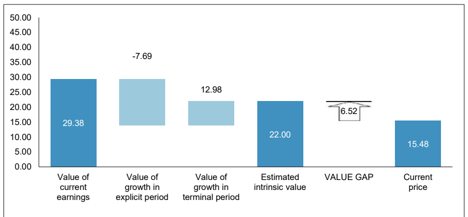
We update our model to reflect the latest aluminum and alumina price estimates from our global commodity team. As a result of lower price estimates, our EPS in 2027-2028 lower by 44% and 42%, respectively, compare to previous estimates. Reflected the lower earnings, our PT decreases to HK$22/share.

The construction for the 500kt capacity expansion in the Middle East is progressing and will commence production in 1Q27. This would implies a high volume increase for the company in 2027 and still some volume growth in 2028. Meanwhile, the green power installation in Inner Mongolia is gradually ramping up, leading to a cheaper production costs for Chuangxin. The cost for wind power is below Rmb0.1/kWh vs. Rmb0.32-33/kWh for captive thermal power. The declining costs, volume increase, and the cheaper valuation warrant an OW for the company.

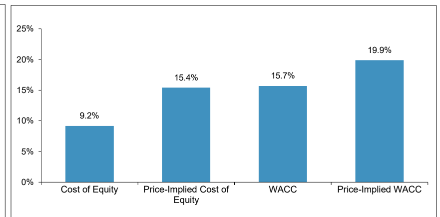
Exhibit 73: Intrinsic Valuation

INTRINSIC VALUE

2788.HK



Source: Morgan Stanley Research, ModelWare



Source: Morgan Stanley Research, ModelWare

Intrinsic Value Model

Inputs:		**Outputs:**		
Cost of equity	9.2%	**Price**		**15.48**
Long-term ROE on new investments	11.0%	**IV Per Share (12 Month), Ex. Div**	**Jul-27**	**22.00**
Years to reach steady-state growth	10	Expected share price return		42.12%
Steady-state revenue growth rate (%)	4%	Expected dividend yield		7.56%
Shares Outstanding	2,075	**Expected total return**		**49.68%**
Steady-state borrowing cost (net of tax)	3.0%			
Steady-state leverage (Net debt/Equity)	50%	WACC		15.7%
Price target horizon (months)	12	Long-term RNOA on new investments		8.3%
Conv. factor - Model to traded Ccy	1.15	Explicit forecast period (years)		12
Decimals	2	Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):		**DCF (operating):**	
Beginning NOA	**14,451**	Sum of PVFCFO	42,891
Sum of PVRI	28,440	Beginning NNOAL	(4,828)
Beginning NNOAL	(4,828)	**DCF Equity Value**	**38,063**
ROI equity value	**38,063**		
Residual Income (equity):		**DDM (Equity):**	
Beginning equity	9,623	Sum of div & net rep	38,063
Sum of PVRI	28,440	**DDM equity value**	**38,063**
RI equity value	**38,063**		

Source: Morgan Stanley Research estimates

Shenhuo Aluminum

Exhibit 74: Financial Summary

Key Drivers	2024	2025	2026e	2027e	2028e
Sales Volume					
Aluminium (mnt)	1.6	1.7	1.7	1.7	1.7
Coal (mnt)	6.7	7.2	6.8	6.9	6.9
ASP					
Aluminium (Rmb/t)	15956	17046	20861	18029	18218
Coal (Rmb/t)	1019	775	822	814	806
Cost per ton					
Aluminium (Rmb/t)	(11949)	(11922)	(12211)	(12097)	(12168)
Coal (Rmb/t)	(849)	(716)	(745)	(738)	(738)

Income Statement (Rmb mn)	2024	2025	2026e	2027e	2027e
Net Sales	38,373	41,241	47,924	43,178	43,444
Gross Profit	8,146	9,633	15,801	11,186	11,331
EBITDA	8,013	8,341	16,078	11,773	11,981
Consensus EBITDA			14,728	14,994	16,000
Diff. to Consensus EBITDA			9%	-21%	-25%
EBIT	6,464	6,641	13,391	9,015	9,146
Pre-tax Profit	6,413	6,549	13,251	9,455	10,031
Net Profit	4,307	4,005	8,723	6,224	6,603
EPS (Rmb)	1.93	1.81	3.94	2.81	2.98
DPS (Rmb)	0.80	0.80	1.74	1.24	1.32
BVPS (Rmb)	9.73	11.02	13.19	14.74	16.39

Cash Flow (Rmb mn)	2024	2025	2026e	2027e	2027e
EBITDA	8,013	8,341	16,078	11,773	11,981
-Taxes Paid	(1615)	(1985)	(3313)	(2364)	(2508)
-Working Capital	1062	668	(382)	613	875
-Others	258	1719	0	0	0
Operating cash flow	7719	8743	12384	10022	10349
-Capex.	(1328)	(861)	(1001)	(1079)	(1086)
FCF	6390	7882	11383	8943	9263

Balance Sheet (Rmb mn)	2024	2025	2026e	2027e	2027e
Cash & Equivalents	3,283	3,017	10,520	16,692	23,015
Receivables	920	1,102	1,281	1,154	1,161
Inventories	3,564	3,935	3,999	3,983	3,998
Total Assets	50,603	48,860	54,982	59,285	63,884
Payables	7,760	7,636	7,703	7,686	7,702
Borrowings	13,437	8,951	8,951	8,951	8,951
Total Liabilities	24,578	20,613	20,680	20,663	20,679
Shareholders Equity	21,682	24,567	29,407	32,860	36,523
Total Liabilities and Equity	50,603	48,860	54,982	59,285	63,884

Per Share Data (Rmb)	2024	2025	2026e	2027e	2027e
EPS for consensus	1.93	1.81	3.94	2.81	2.98
Consensus EPS			3.22	3.70	4.14
Diff. to Consensus EPS (%)			22%	-24%	-28%
DPS	0.80	0.80	1.74	1.24	1.32
BVPS	9.73	11.02	13.19	14.74	16.39

Profitability Ratios %	2024	2025	2026e	2027e	2027e
ROE	20%	16%	30%	19%	18%
EBITDA Margin	21%	20%	34%	27%	28%
EBIT Margin	17%	16%	28%	21%	21%
Pre-tax Profit Margin	17%	16%	28%	22%	23%
Net Profit Margin	11%	10%	18%	14%	15%
Net Debt / Equity %	47%	24%	-5%	-24%	-39%
Interest Cover (EBITDA) (x	156	90	115	-27	-14
Working Capital % of Sales	2%	5%	4%	4%	4%
Days Sales AR outstanding	9	10	10	10	10
Days in Inventory	43	45	45	45	45
Days Payable outstanding	57	47	47	47	47
Cash Conversion	-5	8	8	8	8

Valuation	2024	2025	2026e	2027e	2027e
P/E	8.8	10.9	7.9	11.0	10.4
P/BV	1.7	1.8	2.3	2.1	1.9
EV/Sales	1.4	1.3	1.5	1.6	1.4
EV/EBITDA	6.5	6.4	4.5	5.7	5.2
FCF Yield %	17%	18%	16%	13%	13%
Dividend Yield %	5%	4%	6%	4%	4%

EV/EBITDA Analysis	2024	2025	2026e	2027e	2027e
Stock price (RMB)	16.90	19.73	31.00	31.00	31.00
Eq Shares outstanding (CN	2,229	2,229	2,229	2,229	2,229
Equity Value	37,670	43,978	69,099	69,099	69,099
Net Debt	10,155	5,934	(1,568)	(7,740)	(14,063)
Minority Interest	4,343	3,680	4,895	5,763	6,683
Enterprise Value	52,168	53,592	72,426	67,122	61,719
EBITDA (in Millions)	8,013	8,341	16,078	11,773	11,981

Sensitivity Analysis				
2025e				
(Rmb mn, EPS in Rmb)	EBITDA	EBIT	Net Income	EPS
Aluminum vol (up 1%)	4%	5%	5%	5%
Aluminum price (up 1%)	4%	5%	5%	5%
Coal price (up 1%)	2%	3%	3%	3%
Aluminum cost (up 1%)	1%	1%	1%	1%
Coal cost (up 1%)	2%	2%	2%	2%

Source: Company data, Morgan Stanley Research estimates

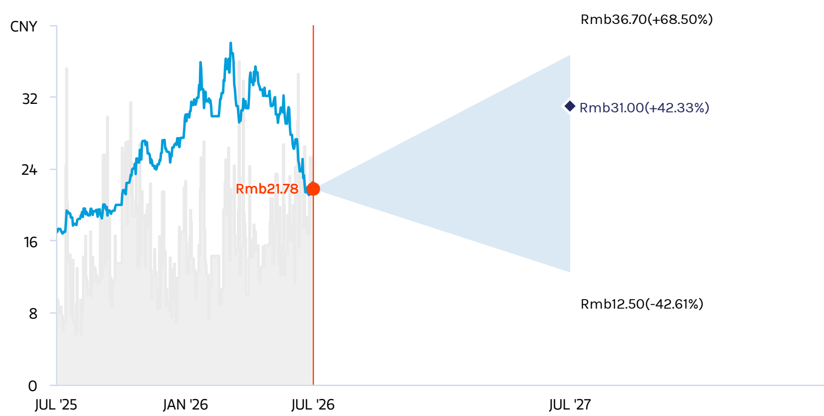
Risk Reward – Shenhuo Coal and Power (000933.SZ)

Integrated aluminum and coal producer with strong earnings outlook

PRICE TARGET Rmb31.00

Residual income model. We apply a cost of equity of 8.4%, calculated from a beta of 0.76, 2.3% risk-free rate and 8% equity risk premium. We assume a steady state revenue growth rate of 2% for Shenhuo and long-term ROE of 11.0%.

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- An integrated business model spanning coal mining, power generation and aluminum processing enhances operating efficiency and lowers production costs.
- The aluminum segment provides strong earnings, supported by higher prices and a government-imposed capacity cap of 45mnt that limits supply.
- Additional growth potential exists through the acquisition of aluminum quotas and the development of the Wucaiwan coal mine.
- We view valuation as attractive.

Consensus Rating Distribution

100%	Overweight
0%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb36.70

9.3x 2026e P/E

Strong demand with margin expansion: Aluminum price is higher than expected with slow volume growth overseas. Meanwhile, coal prices also rebound in response to constrained supply in the domestic market and less imports, while demand for aluminum and coal remains strong.

BASE CASE Rmb31.00

7.9x 2026e P/E

Aluminum prices increase 22% YoY in 2026 to Rmb23,573/t, while coal prices increase slightly to Rmb822/t.

BEAR CASE Rmb12.50

3.2x 2026e P/E

Weaker-than-expected aluminum demand and higher-than-expected aluminum supply overseas, pressure aluminum prices, weighing on earnings for Shenhuo.

Risk Reward – Shenhuo Coal and Power (000933.SZ)

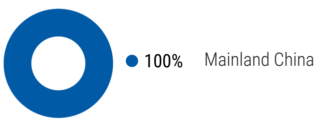
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue (Rmb mn) (Rmb, mn)	41,241	47,924	43,178	43,444

INVESTMENT DRIVERS

- Aluminum volume growth through the acquisition of production quotas in Xinjiang
- Cost savings through lower power tariffs

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected demand
- Further cost saving from power tariff
- Aluminum and coal capacity expansion

RISKS TO DOWNSIDE

- Slowdown in demand globally
- Rising raw material and energy prices
- Industry overcapacity

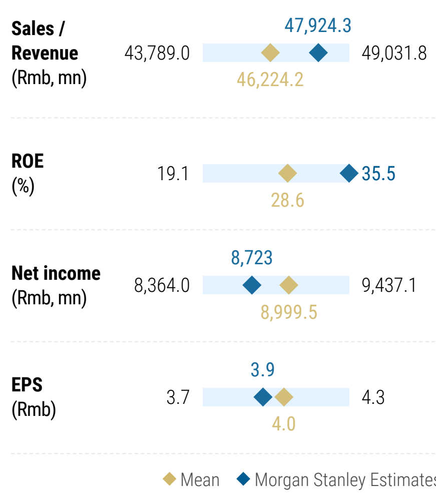
OWNERSHIP POSITIONING

Inst. Owners, % Active 89.2%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e



Source: Refinitiv, Morgan Stanley Research

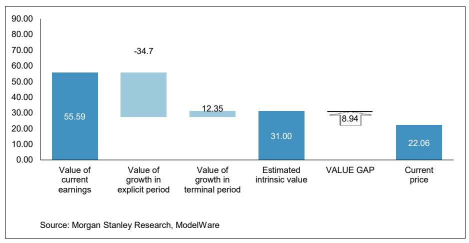
Valuation and earnings changes

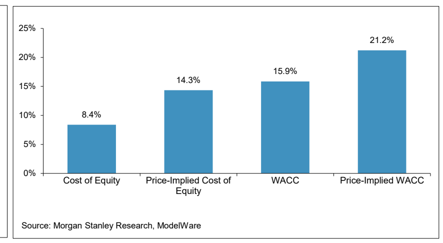
We update our model to reflect the latest aluminum price estimates from our global commodity team. The lower aluminum price estimates in 2027-2028 resulted in 32% and 33% decline in EPS compared to our previous estimates. However, besides the aluminum business, Shenhuo is a major coal producer in Henan province. Coal price in domestic market would see the constrained supply due to the tightened safety inspection nationwide triggered by the mine accidents in late May in Shanxi. We stay OW on the name.

Exhibit 75: Intrinsic Valuation

INTRINSIC VALUE

000933.SZ





Intrinsic Value Model

Inputs:	
Cost of equity	**8.4%**
Long-term ROE on new investments	**11.0%**
Years to reach steady-state growth	**10**
Steady-state revenue growth rate (%)	**2%**
Shares Outstanding	**2,229**
Steady-state borrowing cost (net of tax)	**3.0%**
Steady-state leverage (Net debt/Equity)	**50%**
Price target horizon (months)	**12**
Conv. factor - Model to traded Ccy	**1.15**
Decimals	**2**

Outputs:		
Price		**22.06**
IV Per Share (12 Month), Ex. Div	**Jul-27**	**31.00**
Expected share price return		40.53%
Expected dividend yield		7.95%
Expected total return		**48.48%**
WACC		15.9%
Long-term RNOA on new investments		8.3%
Explicit forecast period (years)		9
Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):	
Beginning NOA	**28,296**
Sum of PVRI	32,373
Beginning NNOAL	(3,729)
ROI equity value	**56,940**
Residual Income (equity):	
Beginning equity	24,567
Sum of PVRI	32,373
RI equity value	**56,940**

DCF (operating):	
Sum of PVFCFO	60,669
Beginning NNOAL	(3,729)
DCF Equity Value	**56,940**
DDM (Equity):	
Sum of div & net rep	56,940
DDM equity value	**56,940**

Source: Morgan Stanley Research estimates

Tianshan Aluminum

Exhibit 76: Financial Summary

Key Drivers	2024	2025	2026e	2027e	2028e
Sales Volume (kt)					
Primary Aluminum	1,073	1,046	1,344	1,400	1,400
High purity aluminum	26	26	30	30	30
Alumina	2,087	2,280	2,500	2,500	2,500
Aluminum foil	22	106	106	106	107
ASP (Rmb/t)					
Primary Aluminum	17,493	18,220	20,861	18,029	18,218
High purity aluminum	23,123	23,796	27,344	23,633	24,087
Alumina	3,505	2,994	2,353	2,340	2,352
Aluminum foil	23,223	21,616	23,777	23,064	22,603
GPM (%)					
Primary Aluminum	22%	30%	44%	33%	34%
High purity aluminum	28%	35%	40%	33%	33%
Alumina	29%	10%	-11%	-5%	-5%
Aluminum foil	-2%	20%	29%	27%	24%

Income Statement (Rmb mn)	2024	2025	2026e	2027e	2028e
Net Sales	28,089	29,502	37,967	34,951	35,248
Gross Profit	6,536	7,199	12,681	8,985	9,217
EBITDA	7,637	7,874	13,328	9,780	10,034
Consensus EBITDA			NA	NA	NA
Diff. to Consensus EBITDA			NA	NA	NA
EBIT	5,930	6,062	11,372	7,779	8,001
Pre-tax Profit	5,223	5,593	10,906	7,367	7,623
Net Profit	4,455	4,818	8,191	5,533	5,725
EPS (Rmb)	0.97	1.05	1.78	1.21	1.25
DPS (Rmb)	0.40	0.55	0.94	0.63	0.65
BVPS (Rmb)	5.77	6.39	7.23	7.80	8.39

Cash Flow (Rmb mn)	2024	2025	2026e	2027e	2028e
EBITDA	7,637	7,874	13,328	9,780	10,034
-Taxes Paid	-767	-781	-2,726	-1,842	-1,906
-Working Capital	45	-1,561	-878	-121	-21
-Others	-1,696	2,522	-101	540	105
Operating cash flow	5,220	8,053	9,623	8,358	8,212
-Capex.	-1,278	-1,534	-949	-699	-705
FCF	3,943	6,519	8,673	7,659	7,507

Balance Sheet (Rmb mn)	2024	2025	2026e	2027e	2028e
Cash & Equivalents	9,108	6,540	10,402	14,665	18,675
Receivables	899	2,240	2,883	2,654	2,677
Inventories	8,824	7,187	8,149	8,368	8,389
Total Assets	56,780	54,150	58,942	61,774	64,512
Payables	5,853	4,527	4,746	4,796	4,801
Borrowings	11,101	8,845	8,845	8,845	8,845
Total Liabilities	29,946	24,586	25,491	25,697	25,717
Shareholders Equity	26,831	29,567	33,466	36,100	38,825
Total Liabilities and Equity	56,780	54,150	58,942	61,774	64,512

Per Share Data (Rmb)	2024	2025	2026e	2027e	2028e
EPS for consensus	0.97	1.05	1.78	1.21	1.25
Consensus EPS			NA	NA	NA
Diff. to Consensus EPS (%)			NA	NA	NA
DPS	0.40	0.55	0.94	0.63	0.65
BVPS	5.77	6.39	7.23	7.80	8.39

Profitability Ratios %	2024	2025	2026e	2027e	2028e
ROE	17%	16%	24%	15%	15%
EBITDA Margin	27%	27%	35%	28%	28%
EBIT Margin	21%	21%	30%	22%	23%
Pre-tax Profit Margin	19%	19%	29%	21%	22%
Net Profit Margin	16%	16%	22%	16%	16%
Net Debt / Equity %	23%	22%	8%	-5%	-15%
Interest Cover (EBITDA) (x)	11x	17x	29x	24x	27x
Working Capital % of Sales	27%	20%	18%	20%	20%
Days Sales AR outstanding	12	28	28	28	28
Days in Inventory	149	118	118	118	118
Days Payable outstanding	150	111	111	111	111
Cash Conversion	11	35	35	35	35

Valuation	2024	2025	2026e	2027e	2028e
P/E	8.1	10.5	7.5	11.0	10.7
P/BV	1.4	1.7	1.8	1.7	1.6
EV/Sales	1.5	1.8	1.7	1.7	1.6
EV/EBITDA	5.6	6.7	4.8	6.1	5.6
FCF Yield %	11%	14%	14%	12%	12%
Dividend Yield %	5%	6%	7%	5%	5%

EV/EBITDA Analysis	2024	2025	2026e	2027e	2028e
Stock price (RMB)	7.87	9.97	13.30	13.30	13.30
Eq Shares outstanding	4,652	4,629	4,629	4,629	4,629
Equity Value	36,610	46,170	61,562	61,562	61,562
Net Debt	6,133	6,403	2,540	-1,722	-5,732
Minority Interest	3	-4	-15	-23	-31
Enterprise Value	42,746	52,569	64,087	59,817	55,799
EBITDA (in Millions)	7,637	7,874	13,328	9,780	10,034

Sensitivity analysis: Impact of a 1% change in the following				
2026e Rmb				
(Rmb mn, EPS in Rmb)	Revenue	EBIT	NI	EPS
Volume (up 1%)	1%	1%	1%	1%
Price (up 1%)	1%	2%	3%	3%
Cost (up 1%)	0%	-1%	-1%	-1%

Source: Company data, Morgan Stanley Research estimates

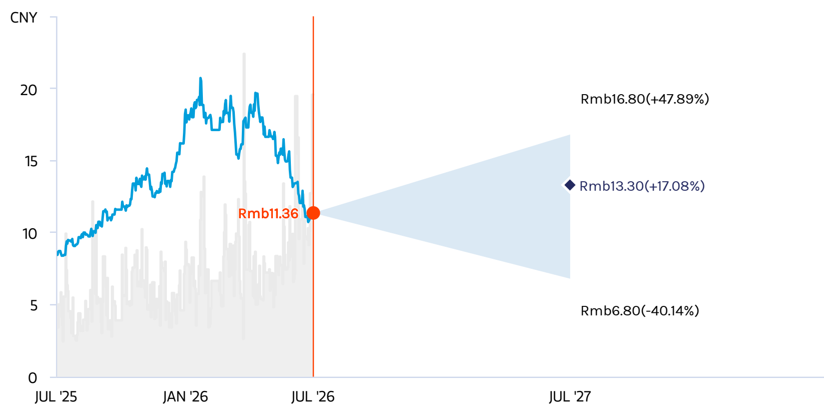
Risk Reward – Tianshan Aluminum (002532.SZ)

Cost competitiveness riding on high self-sufficiency and upstream expansion

PRICE TARGET Rmb13.30

Base case, derived from residual income valuation model. We discount our earnings forecasts through 2037 and then normalize earnings. Key assumptions include a cost of equity of 9.1% (risk-free rate of 2.3%, risk premium of 7%, beta of 0.97), a long-term ROE of 10% and a steady-state growth rate of 4%.

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We expect slight aluminum volume growth in the domestic market this year to lead to solid earnings contribution.
- Higher raw material self-sufficiency rate secures cost competitiveness vs. industry peers.
- Upstream expansion overseas ensures a stable raw material supply.
- Aluminum prices should see downward pressure from increasing supply overseas and restarts in the Middle East. However, the volume increase and cost efficiency offer some buffer.

Consensus Rating Distribution

	100%	Overweight
●	0%	Equal-weight
	0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb16.80

9.4x 2026e P/E

Strong demand and severe supply disruption: For 2026, we assume aluminum prices increase by 32% to Rmb27,134/t, while alumina prices decline by 10% to Rmb3,061/t.

BASE CASE Rmb13.30

7.5x 2026e P/E

Demand shift and continued supply disruption on power constraints: For 2026, we estimate aluminum prices increase by 14% YoY to Rmb23,573/t, while alumina prices fall by 22% to Rmb2,659/t.

BEAR CASE Rmb6.80

3.8x 2026e P/E

Sluggish demand and sufficient supply: For 2026, we assume aluminum prices fall by 7% to Rmb19,121/t, while alumina prices decrease by 34% to Rmb2,115/t.

Risk Reward – Tianshan Aluminum (002532.SZ)

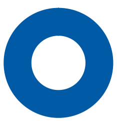
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
EBITDA margin %	0.3	0.4	0.3	0.3

INVESTMENT DRIVERS

- Volume release in 2026 and 2027
- Aluminum prices
- Costs, mainly power-related costs

GLOBAL REVENUE EXPOSURE



● 100% Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 **MOST**	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected demand
- Decreasing raw material and energy prices
- More plants in maintenance than expected

RISKS TO DOWNSIDE

- Slowdown in global aluminum demand
- Rising raw material and energy prices
- Industry overcapacity

OWNERSHIP POSITIONING

Inst. Owners, % Active 88.4%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

EBIT (Rmb, mn)
◆ 11,042
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (Rmb, mn)
◆ 12,998
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (Rmb)
1.8 ◆ 2.3
2.0

DPS (Rmb)
0.9 1.2
1.0

ROE (%)
27.7
25.5 31.4
28.1

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation and earnings changes

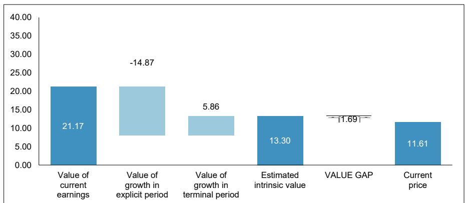
We derive our PT from residual income valuation model. We discount our earnings forecasts through 2037 and then normalize earnings. Key assumptions include a cost of equity of 9.1% (risk-free rate of 2.3%, risk premium of 7%, beta of 0.97), a long-term ROE of 10% and a steady-state growth rate of 4%.

Reflecting the weaker aluminum price in 2027 and 2028, our earnings estimates in 2027-28e down 47% and 46%, respectively, compare to our previous estimates. This also reflects the higher tax rate in Xinjiang after the review from tax authority. The lower aluminum price would pressure on earnings. However, we think the integrated supply chain and the production cost efficiency would limits the downside. We move the stock to EW.

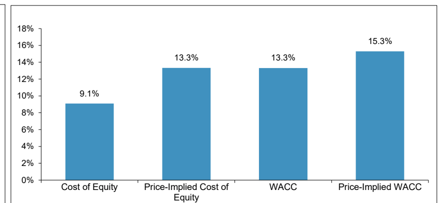
Exhibit 77: Intrinsic Valuation

INTRINSIC VALUE

002532.SZ



Source: Morgan Stanley Research, ModelWare



Source: Morgan Stanley Research, ModelWare

Intrinsic Value Model

Inputs:	
Cost of equity	**9.1%**
Long-term ROE on new investments	**10.0%**
Years to reach steady-state growth	**10**
Steady-state revenue growth rate (%)	**4%**
Shares Outstanding	**4,629**
Steady-state borrowing cost (net of tax)	**3.0%**
Steady-state leverage (Net debt/Equity)	**50%**
Price target horizon (months)	**12**
Conv. factor - Model to traded Ccy	**1.00**
Decimals	**2**

Outputs:		
Price		**11.61**
IV Per Share (12 Month), Ex. Div	**Jul-27**	**13.30**
Expected share price return		14.56%
Expected dividend yield		0.00%
Expected total return		**14.56%**
WACC		13.3%
Long-term RNOA on new investments		7.7%
Explicit forecast period (years)		10
Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):	
Beginning NOA	**31,869**
Sum of PVRI	24,243
Beginning NNOAL	(2,301)
ROI equity value	**53,811**

DCF (operating):	
Sum of PVFCFO	56,112
Beginning NNOAL	(2,301)
DCF Equity Value	**53,811**

Residual Income (equity):	
Beginning equity	29,567
Sum of PVRI	24,243
RI equity value	**53,811**

DDM (Equity):	
Sum of div & net rep	53,811
DDM equity value	**53,811**

Source: Morgan Stanley Research estimates

Nanshan Aluminum

Exhibit 78: Financial Summary

Key assumptions (Rmb/t)	2024	2025	2026e	2027e	2028e
Aluminum price	17,625	18,344	20,969	18,029	18,218
Flat rolled process fee	4,514	4,376	3,291	3,291	3,291
Aluminum profile process fee	3,300	2,805	2,805	2,805	2,805
Income Statement (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Under PRC GAAP					
Net Sales	33,477	34,620	36,915	35,681	36,121
Gross Profit	8,814	8,476	9,295	7,749	7,969
EBITDA	8,643	8,528	9,326	7,999	8,286
Consensus EBITDA			9,712 e	10,704 e	11,453 e
Diff. to Consensus EBITDA (%)			-4.0 %	NA	NA
EBIT	6,310	6,328	6,987	5,527	5,717
Pre-tax Profit	6,654	6,538	7,245	5,794	6,019
Net Profit	4,830	4,736	5,247	4,196	4,359
EPS (Rmb)	0.40	0.40	0.44	0.35	0.36
DPS (Rmb)	0.17	0.37	0.42	0.33	0.34
BVPS (Rmb)	4.21	4.15	4.17	4.19	4.21
Cash Flow (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
EBITDA	8,643	8,528	9,326	7,999	8,286
-Taxes Paid	(697)	(733)	(813)	(650)	(675)
-Working Capital	(981)	(609)	(494)	208	(92)
-Others	(981)	(609)	(494)	208	(92)
Operating cash flow	**7,617**	**7,429**	**8,105**	**7,644**	**7,606**
-Capex.	(2,795)	(3,883)	(2,953)	(2,141)	(2,167)
FCF	4,822	3,546	5,152	5,503	5,438
Balance Sheet (Rmb mn)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Cash & Equivalents	21,422	21,199	21,650	23,453	25,073
Receivables	4,104	3,543	3,777	3,651	3,696
Inventories	6,470	8,131	8,589	8,690	8,758
Total Assets	**70,264**	**71,917**	**73,662**	**74,899**	**76,163**
Payables	3,584	4,512	4,767	4,822	4,860
Borrowings	5,714	5,029	5,029	5,029	5,029
Total Liabilities	**14,040**	**13,864**	**14,136**	**14,196**	**14,237**
Shareholders Equity	50,317	49,587	49,875	50,105	50,344
Minority Interest	5,907	8,465	9,650	10,598	11,583
Total Liabilities and Equity	**70,264**	**71,917**	**73,662**	**74,899**	**76,163**
Per Share Data (Rmb)	**2024**	**2025**	**2026e**	**2027e**	**2028e**
ModelWare EPS	0.40	0.40	0.44	0.35	0.36
Consensus EPS			0.41	0.41	0.41
Diff. to Consensus EPS (%)			7.6	(13.9)	(10.6)
DPS	0.17	0.37	0.42	0.33	0.34
BVPS	4.21	4.15	4.17	4.19	4.21

Profitability Ratios %	2024	2025	2026e	2027e	2028e
ROE	10.0%	9.1%	12.3% e	11.4% e	9.4% e
EBITDA Margin	25.8%	24.6%	25.3%	22.4%	22.9%
EBIT Margin	18.8%	18.3%	18.9%	15.5%	15.8%
Pre-tax Profit Margin	19.9%	18.9%	19.6%	16.2%	16.7%
Net Profit Margin	14.4%	13.7%	14.2%	11.8%	12.1%
Net Debt / Equity %	-35.5%	-33.3%	-33.2%	-35.5%	-37.5%
Z score (<1.8, higher risk)	3.7	3.8	3.7	3.6	3.6
Interest Cover (EBITDA) (x)	48.09	49.36	61.33	52.61	54.49
Working Capital % of Sales	(2.9)	(1.8)	(1.3)	0.6	(0.3)
Days Sales AR outstanding	39	40	36	37	37
Days in Inventory	94	101	110	112	113
Days Payable outstanding	(42)	(49)	(55)	(56)	(56)
Cash Conversion	91	92	91	94	93
Valuation	**2024**	**2025**	**2026e**	**2027e**	**2028e**
P/E	9.7	8.6	5.9 e	5.7 e	12.3 e
P/BV	0.9	0.7	0.7 e	0.6 e	1.1 e
EV/Sales	0.9	0.5	0.4 e	0.2 e	1.0 e
EV/EBITDA	3.4	2.2	1.2 e	0.7 e	3.6 e
FCF Yield %	10.7%	12.3%	16.0% e	18.4% e	9.1% e
Dividend Yield %	6.0%	1.2%	1.7% e	1.8% e	0.8% e
EV/EBITDA Analysis	**2024**	**2025**	**2026e**	**2027e**	**2028e**
Average Stock price (CNY)	5.00	5.00	5.00	5.00	5.00
Eq Shares outstanding (in Mn)	11,950	11,950	11,950	11,950	11,950
Equity Value	67,520	67,520	67,520	67,520	67,520
Net Debt	(19,946)	(19,321)	(19,772)	(21,576)	(23,196)
Minority Interest	5,907	8,465	9,650	10,598	11,583
Enterprise Value	53,482	56,664	57,398	56,543	55,907
EBITDA (in Millions)	8,643	8,528	9,326	7,999	8,286
Quarterly Financials	**2Q24**	**3Q24**	**4Q24**	**2024**	**1Q25**
Revenue	8,443	8,559	9,251	33,477	8,981
Gross Profit	2,283	2,098	2,884	8,814	2,858
EBITDA	2,272	2,425	1,989	8,347	3,138
EBIT	1,586	1,613	1,840	6,014	2,285
Net Income	1,338	1,301	1,340	4,830	1,704
EPS	0.11	0.11	0.11	0.40	0.14

Source: Company data, Morgan Stanley Research

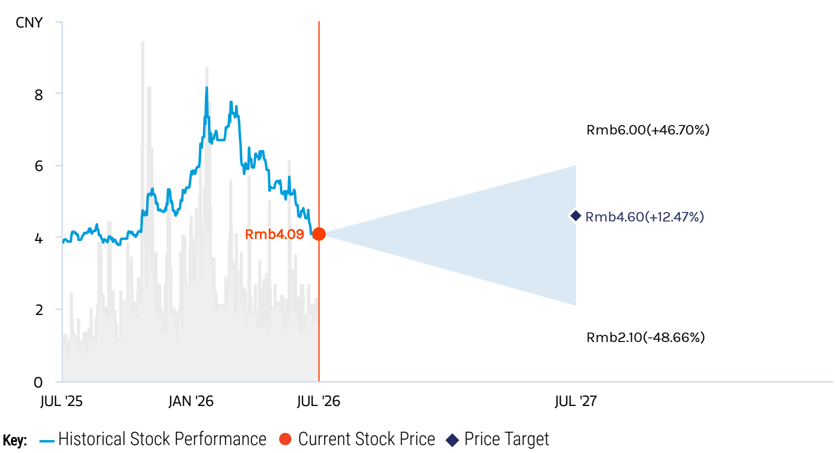
Risk Reward – Shandong Nanshan Aluminium Co. (600219.SS)

Lower aluminum price expected on increasing supply in overseas

PRICE TARGET Rmb4.60

Base case, residual income valuation model. We discount our earnings forecast through 2030, and then normalize earnings. We assume a cost of equity of 10.2% (risk-free rate of 2.3%, risk premium of 7%, beta of 1.126), a long-term ROE of 11% and a steady-state growth rate of 2%.

RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Aluminum price is expected to correct as the market moves into a surplus in 2027 and 2028 along with the volume increase from new scheduled capacities in overseas and capacity restarts in Middle East, dragging on earnings.
- Lower aluminum capacity expected as 200kt capacity quota will eventually be transferred and suspended production at Nanshan side, implying lower earnings.
- High end aluminum processing products and the aluminum capacity expansion in Indonesia would offer some buffer.

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb6.00

1.4x 2026e BVPS

Bull-case aluminum price assumption and higher product processing fees: Demand recovery is stronger than expected. Together with slower-than-expected aluminum production release, we expect aluminum prices to reach US$1.70/lb in 2026. Overseas aluminum production cuts could increase processing fees and Nanshan's overall profit.

BASE CASE Rmb4.60

1.1x 2026e BVPS

Base-case aluminum price assumption and stable processing mark-up fees: We expect aluminum prices in China to remain high thanks to overall capacity restriction. Nanshan Aluminum, as an aluminum products producer, should benefit from higher processing fees in response to potential overseas aluminum production cuts. Furthermore, an increased alumina self-sufficiency rate should further lower the company's production costs.

BEAR CASE Rmb2.10

0.5x 2026e BVPS

Bear-case aluminum price assumption and lower product processing fees: Aluminum supply is higher than expected, bringing down aluminum prices and processing fees for Nanshan. In addition, alumina price rises in response to production cuts, which we expect to increase Nanshan's overall production costs.

Risk Reward – Shandong Nanshan Aluminium Co. (600219.SS)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Alunimum selling price (ex. VAT) (Rmb/t)	18,344.5	20,969.1	18,029.1	18,218.4
ALUMINA Prices (Rmb/t) (Ex. VAT)	2,856.9	2,365.6	2,453.5	2,502.5
Qinghuangdao coal price (Ex. VAT) (Rmb/t)	621.2	654.9	917.4	884.8

INVESTMENT DRIVERS

- Aluminum supply-side reform
- Ramp-up of its auto sheet and aviation sheet capacity
- Share placement plans

GLOBAL REVENUE EXPOSURE

0-10%	North America
10-20%	APAC, ex Japan, Mainland China and India
10-20%	Europe ex UK
50-60%	Mainland China

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

3/5 MOST 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Greater-than-expected capacity cuts
- Stronger-than-expected volume growth of aviation and auto sheets, driving earnings upside

RISKS TO DOWNSIDE

- Rising capacity, leading to increased competitive intensity and margin erosion
- Slowdown in demand and infrastructure spending

OWNERSHIP POSITIONING

Inst. Owners, % Active 93.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (Rmb, mn) ◆ 36,960.0
Note: There are not sufficient brokers supplying consensus data for this metric

EBITDA (Rmb, mn) ◆ 9,099
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (Rmb, mn) ◆ 5,247
Note: There are not sufficient brokers supplying consensus data for this metric

EPS (Rmb) ◆ 0.4
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation and earnings changes

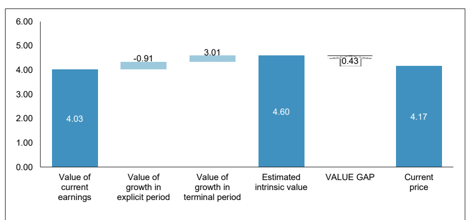
We derive our PT using residual income valuation mode. We discount our earnings forecast through 2032 and then normalize earnings. We assume a cost of equity of 10.2% (risk-free rate of 2.3%, risk premium of 7%, beta of 1.126), a long-term ROE of 11% and a steady-state growth rate of 2%.

Aluminum price is expected to correct as the market moves into a surplus in 2027 and 2028 along with the volume increase from new scheduled capacities in overseas and capacity restarts in Middle East, dragging on earnings. Meanwhile, the company transferred 200kt aluminum production quota to a smelter in Guizhou. These capacities are still running in Nanshan. However, it will eventually be transferred out and lead to production suspension and lower aluminum volume contribution at Shandong Nanshan, implying a lower earnings. However, we think the high end stable earnings aluminum processing products and the aluminum capacity expansion in Indonesia would offer some buffer. We downgrade the stock to EW.

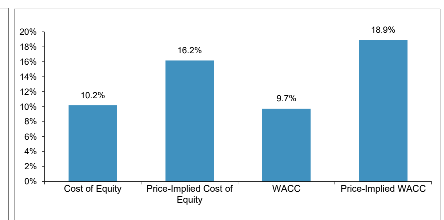
Exhibit 79: Intrinsic valuation

INTRINSIC VALUE

600219.SS



Source: Morgan Stanley Research, ModelWare



Source: Morgan Stanley Research, ModelWare

Intrinsic Value Model

Inputs:	
Cost of equity	10.2%
Long-term ROE on new investments	11.0%
Years to reach steady-state growth	10
Steady-state revenue growth rate (%)	2%
Shares Outstanding	11,950
Steady-state borrowing cost (net of tax)	3.0%
Steady-state leverage (Net debt/Equity)	50%
Price target horizon (months)	12
Conv. factor - Model to traded Ccy	1.00
Decimals	2

Outputs:		
Price		**4.17**
IV Per Share (12 Month), Ex. Div	**Jul-27**	**4.60**
Expected share price return		10.31%
Expected dividend yield		10.78%
Expected total return		**21.09%**
WACC		9.7%
Long-term RNOA on new investments		8.3%
Explicit forecast period (years)		8
Fiscal Year Ending		0

Value Breakdown

Residual Income (operating):	
Beginning NOA	**33,231**
Sum of PVRI	13,308
Beginning NNOAL	17,085
ROI equity value	**63,625**

DCF (operating):	
Sum of PVFCFO	46,539
Beginning NNOAL	17,085
DCF Equity Value	**63,625**

Residual Income (equity):	
Beginning equity	50,317
Sum of PVRI	13,308
RI equity value	**63,625**

DDM (Equity):	
Sum of div & net rep	63,625
DDM equity value	**63,625**

Source: Morgan Stanley Research

Hindalco

What's changed?

Given changes in commodity prices by our global Commodities team, we revise Hindalco earnings estimates for the India business, while Novelis estimates remain largely unchanged.

LME aluminum prices estimates are cut by ~7% for F27, ~12% for F28, and ~10% for F29. Consequently, we cut our F28-29 EBITDA estimates by ~10% and 5%, respectively; and EPS estimates by ~12% and ~7%, respectively.

see our latest commodity price deck

Exhibit 80: Snapshot of Commodity price changes

		NEW			OLD			% Change		
Commodity Price Estimates	**F2026**	**F2027E**	**F2028E**	**F2029E**	**F2027E**	**F2028E**	**F2029E**	**F2027E**	**F2028E**	**F2029E**
LME Aluminium Average Price (US$/t)	2773	3191	2825	2850	3437	3200	3162	-7%	-12%	-10%
LME Premium (Japanese) (US$/t)	162	395	200	144	288	150	144	37%	33%	0%
LME Copper Average Price (US$/t)	10819	13591	12699	12374	12337	11874	11499	10%	7%	8%

Source: Bloomberg, Morgan Stanley Research estimates

Exhibit 81: Hindalco: Snapshot of earnings changes

Old vs New		NEW			OLD			% Change		
Rs Bn	**F2026**	**F2027E**	**F2028E**	**F2029E**	**F2027E**	**F2028E**	**F2029E**	**F2027E**	**F2028E**	**F2029E**
Standalone										
Revenues	1126	1457	1576	1769	1407	1559	1717	3.6%	1.1%	3.0%
EBITDA	163	221	202	250	212	239	273	4.6%	-15.5%	-8.5%
PAT	101	148	130	162	141	158	179	5.2%	-17.6%	-9.7%
Consol										
Revenues	2749	3516	3523	3660	3593	3696	3746	-2.2%	-4.7%	-2.3%
EBITDA	349	425	429	514	423	474	541	0.4%	-9.5%	-5.1%
PAT	204	258	256	282	256	292	304	0.5%	-12.4%	-7.3%
EPS (Rs/share)	91.68	116.16	115.35	126.91	115.54	131.70	136.95	0.5%	-12.4%	-7.3%

Source: Company data, Morgan Stanley Research estimates

Valuation methodology

We use a sum-of-the-parts valuation methodology to arrive at our price target: We value Hindalco's core businesses (India business and Novelis) by applying a 1-year forward EV/EBITDA multiple. We also attribute value to Hindalco's investments in group companies based on current share prices.

We value India business at a 7x Jun-28e EV/EBITDA multiple (vs. 7.5x June-28e EV/EBITDA multiple previously). We value Novelis at 8x Jun-28e EV/EBITDA multiple (unchanged). We also assume a 20% holding company discount.

Investment in Group companies

We value Hindalco's investments in group companies based on current share prices. We have applied a discount of 20% to arrive at the value of investments attributable to our price target. The discount reflects our view that there is limited visibility on the company realizing the value by selling the investments.

Exhibit 82: Hindalco: Core business base case valuation discussion

Rs Bn	India Business	Novelis	Total
EBITDA (12M ending Jun-2028e)	224	226	450
EV/EBITDA Multiple (x)	7.0x	8.0x	
EV (June-2027e)	1581	1806	3387
Less: Net Debt, adjusted for CWIP (June-2027e)	218	491	709
Equity Value (June-2027e)	**1363**	**1315**	**2678**
Less: 20% for HoldCo Discount	0	263	263
Equity Value (June-2027e), Adjusted	**1363**	**1052**	**2415**
No. of Shares (Bn)	*2.2*	*2.2*	*2.2*
Equity Value Per Share (Rs)	**615**	**475**	**1090**

Source: Morgan Stanley Research

Exhibit 83: Hindalco: Value of investments in group companies

Companies	Total Value (Rs Bn)	Value Per Share of Hindalco (Rs)
Grasim	94	42
Ultratech	15	7
ABCL	16	7
Vodafone Idea	10	5
ABLBL	5	2
ABFRL	3	1
Total	**143**	**63**
Less: 20% Holding Company Discount		13
Total Attributable Value (Rounded-off)		**50**

Source: Company data, Morgan Stanley Research. Note: Prices as of May 25, 2026.

Exhibit 84: Hindalco: Consol core business scenario values

Consol (Rs Bn)	Bull	Base	Bear
Probability Weight	0%	100%	0%
LME Aluminium Price (US$/t), Avg. F27-28E	3610	3008	2352
1Y Forward P/B Multiple (x)	1.5	1.3	1.0
Book Value (June-2028e)	2310	1925	1444
Equity Value (June-2027e)	**3465**	**2415**	**1444**
No. of Shares (Bn)	2.2	2.2	2.2
Equity Value Per Share (Rs)	**1560**	**1090**	**650**

Source: Morgan Stanley Research estimates

Exhibit 85: Hindalco: Scenario values and price target snapshot

	NEW				OLD			
SOTP Valuation	**Bull**	**Base**	**Bear**	**PT**	**Bull**	**Base**	**Bear**	**PT**
Hindalco Consol Core Business	1560	1090	650	1090	1810	1275	730	1275
Investments	50	50	50	50	50	50	50	50
Total Value Per Share	**1610**	**1140**	**700**	**1140**	**1860**	**1325**	**780**	**1325**

Source: Morgan Stanley Research estimates

Exhibit 86: Hindalco: Financial Summary

Income Statement (Rs mn; YE Mar)	F2026	F2027E	F2028E	F2029E
Net Sales	**2,749,440**	**3,515,667**	**3,523,148**	**3,659,514**
Chg in stock Inc/(Dec)	(140,820)	(155,054)	(156,693)	(181,639)
Raw Materials	1,911,200	2,532,592	2,487,355	2,472,276
Power and Fuel	136,220	140,931	149,009	172,783
Salaries	171,480	195,990	213,007	237,926
Other Expenses	322,520	375,979	401,651	444,410
Total op exp	2,400,600	3,090,439	3,094,328	3,145,757
EBITDA	**348,840**	**425,228**	**428,821**	**513,757**
Other Income	28,890	58,938	59,337	12,000
Interest	34,800	38,394	33,795	29,324
Depreciation	88,300	123,427	134,266	144,252
PBT (Before extraordinary)	**254,630**	**322,345**	**320,097**	**352,180**
Tax	51,050	64,469	64,019	70,436
Minority Interest & Profits/loss in equity accounted investments	40	-	-	-
PAT (Before extraordinary)	**203,540**	**257,876**	**256,078**	**281,744**
Extraordinary items	69,630	-	-	-
Reported PAT	**133,910**	**257,876**	**256,078**	**281,744**

Balance Sheet (Rs mn; YE Mar)	F2026	F2027E	F2028E	F2029E
Sources of Funds				
Equity Capital	2,220	2,220	2,220	2,220
Reserves and Surplus	1,363,610	1,610,386	1,855,364	2,126,008
Net Worth	1,365,830	1,612,606	1,857,584	2,128,228
Debt	991,610	1,034,950	849,195	666,389
Deferred Tax Liability	31,800	44,810	44,810	44,810
Minority	120	127	128	131
Other LT Liabilities	54,430	53,723	53,962	54,535
Total	**2,443,790**	**2,746,215**	**2,805,679**	**2,894,092**
Application of funds				
Gross Block	2,270,009	2,638,245	2,817,695	3,062,223
Less : Dep & Amort.	967,499	1,128,318	1,269,729	1,433,115
Capital WIP	495,260	572,054	625,905	635,092
Net Fixed Assets	1,797,770	2,081,981	2,173,871	2,264,200
Invt/LT assets	317,580	325,915	330,115	333,208
Other LT Assets	68,000	80,000	85,000	90,000
Current assets	1,278,540	1,389,296	1,330,657	1,305,179
Inventories	755,170	814,134	792,946	753,869
Sundry Debtors	272,220	340,628	331,197	330,827
Other current assets	76,140	79,824	81,673	84,203
Cash & Bank	148,080	121,353	88,291	94,584
Loans & Advances	26,930	33,358	36,550	41,696
Current Liabilities	950,100	1,050,977	1,028,964	1,008,494
Sundry Creditors	616,240	689,802	660,710	630,280
Other Liabilities	305,050	334,688	341,687	351,456
Provisions	28,810	26,488	26,567	26,758
Net Current Assets	328,440	338,319	301,693	296,685
Total	**2,443,790**	**2,746,215**	**2,805,679**	**2,894,092**

Cash Flow (Rs mn; YE Mar)	F2026	F2027E	F2028E	F2029E
PAT	133,910	257,876	256,078	281,744
Depreciation	88,300	123,427	134,266	144,252
Chg in Debtors	(73,880)	(68,408)	9,430	370
Chg in Inv	(267,160)	(58,964)	21,188	39,077
Chg in Other C. Assets	(24,420)	(3,684)	(1,849)	(2,529)
Chg in Loans/Advances	(8,120)	(6,428)	(3,192)	(5,146)
Chg in Creditors	192,790	73,562	(29,092)	(30,430)
Chg in Other C. Liab.	154,150	29,638	6,999	9,769
Chg in Provisions	6,910	(2,323)	80	191
Net Operating CF	**202,480**	**344,696**	**393,908**	**437,298**
Capex	(446,540)	(407,638)	(226,155)	(234,581)
Other Investing CFs	(28,380)	(8,335)	(4,200)	(3,092)
CF from Investing	**(474,920)**	**(415,973)**	**(230,356)**	**(237,674)**
Chg in Equity	5,930	-	-	-
Dividends Paid	(11,100)	(11,100)	(11,100)	(11,100)
Chg in Borrowings	352,320	43,340	(185,755)	(182,806)
Chg in Other LT Liabilities	(35,090)	12,310	241	575
CF from Financing	**312,060**	**44,549**	**(196,614)**	**(193,331)**
Change in Cash	**39,620**	**(26,727)**	**(33,062)**	**6,293**
Closing Cash Balance	**148,080**	**121,353**	**88,291**	**94,584**

RATIOS	F2026	F2027E	F2028E	F2029E
EPS (Adjusted)	91.68	116.16	115.35	126.91
EPS (Reported)	60.32	116.16	115.35	126.91
Book Value Per Share	615.2	726.4	836.7	958.7
Valuation				
P/E	10.1	8.4	8.5	7.7
EV/EBITDA	8.3	7.2	6.8	5.3
Price to Book Value	1.5	1.3	1.2	1.0
Dividend Yield (%)	1%	1%	1%	1%
Profitability Ratios (%)				
EBITDA Margin	13%	12%	12%	14%
Net Profit Margin	7%	7%	7%	8%
Average RoE	16%	17%	15%	14%
Average RoCE	13%	14%	13%	13%
Growth (%)				
Net Sales	15%	28%	0%	4%
EBITDA	10%	22%	1%	20%
Net Profit	21%	27%	-1%	10%
EPS	21%	27%	-1%	10%
Leverage Ratio				
Net debt/Equity (x)	0.6	0.6	0.4	0.3
Net debt/EBITDA (x)	2.4	2.1	1.8	1.1

Source: Company data, Morgan Stanley Research estimates

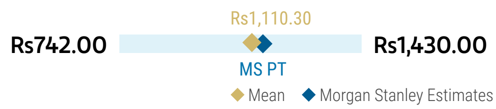
Risk Reward – Hindalco Industries Ltd (HALC.NS)

Well positioned for the medium term; OW

PRICE TARGET Rs1,140.00

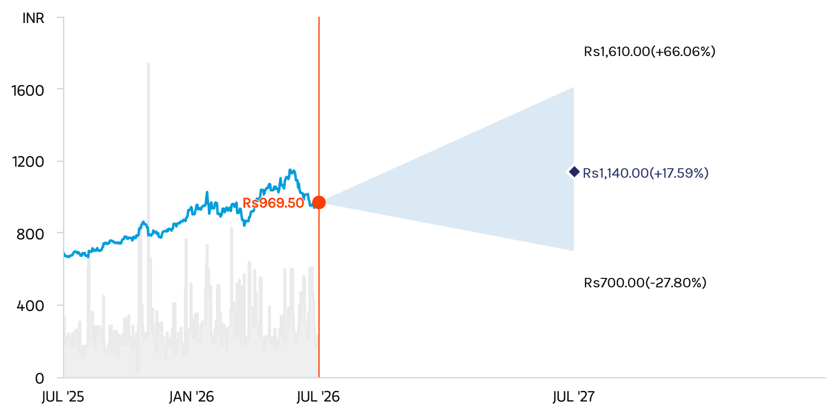
Base case, SOTP
India business: 7x Jun-28e EV/EBITDA.
Novelis: 8x Jun-28e EV/EBITDA. We assume a 20% holdco discount.
Investments in group cos based on current share prices

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



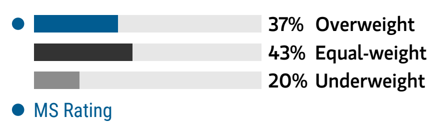
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We believe global aluminium balance should move from deficit to oversupply 2028 onwards, impacting medium-term prices, we believe.
- However, backward integration, a shift toward captive energy sourcing, and capacity expansion should support medium-term profitability in domestic business.
- Novelis' profitability should improve materially from 2HF27 onward, driven by normalization of the Oswego fire impact, the Bay Minette ramp-up, and the realization of cost-saving initiatives.
- Hindalco's valuation multiples are closely correlated with LME aluminium prices, and we expect current valuation multiples to sustain supported by a strong earnings growth outlook.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Electric Vehicles:	*Positive*
Renewable Energy:	*Positive*
Self-help:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rs1,610.00

~10x Jun-28e consol base EV/EBITDA

Higher aluminium prices, sharp pickup in volumes, faster execution on capex plans: 1) LME aluminium prices at ~20% higher than our base case in F27 and F28, on average; and 2) power costs see sharp moderation, helped by a sharp decline in spot prices and faster ramp-up of captive coal blocks.

BASE CASE Rs1,140.00

~7.5x Jun-28e consol base EV/EBITDA

Aluminium prices should moderate, but medium-term earnings should expand: 1) LME aluminium prices of ~US$3191/ton in F27e, US$2825/ton in F28e, and US$2850/ton in F29e; 2) power costs moderates from F28 onward with coal mines commissioning; and 3) improved volumes from normalization of operations and capacity ramp-up at Novelis, and expansion in high-margin products in India should help drive consol EBITDA CAGR of ~14% over F26-29e.

BEAR CASE Rs700.00

~5.5x Jun-28e consol base EV/EBITDA

Lower aluminium prices, muted volumes given stalled China recovery, weak beverage can demand for Novelis: 1) LME aluminum prices at ~25% lower than our base case over F27-28e; and 2) sharp rise in global energy prices keep volumes and margins under pressure for longer.

Risk Reward – Hindalco Industries Ltd (HALC.NS)

KEY EARNINGS INPUTS

Drivers	2026	2027e	2028e	2029e
Standalone Aluminium Volumes (kt)	1,367	1,483	1,555	1,740
Standalone Aluminium Realization (US$/t) (Rs)	3,627	4,019	3,623	3,636
Standalone Aluminium EBITDA (US$/t) (Rs)	1,182	1,440	1,212	1,316
Novelis Volumes (kt)	3,557	3,800	4,100	4,400
Novelis EBITDA (US$/t) (Rs)	578	525	560	590

INVESTMENT DRIVERS

- Aluminium prices
- Global energy prices
- FRP demand and hence volumes for Novelis, apart from conversion margins.

GLOBAL REVENUE EXPOSURE

0-10%	APAC, ex Japan, Mainland China and India
0-10%	Latin America
0-10%	UK
10-20%	Europe ex UK
20-30%	North America
30-40%	India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Sharp global macro recovery driving demand amid limited supply, supporting higher aluminium prices
- Faster recovery in Novelis' volumes/margins, with normalization of Oswego operations and ramp-up at Bay Minette expansion

RISKS TO DOWNSIDE

- Weak global demand and high supply from Indonesia may lead to an unfavorable demand-supply balance, keeping aluminium prices weak
- Substantial delay in Bay Minette commissioning

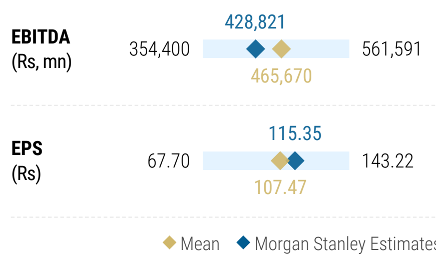
OWNERSHIP POSITIONING

Inst. Owners, % Active 65%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Mar 2028e



Source: Refinitiv, Morgan Stanley Research

Alcoa

Moving to Equal-weight on Alcoa (from Overweight) as the aluminum S/D picture is rapidly becoming less compelling and given the company's high degree of operating leverage to aluminum prices, we expect sizeable downward revisions to consensus estimates in the coming months, which in our view will weigh on shares. As a result of our lower aluminum forecasts, our new 2027-28 estimates are 33-49% below sell-side consensus for EBITDA, and 42-64% below for EPS. That said, AA's stock price has already pulled back significantly in the last month (-36%) suggesting to us the buy-side may already be closer to our updated estimates, leaving the risk-reward more balanced and driving our EW rating. For more details see: Supply Drives the Narrative: Downgrading AA & Vale to EW; Upgrading IVN to OW.

Valuation methodology: Our mid-27 PT (YE26 previously) is derived using an EV/EBITDA multiple approach at a target multiple of 7.0x (vs. 5.3x), which is 0.25x standard devs above the 5yr average EV/EBITDA multiple on our 2H27-1H28e EBITDA. We raise our target multiple as we expect earnings will normalize at a higher level longer-term than our new anchor EBITDA as the company completes its mine move in WA and sees a step change in alumina segment profitability.

Exhibit 87: Financial Summary

Operating Statistics	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Bauxite production, bdmt	47	48	48	42	41	38	38	38	40	42
Alumina production, kt	13,302	13,475	13,259	12,544	10,908	10,034	9,549	9,676	9,788	10,375
Aluminum production, kt	2,135	2,263	2,193	2,010	2,114	2,215	2,319	2,478	2,499	2,506
Third-party aluminum shipments, kt	2,859	3,016	3,007	2,570	2,491	2,590	2,522	2,679	2,719	2,726
Bauxite Prices, CIF China, $/t	44	44	43	54	59	70	67	64	60	60
Realized 3rd party Bauxite Price, $/t	48	42	41	58	64	60	74	61	60	60
Alumina Price, $/t	330	271	332	376	341	493	362	314	325	350
Realized 3rd party Alumina Price $/t	343	272	326	384	360	475	420	327	328	349
LME Aluminum Price, $/lb	0.81	0.77	1.13	1.24	1.02	1.07	1.20	1.48	1.29	1.27
Realized Aluminum Price, $/lb	0.97	0.87	1.31	1.57	1.28	1.29	1.53	2.06	1.82	1.56
Aluminum costs, $/lb	1.00	0.83	1.04	1.30	1.20	1.18	1.35	1.44	1.37	1.27
Alumina cash costs, $/t	237	208	254	341	347	381	362	356	335	336

Segment EBITDA	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Alumina	1,601	992	1,174	788	273	1,408	882	(223)	20	284
Aluminum	25	325	1,879	1,492	461	657	1,058	3,738	2,794	1,824
Others	30	(163)	(290)	(51)	(198)	(476)	25	(264)	(244)	(260)
Total	1,656	1,154	2,763	2,229	536	1,589	1,965	3,251	2,569	1,847

Quarterly Data	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e
Production										
Bauxite (mn bdmt)	9	9	10	9	9	9	9	10	10	10
Alumina (kt)	2,435	2,390	2,355	2,351	2,453	2,390	2,355	2,387	2,467	2,467
Aluminum (kt)	559	571	564	572	579	604	607	612	630	630
Segment EBITDA										
Alumina	367	716	664	139	67	12	(40)	(110)	(47)	(26)
Aluminum	180	194	134	97	307	520	694	1,131	1,056	856
Operating Revenues	2,904	3,486	3,369	3,018	2,995	3,449	3,201	3,997	3,882	3,632
Quarterly EBITDA	455	677	855	313	270	527	595	956	934	766
Normalized Net Income	135	276	568	103	(6)	335	373	641	518	401
Normalized EPS ($/sh)	$0.58	$1.06	$2.18	$0.40	($0.02)	$1.27	$1.40	$2.41	$1.95	$1.51
Total Capex	146	169	93	131	151	243	119	170	215	245

Indebtedness	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Net Debt + Employee benefits	3,213	3,171	1,069	1,176	1,645	2,124	1,531	478	(317)	(756)
Net Debt + Employee benefits / EBITDA	2.0x	2.8x	0.4x	0.5x	3.1x	1.3x	0.8	0.1	(0.1)	(0.4)
Net Debt + Employee benefits / Equity	78.7%	95.8%	23.0%	23.3%	38.7%	41.2%	25.0%	5.7%	-3.3%	-7.3%
EBITDA / Interest	13.1	7.9	14.2	21.0	5.0	10.2	12.4	24.1	18.8	13.5

Valuation	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Tang. Book Value per Share	22.0	17.8	24.4	28.6	23.8	19.8	23.3	31.6	36.3	39.1
Cash Flow/ Share	$3.70	$2.12	$4.84	$4.65	$0.51	$2.39	$4.50	$7.15	$6.35	$5.05
P/E	**(21.7)**	**(19.9)**	**8.7**	**9.7**	**(15.0)**	**38.0**	**13.9**	**6.8**	**10.6**	**18.5**
EV - pension adjusted	8,084	8,337	12,830	9,891	8,437	10,931	14,696	12,709	11,901	11,451
EV/EBITDA - pension adjusted	**5.1x**	**7.3x**	**4.6x**	**4.4x**	**15.7x**	**6.9x**	**7.5x**	**3.9x**	**4.6x**	**6.2x**
EV - unadjusted	6,731	6,947	12,799	10,230	8,744	11,297	15,073	13,112	12,318	11,881
EV/EBITDA- unadjusted	4.2x	6.1x	4.6x	4.6x	16.3x	7.1x	7.7x	4.0x	4.8x	6.4x
Price to Book Value	1.0x	1.3x	2.4x	1.6x	1.4x	1.9x	2.3x	1.6x	1.4x	1.3x
ROE	-3.8%	-5.8%	32.6%	18.4%	-8.7%	6.3%	17.7%	26.6%	13.8%	7.1%
FCF Yield	-2.9%	-3.3%	3.0%	2.2%	-4.6%	0.6%	4.2%	8.8%	6.8%	4.1%
FCF Per Share	($0.61)	($0.76)	$1.77	$1.00	($1.58)	$0.22	$2.26	$4.40	$3.39	$2.05
Dividend Per Share	$0.00	$0.00	$0.10	$0.40	$0.40	$0.40	$0.40	$0.40	$0.40	$0.40
Dividend Yield	n/a	n/a	0.2%	0.9%	1.2%	1.1%	0.8%	0.8%	0.8%	0.8%

Income Statement, US$mn, YE Dec.	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Operating Revenues	10,433	9,286	12,152	12,451	10,551	11,895	12,831	14,704	13,649	12,303
Operating costs or COGS	8,537	7,969	9,153	10,212	9,813	10,044	10,639	11,097	10,759	10,142
Depreciation and Amortization	713	653	664	617	632	642	623	657	640	660
SG&A and R&D Expeses	307	233	258	236	265	332	323	364	321	314
Interest expense	121	146	195	106	107	156	158	135	137	137
Others	(984)	7	(1,272)	(893)	(121)	(287)	(1,025)	(44)	30	30
EBITDA	**1,589**	**1,147**	**2,763**	**2,229**	**536**	**1,589**	**1,965**	**3,251**	**2,569**	**1,847**
Pre Tax Op Income	1,739	278	3,154	2,173	(145)	1,008	2,113	2,495	1,762	1,021
Tax	501	226	522	444	156	351	228	567	529	306
Effective tax rate	70.8%	NM	25.8%	30.1%	-47.4%	52.6%	19.1%	22.9%	30.0%	30.0%
Minority Interest	391	163	207	143	(80)	20	(33)	(22)	(13)	0
Extraordinary and Others	(941)	45	(868)	(992)	(246)	(236)	170	52	0	0
Reported Net Income	(1,125)	(170)	429	(102)	(651)	60	1,170	1,984	1,246	715
Normalized Net Income	(184)	(215)	1,297	890	(405)	296	1,000	1,932	1,246	715
Normalized EPS	**($0.99)**	**($1.16)**	**$6.82**	**$4.70**	**($2.27)**	**$0.99**	**$3.83**	**$7.27**	**$4.69**	**$2.69**
Avg. Diluted shares out (mn)	185.6	185.9	189.9	177.0	178.5	260.5	263.3	265.9	265.9	265.9

Balance Sheet	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Cash and Equivalents	879	1,607	1,814	1,363	944	1,138	1,597	2,523	3,319	3,759
Inventories	1,644	1,398	1,956	2,427	2,158	1,998	2,177	2,164	2,193	2,067
Trade Receivables	660	556	884	909	808	1,239	1,268	1,313	1,292	1,206
Net PP&E	7,916	7,190	6,623	6,493	6,785	6,389	6,700	6,996	7,155	7,295
Deferred Tax Assets	649	655	504	308	333	283	771	663	663	663
Other Assets	2,892	3,454	3,222	3,283	3,130	3,018	2,302	2,580	2,580	2,580
Total Assets	14,640	14,860	15,003	14,783	14,158	14,065	14,815	16,238	17,202	17,570
Short term borrowings	1	79	76	1	135	125	10	110	110	110
Long term borrowings	1,799	2,463	1,726	1,806	1,732	2,470	2,438	2,222	2,222	2,222
Trade Payables	1,484	1,403	1,674	1,757	1,714	1,805	1,938	1,841	1,792	1,677
Taxes	380	300	564	450	282	111	450	341	341	341
Employee Benefits	2,292	2,236	1,081	732	722	667	680	669	670	672
Other Liabilities	2,828	3,363	3,633	3,465	3,727	3,728	4,487	4,080	3,822	3,564
Minority Interests	1,774	1,705	1,611	1,514	1,595	0	76	68	70	73
Total shareholders equity	4,082	3,311	4,638	5,058	4,251	5,159	6,133	8,392	9,661	10,397
Liabilities & Equity	14,640	14,860	15,003	14,783	14,158	14,065	16,136	17,655	18,618	18,982

Cashflow Statement	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Net Income	(1,125)	(170)	429	(102)	(651)	60	1,170	1,984	1,246	715
Depreciation and Amortization	713	653	664	617	632	642	623	657	640	660
Chg in Working Capital	(92)	239	(419)	(700)	65	(282)	(7)	(372)	(57)	98
Other Op cash flows	1,190	(328)	246	1,007	45	202	(601)	(369)	(142)	(130)
Net Cash from Operations	**686**	**394**	**920**	**822**	**91**	**622**	**1,185**	**1,901**	**1,688**	**1,342**
Acquisitions	0	0	0	0	0	0	0	0	0	0
Capex	(379)	(353)	(390)	(480)	(531)	(580)	(618)	(749)	(800)	(800)
Asset sale, investment, etc.	(89)	186	955	(15)	(54)	(28)	116	(10)	0	0
Net Cash (used in) Investing	**(468)**	**(167)**	**565**	**(495)**	**(585)**	**(608)**	**(502)**	**(759)**	**(800)**	**(800)**
Share issue/repurchase	2	1	(125)	(478)	0	0	0	0	0	0
Debt issue/repayment	(7)	738	(799)	3	55	353	(164)	(119)	0	0
Dividend paid	0	0	(19)	(72)	(72)	(90)	(105)	(106)	(106)	(106)
Minority Interest & others	(439)	(225)	(215)	(221)	74	(62)	8	5	14	3
Net cash (used in) Financing	**(444)**	**514**	**(1,158)**	**(768)**	**57**	**201**	**(261)**	**(220)**	**(91)**	**(102)**
Change in FX	(7)	(14)	(13)	(9)	10	(28)	36	3	0	0
Net Cash Inflow	(233)	727	314	(450)	(427)	187	458	925	796	440
Closing Cash Balance	879	1,607	1,814	1,363	944	1,138	1,597	2,523	3,319	3,759

E = Morgan Stanley Research estimates. Source: Company data, Morgan Stanley Research

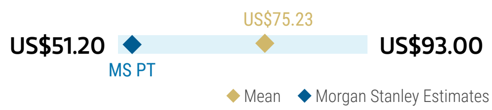
Risk Reward – Alcoa Corp (AA.N)

Aluminum Moving into Surplus while S32 Acquisition May Increase Leverage

PRICE TARGET US$53.00

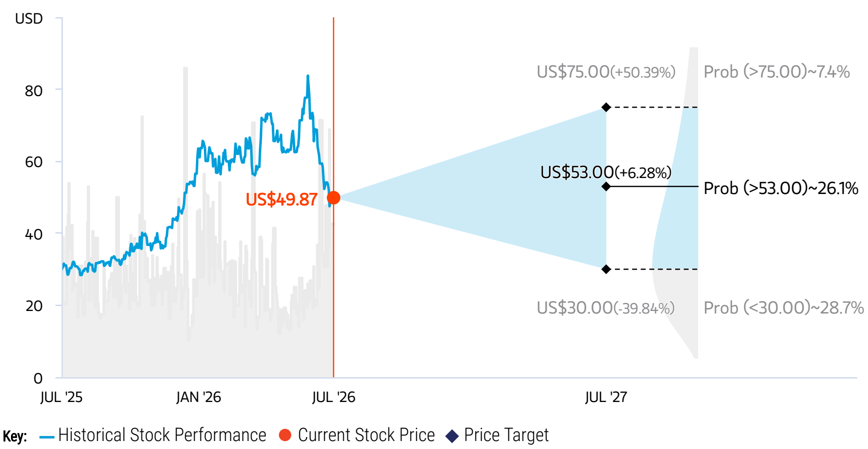
Our mid-2027 price target (was YE-2026) is derived using an EV/EBITDA multiple approach at a target multiple of 7.0x (vs. 5.3x), 0.25x SD above the five-year average EV/EBITDA multiple on our 2H27-1H28e EBITDA. Our target multiple reflects earnings normalizing at a higher level longer term than our latest anchor EBITDA as the company completes its mine move in WA and sees a step change in alumina segment profitability.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)

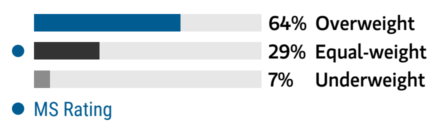


Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 6 Jul 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology here

EQUAL-WEIGHT THESIS

Alcoa has benefitted from elevated aluminum prices supported by recent disruptions in the Middle East. However, with the conflict subsiding and new supply coming online from RoW, particularly Indonesia, we expect aluminum prices to decline in 2027 and 1H28. AA's relatively high-cost smelters provide significant operating leverage to aluminum prices, driving sizeable downward revisions to earnings estimates, which we do not believe is reflected in consensus estimates. Moreover, we expect declining FCF generation in 2027 and 2028, which may strain the balance sheet following the S32 acquisition and limit any upside to shareholder returns via dividends and/or buybacks.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: *Positive*
Electric Vehicles: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE US$75.00

26e Aluminum: $1.71/lb; 7.0x 2H27-1H28 Bull EBITDA

Aluminum prices overshoot MSe as supply-side reform in China tightens the global market at a faster pace than expected. Disruptions in Middle East resume limiting supply from the region and pushing aluminum prices higher.

BASE CASE US$53.00

26e Aluminum: $1.48/lb; 7.0x 2H27-1H28 Base EBITDA

Downside to Aluminum prices driven by new supply from Indonesia and RoW, and a normalization of prices closer to the cash cost curve in line with historical levels. Aluminum supply in 2027 and 2028 tips into surplus

BEAR CASE US$30.00

26e Aluminum: $1.20/lb; 6.4x 2H27-1H28 Bear EBITDA

US enters a mild recession and aluminum prices undershoot MSe as demand softens. Moreover, the effects of supply-side reform in China take longer to materialize. Aluminum and Alumina markets are over-supplied through 2027/2028.

Risk Reward – Alcoa Corp (AA.N)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Aluminum ton sold (kmt)	2,522	2,679	2,719	2,726
Aluminum Adj. EBITDA/t (US$)	419.5	1,395.0	1,027.4	669.1
Alumina tons sold (kmt)	13,300	11,748	12,188	12,375
Alumina benchmark price (US$/t)	362.05	313.66	325.00	350.00
Alumina Adj.EBITDA/t (US$)	66.3	(19.0)	1.6	22.9

INVESTMENT DRIVERS

- Underlying commodity prices (i.e. aluminum, alumina)
- Energy and raw material prices, in particular natural gas and caustic soda

GLOBAL REVENUE EXPOSURE

- 0-10% Latin America
- 20-30% APAC, ex Japan, Mainland China and India
- 20-30% Europe ex UK
- 40-50% North America

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

2/5 BEST	24 Month Horizon	1/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Stronger-than-expected demand for aluminum and alumina; commodity prices overshoot MS forecasts
- Refining and smelting capacity rationalization in China
- Operational setbacks in the bauxite/alumina businesses are resolved

RISKS TO DOWNSIDE

- Bauxite and alumina operational setbacks continue
- Demand softens and metal prices undershoot MS estimates
- Additional smelting capacity is restarted faster than expected

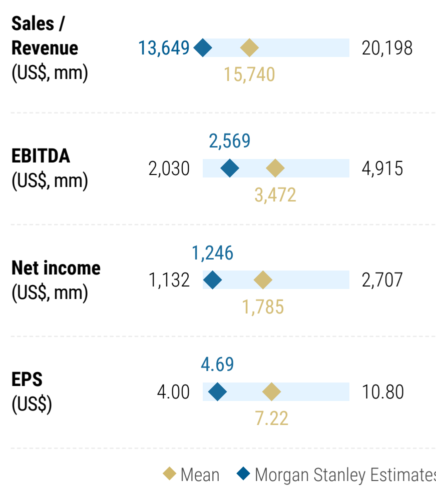
OWNERSHIP POSITIONING

Inst. Owners, % Active	47%
HF Sector Long/Short Ratio	2x
HF Sector Net Exposure	5.2%

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of June 30, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: Greater China Materials

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/07/2026)
Chris Jiang		
Anhui Honglu Steel Construction (002541.SZ)	U (12/16/2025)	Rmb18.10
CGN Mining Co Ltd (1164.HK)	O (01/18/2023)	HK$2.52
Chengxin Lithium Group Co. Ltd. (002240.SZ)	E (12/16/2025)	Rmb42.38
GEM Co Ltd (002340.SZ)	U (04/20/2026)	Rmb6.66
Shenzhen Kedali Industry Co Ltd (002850.SZ)	O (08/21/2023)	Rmb214.50
Sinomine Resource Group Co Ltd (002738.SZ)	O (12/16/2025)	Rmb61.05
Yongxing Special Materials Technology (002756.SZ)	E (11/25/2022)	Rmb58.83
Zhejiang Huayou Cobalt Co Ltd (603799.SS)	O (10/08/2025)	Rmb44.89
Hannah Yang, CFA		
China Hongqiao Group (1378.HK)	O (09/15/2023)	HK$20.92
Chuangxin Industries Holdings Ltd. (2788.HK)	O (03/19/2026)	HK$15.72
Flat Glass Group Co Ltd (6865.HK)	O (07/30/2020)	HK$6.19

Flat Glass Group Co Ltd (601865.SS)	O (07/30/2020)	Rmb9.80
MMG Ltd (1208.HK)	O (12/16/2024)	HK$7.02
Shandong Pharmaceutical Glass Co. Ltd. (600529.SS)	U (04/20/2026)	Rmb18.05
Shenhuo Coal and Power (000933.SZ)	O (09/02/2025)	Rmb21.78
Tianshan Aluminum (002532.SZ)	E (07/07/2026)	Rmb11.36
Xinyi Glass Holding Limited (0868.HK)	U (05/14/2024)	HK$8.47
Zhongfu Shenying Carbon Fiber Co Ltd (688295.SS)	O (08/25/2023)	Rmb48.34
Rachel L Zhang		
Aluminum Corp. of China Ltd. (601600.SS)	O (11/30/2020)	Rmb8.26
Aluminum Corp. of China Ltd. (2600.HK)	O (11/30/2020)	HK$7.58
Baoshan Iron & Steel (600019.SS)	O (01/16/2016)	Rmb5.56
Beijing New Building Materials (000786.SZ)	O (04/09/2024)	Rmb17.68
Beijing Oriental Yuhong Waterproof Techn (002271.SZ)	E (09/25/2024)	Rmb11.54
China Jushi (600176.SS)	O (12/22/2020)	Rmb60.00
China Lesso Group Holdings Ltd (2128.HK)	U (10/08/2025)	HK$3.82
China Steel Corp. (2002.TW)	U (12/16/2025)	NT$18.60
CMOC Group Ltd (3993.HK)	O (09/24/2019)	HK$15.20
CMOC Group Ltd (603993.SS)	O (06/21/2024)	Rmb17.69
CNGR Advanced Material Co., Ltd. (300919.SZ)	E (01/12/2026)	Rmb41.97
CNGR Advanced Material Co., Ltd. (2579.HK)	O (01/12/2026)	HK$25.02
FangDa Carbon New Material Co. Ltd. (600516.SS)	U (12/16/2024)	Rmb5.40
Ganfeng Lithium Co. Ltd. (1772.HK)	O (12/16/2025)	HK$47.18
Ganfeng Lithium Co. Ltd. (002460.SZ)	O (04/20/2026)	Rmb61.44
Henan Liliang Diamond Co. Ltd (301071.SZ)	U (04/20/2026)	Rmb85.13
Jiangsu Dingsheng New Materials (603876.SS)	U (04/20/2026)	Rmb24.45
Jiangxi Copper (0358.HK)	O (10/08/2025)	HK$30.50
Jiangxi Copper (600362.SS)	O (10/08/2025)	Rmb40.74
JL Mag Rare-Earth Co. Ltd (6680.HK)	O (04/23/2025)	HK$18.58
JL Mag Rare-Earth Co. Ltd (300748.SZ)	O (04/23/2025)	Rmb31.69
Nine Dragons Paper (2689.HK)	E (01/04/2023)	HK$7.36
Shandong Gold Mining Co. Ltd (600547.SS)	O (04/23/2025)	Rmb24.77
Shandong Gold Mining Co. Ltd (1787.HK)	O (12/12/2024)	HK$19.06
Shandong Nanshan Aluminium Co. (600219.SS)	E (07/07/2026)	Rmb4.09
Tianqi Lithium Industries Inc. (9696.HK)	O (12/16/2025)	HK$39.54
Tianqi Lithium Industries Inc. (002466.SZ)	O (04/20/2026)	Rmb59.19
Weixing New Building Materials (002372.SZ)	U (10/08/2025)	Rmb8.48
Zhaojin Mining Industry (1818.HK)	O (06/21/2024)	HK$18.44
Zijin Gold International (2259.HK)	O (11/06/2025)	HK$105.60
Zijin Mining Group (2899.HK)	O (11/06/2025)	HK$30.20
Zijin Mining Group (601899.SS)	O (11/06/2025)	Rmb27.62

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.